arXiv:2603.11896v1 [cs.CV] 12 Mar 2026

# Think While Watching: Online Streaming Segment-Level Memory for Multi-Turn Video Reasoning in Multimodal Large Language Models

**Lu Wang**[1] **Zhuoran Jin**[1] **Yupu Hao**[1] **Yubo Chen**[1]
Kang Liu[1] Yulong Ao[2] Jun Zhao[1]
[1]The Key Laboratory of Cognition and Decision Intelligence for Complex Systems, Institute of Automation, Chinese Academy of Sciences, Beijing, China
[2]Beijing Academy of Artificial Intelligence (BAAI), Beijing, China
wanglu2026@ia.ac.cn, zhuoran.jin@nlpr.ia.ac.cn, haoyupu2023@ia.ac.cn
yubo.chen@nlpr.ia.ac.cn, kliu@nlpr.ia.ac.cn
aoyulong@outlook.com, jzhao@nlpr.ia.ac.cn

## Abstract

Multimodal large language models (MLLMs) have demonstrated strong performance in offline video understanding tasks, but most remain constrained to offline inference or exhibit weak online reasoning ability, rendering online multi-turn interaction over continuously arriving video streams challenging. Existing streaming approaches adopt an interleaved perception-generation paradigm, which precludes concurrent perception and generation and induces early memory decay with growing streams, degrading long-range dependency modeling. We propose **Think While Watching**, a memory-anchored streaming video reasoning framework that maintains continuous segment-level memory during multi-turn interaction. We construct a three-stage, multi-round, chain-of-thought (CoT) dataset with a stage-matched training strategy while enforcing strict causality in streaming reasoning via a segment-level streaming causal mask and streaming positional encoding. At inference, we design an efficient pipeline that overlaps watching and thinking and adaptively selects the optimal attention backend. We evaluate our method under single-round and multi-round streaming input protocols. Based on Qwen3-VL, we improve single-round accuracy by 2.6% on StreamingBench and 3.79% on OVO-Bench. In the multi-round protocol, we maintain performance while reducing output tokens by 56%. Our code is available at GitHub.

## 1 Introduction

Video understanding and reasoning are becoming central capabilities for multimodal assistants. While multimodal large language models (MLLMs) have achieved strong performance in offline video benchmarks where the full video is available before inference in a single-turn setting [1–4], many high-impact scenarios are inherently **streaming**: live broadcasting [5, 6], monitoring, robotic assistants [7], and other streaming scenarios. In these settings, users may ask questions at any time, and the assistant must answer in real time while staying faithful to the visual evidence observed so far, especially under **multi-turn** interaction, where later questions often depend on earlier memories.

A widely adopted approach for streaming MLLMs is to interleave perception and generation [8, 9, 6]. Although this reduces delay compared to offline approaches, it remains fundamentally serialized: text decoding blocks further video ingestion. This interleaved pattern leads to two phenomena. First, **Memory Erosion**: multi-turn subsequent questions frequently refer back to earlier questions or earlier visual cues, but interleaving with generation tends to erode long-range capability. The failure case in Fig. 1(a) makes this explicit: a later query about the first two questions

Preprint.

arXiv:2603.11896v1 [cs.CV] 12 Mar 2026

# 邊觀看邊思考：多模態大型語言模型中的多輪視頻推理線段級記憶

盧王[1] 金卓然[1] 郝宇璞[1] 陳宇博[1]劉康[1] 敖宇龍[2] 趙俊[1]中國科學院自動化研究所複雜系統認知與決策智識實驗室，北京，中國[2]北京人工智能學院 (BAAI)，北京，中國 wanglu2026@ia.ac.cn, zhuoran.jin@nlpr.ia.ac.cn, haoyupu2023@ia.ac.cn yubo.chen@nlpr.ia.ac.cn, kliu@nlpr.ia.ac.cn aoyulong@outlook.com, jzhao@nlpr.ia.ac.cn

## 摘要

多模態大型語言模型 (MLLMs) 在離線視頻理解任務中展現出強勢性能，但大多數仍受限於離線推論或表現出較弱的線上推理能力，導致對持續到來的視頻流進行線上多輪互動變得挑戰重重。現有的流式處理方法採用交錯感知-生成模式，這種模式阻礙了同時感知和生成，並隨著視頻流的增長引發早期記憶衰退，損害了長程依賴建模。我們提出「邊觀看邊思考」，一個記憶錨定的流式視頻推理框架，在多輪互動期間維持持續的線段級記憶。我們構建了一個三階段、多輪次、鏈式思考 (CoT) 數據集，採用與階段匹配的訓練策略，並通過線段級流式因果掩碼和流式位置編碼嚴格確保流式推理中的因果關係。在推論階段，我們設計了一個高效的流程，使觀看和思考重疊進行，並自適應選擇最佳注意力後端。我們在單輪次和多輪次流式輸入協議下評估了我們的方法。基於 Qwen3-VL，我們在 StreamingBench 上將單輪次準確率提升了 2.6%，在 OVO-Bench 上提升了 3.79%。在多輪次協議中，我們在保持性能的同時將輸出 token 數量減少了 56%。我們的代碼已開放到 GitHub。

## 1 前言

視頻理解與推理正逐漸成為多模態助理的核心能力。雖然多模態大型語言模型（MLLMs）在離線視頻基準測試中取得了強勁的表現，這些測試在單輪設定中在推理前提供完整視頻 [1–4]，但許多高影響力場景本質上是流式傳輸的：直播 [5, 6], 監控、機器人助理 [7], 以及其他流式傳輸場景。在這些設定中，用戶隨時可以提問，而助理必須在保持對迄今觀察到的視覺證據忠誠的同時實時回答，尤其是在多輪互動中，後續問題經常依賴早期的記憶。

一種廣泛採用的串流 MLLMs 方法是交錯感知和生成[8, 9, 6]。雖然這與離線方法相比減少了延遲，但它仍然基本上是串行的：文本解碼會阻礙進一步的視頻攝取。這種交錯模式導致兩種現象。首先，記憶侵蝕：多輪後續問題經常參照早期問題或早期的視覺提示，但與生成交錯 tend to 會侵蝕長程能力。圖 1(a) 中的失敗案例使這一點顯而易見：一個關於前兩個問題的後續查詢

草稿。

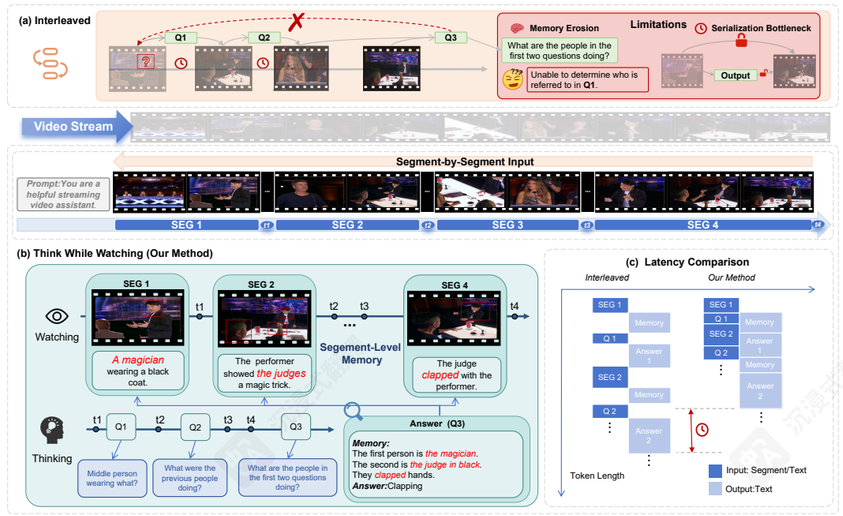



Figure 1: **Overview of Think While Watching. (a) Interleaved baseline.** Video perception and answer generation are executed sequentially, which can cause **memory erosion**, where early memory is forgotten, and a **serialization bottleneck**, where generation stalls further input ingestion. **(b) Think While Watching (ours).** The video frames are processed in segments (SEG 1 to SEG 4) to build a continuous **segment-level memory**. During streaming, questions are answered online by retrieving implicitly relevant memories while continuing to watch. **(c) Latency comparison.** A schematic timeline showing that interleaved processing accumulates queueing delay, while our decoupled design parallelizes segment processing and answering to reduce latency.

becomes unanswerable because the model forgets who Q1 refers to. This issue is also validated by our experimental results: for the Qwen3-VL-4B Thinking model, accuracy in the online multi-round setting drops $40.39\%$ compared to the offline setting, highlighting the severe challenge of maintaining long-term temporal consistency. Second, **Serialization Bottleneck**: as illustrated on the top right of Fig. 1(a), once the model starts generation, the decoder effectively locks the streaming sequence, directly harming responsiveness in dynamic streams. The root cause is that autoregressive models use unified positional encoding, so new inputs must align with generated outputs whose length is unknown, forcing ingestion to pause and causing a serialization bottleneck. Fig. 1(c) further visualizes this effect: under interleaved processing, as the number of rounds accumulates, the input keeps piling up, leading to increasing end-to-end latency. To mitigate **Memory Erosion**, we make memory writing an explicit online behavior: for each observed segment, the model writes a memory note and appends it to a memory bank; when a question arrives, the model answers by implicitly integrating the relevant notes via the attention mechanism in Fig. 1(b). To break the **Serialization Bottleneck**, we assign independent positional encodings to decouple input and output streams at inference time, enabling input-output parallelism so the model can keep watching while thinking and thus reduce latency.

We propose **Think While Watching**, a memory-anchored streaming video reasoning framework for online multi-turn interaction. We represent a continuously arriving video as a sequence of segments and maintain a persistent **segment-level memory** throughout the dialogue. To make the framework practical, we design corresponding training and inference procedures. On the training side, we construct a three-stage, multi-round chain-of-thought (CoT) dataset with training matched to each stage, together with a **segment-level streaming causal mask** and **streaming positional encoding**, which jointly enforce strict causality throughout streaming reasoning. On the inference side, we design an efficient pipeline that overlaps watching with thinking. Our implementation is inspired by CPU process scheduling [10] in operating systems: we organize inference as a **multi-stage pipeline**



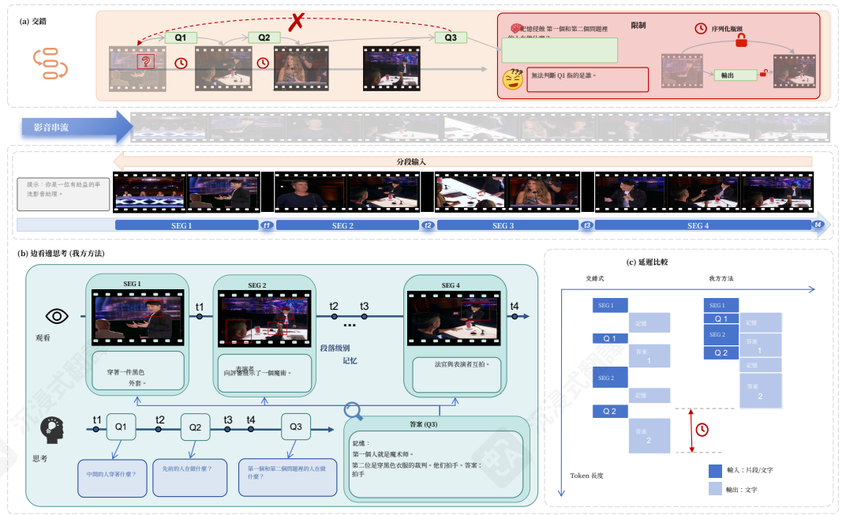



圖1：思考邊觀看（Think While Watching）的概觀。(a)交錯基準線。影音感知與答案產生是依序執行，這可能會導致記憶侵蝕，即早期記憶被遺忘，以及序列瓶頸，即產生過程會停滯進一步的輸入吸收。(b)思考邊觀看（我們的方法）。影音畫面是分段（SEG 1 至 SEG 4）處理以建立連續的段落層級記憶。在串流過程中，透過持續觀看時回撥隱含相關的記憶來線上回答問題。(c)延遲比較。一個示意時間線，顯示交錯處理會累積排隊延遲，而我們的解耦設計並行處理段落與回答以降低延遲。

變得無法回答，因為模型會忘記 Q1 指的是誰。這個問題也由我們的實驗結果驗證：對於 Qwen3-VL-4B 思考模型，在線多輪設置中的準確率與離線設置相比下降了 40.39%，突顯了維持長期時序一致性所面臨的嚴峻挑戰。其次，序列化瓶頸：如圖 1(a) 右上所示，一旦模型開始生成，解碼器會有效地鎖定流式序列，直接損害動態流中的響應性。根本原因是自回歸模型使用統一的位姿編碼，因此新輸入必須與長度未知的最終輸出對齊，迫使攝取操作暫停，並導致序列化瓶頸。圖 1(c) 進一步可視化了這種效應：在交錯處理下，隨著輪數的累積，輸入不斷堆積，導致端到端延遲增加。為了緩解記憶侵蝕，我們將記憶寫入作為一個明確的在線行為：對於每個觀察到的片段，模型寫入一條記憶筆記並附加到記憶庫中；當問題到來時，模型通過在圖 1(b) 中的注意力機制中隱式整合相關筆記來回答。為了打破序列化瓶頸，我們為解碼器分配獨立的位姿編碼，在推理時將輸入和輸出流解耦，實現輸入輸出並行，使模型能在思考時持續觀察，從而降低延遲。

我們提出「邏輯觀看」(Think While Watching)，一個基於記憶的串流視頻推理框架，用於線上多輪互動。我們將持續到來的視頻表示為一系列片段，並在對話中維持持續的片段層記憶。為了讓框架更具實用性，我們設計了對應的訓練和推論程序。在訓練方面，我們構建了一個三階段、多輪的思考链 (CoT) 數據集，每個階段都有對應的訓練內容，以及片段層的串流因果掩碼和串流位置編碼，它們共同強制在串流推理過程中保持嚴格的因果關係。在推論方面，我們設計了一個高效的流程，將觀看與思考重疊。我們的實現受到作業系統中 CPU 處理序程調度的啟發 [10]：我們將推論組織為一個多階段流程

as illustrated in Fig. 1(c) and decouple continuous visual ingestion from text decoding via a dual KV cache [11–13], enabling parallelism between perception and generation and mitigating serialization.

We evaluate Think While Watching under two streaming input protocols: **single-round**, where the input contains many arriving segments but the assistant answers one question, and **multi-round**, where the input contains many arriving segments and the assistant answers multiple questions over time. In the Qwen3-VL family, we improve single-round accuracy by 2.6% on StreamingBench and 3.79% on OVO-Bench. In the multi-round protocol, we preserve performance while reducing output tokens by 56%.

**Our contributions** are as follows:

- We propose **Think While Watching**, which maintains **segment-level memory** as a persistent state and answers each query by implicitly retrieving and integrating relevant memories, improving multi-turn consistency and enabling decoupled perception and generation. We further provide a practical **training and inference** pipeline with **three-stage training**, **streaming segment masking**, and **streaming positional encoding** for causal segment-level modeling, and a dual KV cache at inference time to support parallelism between perception and generation.
- We construct a three-stage, stage-aligned streaming CoT dataset with multi-round dialogues to support the proposed training strategy.
- On Qwen3-VL, we improve single-round accuracy by 2.6% on StreamingBench and 3.79% on OVO-Bench, while in multi-round streaming, we reduce output tokens by 56% without accuracy drop.

## 2 Related Work

### 2.1 Offline Video Understanding

Offline video MLLMs are improved by structured perception and cognition pipelines and temporal reasoning designs [4, 1, 2], and by reinforcement learning for complex temporal reasoning [3, 14]. Most of these methods assume the full video is available before answering, leaving causal online multi-turn interaction less explored.

### 2.2 Online Streaming Video Understanding

**Benchmarks.** StreamingBench evaluates the gap between offline models and streaming video understanding [15], while OVO-Bench emphasizes real-world online video understanding [16]. Recent work further studies streaming along with active perception and multi-turn interaction [17–21].

**Interleaved perception and generation.** Many streaming systems alternate visual ingestion and text decoding, as in VideoLLM-online [8] and StreamChat [9], and scale streaming supervision for online interaction [6, 22]. This coupling limits input-output parallelism and makes it harder to model dependencies over a long horizon across multiple turns.

**Memory and efficient inference for long-horizon streaming.** For efficiency, one line reduces redundant visual tokens in streaming videos [23–26]. Another line reuses historical context via KV cache retrieval and compression [27–32]. Persistent memory and long-term multimodal agent memory further support evidence reuse across long streams [33–35]. Our work emphasizes stable segment-level memory for multi-turn streaming and an inference design that keeps perception and generation decoupled.

## 3 Preliminary

This section introduces the online multi-turn streaming video question answering setting studied in this work. A video is observed sequentially as a stream of segments, while a user may ask questions at arbitrary segment boundaries. The central requirement is strict streaming causality: at each turn, the system must produce its response using only the video content observed so far and the dialogue history, without accessing any future segments.



，如圖1(c)所示，並通過雙 KV 緩存 [11–13], 將連續的視覺輸入與文本解碼解耦，實現感知與生成之間的並行，並減輕序列化問題。

我們在兩種串流輸入協議下評估 Think While Watching：單輪，輸入包含許多到達的片段但輔助工具僅回答一個問題，以及多輪，輸入包含許多到達的片段且輔助工具在時間上回答多個問題。在 Qwen3-VL 系列中，我們在 StreamingBench 上將單輪準確率提升了 2.6%，在 OVO-Bench 上提升了 3.79%。在多輪協議中，我們在降低輸出 token 數量 56% 的同時保留了性能。

**我們的貢獻如下：**

- 我們提出 Think While Watching，它維持片段級別記憶作為持續狀態，透過隱式檢索和整合相關記憶來回答每個查詢，提升多輪一致性並實現感知與生成的解耦。我們進一步

  提供一個實用的訓練和推理管線，包含三階段訓練、串流片段遮罩以及串流位置編碼以支援因果片段級別建模，並在推理時採用雙 KV 緩存以支援感知與生成之間的並行處理。

- 我們構建了一個三階段、階段對齊的串流 CoT 數據集，包含多輪對話以支援所提出的訓練策略。

- 在Qwen3-VL上，我們將單輪準確率在StreamingBench上提升了2.6%，在OVO-Bench上提升了3.79%，而在多輪流式處理中，我們減少了56%的輸出token數量，且準確率不下降。

## 2 相關工作

### 2.1 線下視頻理解

線下視頻MLLMs透過結構化感知和認知流程設計以及時間推理設計 [4, 1, 2], ，並透過強化學習來處理複雜的時間推理 [3, 14]進行了改進。大多數這些方法假設在回答前可以獲得完整視頻，導致因果線上多輪互動探索較少。

### 2.2 線上流式視頻理解

基準測試。StreamingBench評估了線下模型與流式視頻理解之間的差距 [15], ，而OVO-Bench則強調真實世界線上視頻理解 [16]。近期研究進一步探討了流式處理與主動感知和多輪互動 [17–21]。

交錯的感知與生成。許多串流系統交替視覺輸入與文字解碼，例如 VideoLLM-online [8] 和 StreamChat [9], ，並擴展串流監督以實現線上互動 [6, 22]。這種耦合限制了輸入輸出並行性，並使在多個回合間跨長時間範圍內建立依賴關係變得更加困難。

記憶與長時間範圍內高效的推論。為了提高效率，一條路徑在串流影片中減少重複的視覺標記 [23–26]。另一條路徑透過 KV 緩存檢索和壓縮重用歷史上下文 [27–32]。持續記憶和長期多模態代理記憶進一步支持在長串流中重用證據 [33–35]。我們的工作強調穩定的段落級記憶，以實現多回合串流，並採用保持感知與生成解耦的推論設計。

## 3 初步

本節介紹本研究探討的線上多輪串流視頻問答設定。一個視頻會被依序觀察為一系列片段，而使用者可以在任意片段邊界提出問題。核心要求是嚴格的串流因果性：在每一輪，系統必須僅使用到目前觀察到的視頻內容和對話歷史來產生其回應，而不存取任何未來的片段。

### 3.1 Streams and Turns

**Segmented stream.** We represent a video stream as an ordered sequence of segments

$$\mathbf{S}_{1:T} \triangleq \langle \mathbf{S}_1, \ldots, \mathbf{S}_T \rangle, \tag{1}$$

where each $\mathbf{S}_t$ denotes a contiguous chunk of frames. Segments arrive in temporal order, and the system processes them online.

**Multi-turn questioning.** We consider an interaction with $R$ question and answer turns. At turn $r \in \{1, \ldots, R\}$, the user asks a question $q_r$ after the system has observed a prefix of the stream. Let $\tau_r \in \{1, \ldots, T\}$ denote the index of the latest observed segment when $q_r$ is issued. Equivalently, $q_r$ is asked after ingesting the segment prefix

$$\mathbf{S}_{1:\tau_r} \triangleq \langle \mathbf{S}_1, \ldots, \mathbf{S}_{\tau_r} \rangle. \tag{2}$$

Since questions arrive over time, the indices are nondecreasing:

$$1 \le \tau_1 \le \tau_2 \le \cdots \le \tau_R \le T. \tag{3}$$

The dialogue history before turn $r$ is

$$\mathcal{H}_{r-1} \triangleq \langle \langle q_1, a_1 \rangle, \ldots, \langle q_{r-1}, a_{r-1} \rangle \rangle. \tag{4}$$

Under strict causality, the answer $a_r$ at turn $r$ is conditioned only on the observed video prefix $\mathbf{S}_{1:\tau_r}$, the question $q_r$, and the dialogue history $\mathcal{H}_{r-1}$.

### 3.2 Streaming Protocols

We consider two online evaluation protocols that share the same segmented stream $\mathbf{S}_{1:T}$ but differ in the number of question turns.

**Single-round streaming.** Only one question is asked, so $R = 1$. The system processes segments online and produces a single output for the question asked at $\tau_1$. We denote the model output as $\langle \pi_1, a_1 \rangle$, where $\pi_1$ is an optional intermediate rationale such as chain of thought and $a_1$ is the final answer.

**Multi-round streaming.** Multiple questions are asked, so $R > 1$, at different times with nondecreasing $\tau_r$. At each turn $r$, the system must answer online using only the stream prefix $\mathbf{S}_{1:\tau_r}$ and the dialogue history $\mathcal{H}_{r-1}$, producing an output pair $\langle \pi_r, a_r \rangle$.

### 3.3 Streaming Unit Notation

To describe training and inference in a single causal formulation, we serialize a streaming interaction as an interleaved sequence of received units and a one-to-one aligned sequence of generated units.

**Received units.** Let the received unit sequence be

$$\mathbf{R}_{1:U} \triangleq \langle R_1, \ldots, R_U \rangle, \tag{5}$$

where each $R_u$ is either a visual segment unit $S_t$ that contains the content of $\mathbf{S}_t$, or a question unit $Q_r$ that contains the text $q_r$. We write $R_u \in \{S, Q\}$ to indicate the unit type. Let $\mathrm{idx}[\cdot]$ return the arrival index in $\mathbf{R}_{1:U}$, so $\mathrm{idx}[S_t]$ is the index $u$ where segment $t$ appears, and $\mathrm{idx}[Q_r]$ is the index $u$ where question $r$ appears.

**Generated units.** For each received unit $R_u$, the model generates exactly one output unit $C_u$ in the same order, forming

$$\mathbf{C}_{1:U} \triangleq \langle C_1, \ldots, C_U \rangle. \tag{6}$$

If $R_u = S_t$, then $C_u$ is a memory note denoted $m_t$. If $R_u = Q_r$, then $C_u$ is the question answering output that contains the rationale $\pi_r$ and answer $a_r$.

**Token lengths and visual grids.** For any text unit $Y$ in $\{Q_1, \ldots, Q_R, C_1, \ldots, C_U\}$, let $L[Y]$ denote its text token length. For any segment unit $S_t$, let its visual token grid sizes be $\langle T_t, H_t, W_t \rangle$. Here $T_t$ is the number of visual tokens along the temporal axis, $H_t$ is the height axis, and $W_t$ is the width axis, defined by the vision encoder token grid for this segment. We will also use a unit span function $\Delta[R_u]$ that assigns each received unit a nonoverlapping input position span:

$$\Delta[R_u] = \begin{cases} \max\{T_u, H_u, W_u\}, & R_u \in \{S\}, \\ L[R_u], & R_u \in \{Q\}. \end{cases} \tag{7}$$



### 3.1 流與轉換

區段式串流。我們將視頻串流表示為一個有序的區段序列

$$\mathbf{S}_{1:T} \triangleq \langle \mathbf{S}_1, \ldots, \mathbf{S}_T \rangle, \tag{1}$$

其中每個 $S_t$ 代表一個連續的畫面片段。片段會以時間順序到達，系統則會即時處理它們。

多輪問答。我們考慮一個包含 $R$ 問題與回答輪次的互動。在輪次 $r \in \{1, \ldots, R\}$ 時，使用者會在系統觀察到串流前綴後問一個問題 $q_r$。讓 $\tau_r \in \{1, \ldots, T\}$ 代表當 $q_r$ 發出時最新觀察到的片段索引。同義地， $q_r$ 是在消化串流前綴後才被問到的。

$$\mathbf{S}_{1:\tau_r} \triangleq \langle \mathbf{S}_1, \ldots, \mathbf{S}_{\tau_r} \rangle. \tag{2}$$

由於問題會隨時間到達，索引值是非遞減的：

$$1 \le \tau_1 \le \tau_2 \le \cdots \le \tau_R \le T. \tag{3}$$

在輪次 $r$ 之前對話歷史是

$$\mathcal{H}_{r-1} \triangleq \langle \langle q_1, a_1 \rangle, \ldots, \langle q_{r-1}, a_{r-1} \rangle \rangle. \tag{4}$$

在嚴格的因果關係下，輪次 $r$ 的回答 $a_r$ 只受觀察到的視頻前綴 $\mathbf{S}_{1:\tau_r}$、問題 $q_r$ 與對話歷史 $\mathcal{H}_{r-1}$ 的影響。

### 3.2 動態協議

我們考慮兩種線上評估協議，它們共享相同的分段流 $\mathbf{S}_{1:T}$，但在問題輪數上有所不同。

單輪動態。只問一個問題，所以 $R = 1$。系統線上處理段，並在 $\tau_1$ 時為所問問題產生單一輸出。我們將模型輸出表示為 $\langle \pi_1 \ a_1 \rangle$，其中 $\pi_1$ 是一個可選的 intermediate 理由，例如 chain of thought，而 $a_1$ 是最終答案。

多輪動態。問多個問題，所以 $R > 1$，在不同時間以非遞減的 $\tau_r$。在每一輪 $r$，系統必須使用線上流前綴 $\mathbf{S}_{1:\tau_r}$ 和對話歷史 $\mathcal{H}_{r-1}$ 來回答，並產生輸出對 $\langle \pi_r, a_r \rangle$。

### 3.3 動態單位符號

為了單一因果公式中的訓練和推理進行描述，我們將動態互動序列化為接收單位和生成單位的交錯序列，以及一對一對齊的序列。

已接收單位。令接收單位序列為

$$\mathbf{R}_{1:U} \triangleq \langle R_1, \ldots, R_U \rangle, \tag{5}$$

其中每個 $R_u$ 要么是包含 $S_t$ 內容的視覺片段單位 $S_t$，要么是包含文本 $q_r$ 的問答單位 $Q_r$。我們用 $R_u \in \{S, Q\}$ 來標示單位類型。令 $\mathrm{idx}[\cdot]$ 返回在 $\mathbf{R}_{1:U}$ 中的到達索引，所以 $\mathrm{idx}[S_t]$ 是片段 $t$ 出現的索引 $u$，而 $\mathrm{idx}[Q_r]$ 是問答 $r$ 出現的索引 $u$。

生成單位。對於每個接收單位 $R_u$，模型以相同順序生成恰好一個輸出單位 $C_u$，形成

$$\mathbf{C}_{1:U} \triangleq \langle C_1, \ldots, C_U \rangle. \tag{6}$$

若 $R_u = S_t$，則 $C_u$ 是表示為 $m_t$ 的記憶筆記。若 $R_u = Q_r$，則 $C_u$ 是包含推理過程 $\pi_r$ 和答案 $a_r$ 的問答輸出。

Token 長度和視覺網格。對於 $\{Q_1, \ldots, Q_R, C_1, \ldots, C_U\}$ 中的任何文本單元 $Y$，讓 $L[Y]$ 表示其文本 token 長度。對於任何片段單元 $S_t$，讓其視覺 token 網格大小為 $\langle T_t, H_t, W_t \rangle$。這裡 $T_t$ 是沿著時間軸的視覺 token 數量， $H_t$ 是高度軸， $W_t$ 是寬度軸，由視覺編碼器 token 網格為此片段定義。我們還將使用一個單元跨度函數 $\Delta[R_u]$，為每個接收到的單元分配一個非重疊的輸入位置跨度：

$$\Delta[R_u] = \begin{cases} \max\{T_u, H_u, W_u\}, & R_u \in \{S\}, \\ L[R_u], & R_u \in \{Q\}. \end{cases} \tag{7}$$

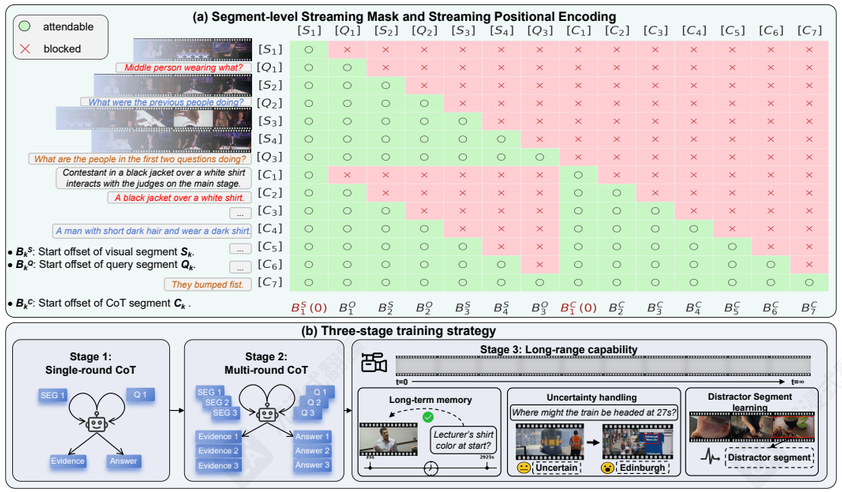



Figure 2: **Training components of Think While Watching.** (a) segment-level streaming attention mask and streaming positional encoding: example input stream $\mathbf{R} = \langle S_1, Q_1, S_2, Q_2, S_3, S_4, Q_3 \rangle$ with generated outputs $\mathbf{C} = \langle C_1, \ldots, C_7 \rangle$. Green indicates the source prefix available at time step $u$, which $C_u$ is allowed to attend to. Red masks all future segments to prevent information leakage. For positional encoding, we use separate position indices for the input and output streams. (b) Three-stage training strategy: single-round CoT for streaming input adaptation, multi-round CoT for multi-turn interaction, and long-range capability training for long-term memory, uncertainty handling, and distractor learning.

## 4 Method

A simple streaming design interleaves perception and generation but is inherently serial: autoregressive decoding halts further input ingestion, and the interleaving pattern mismatches the LLM pretraining format. Think While Watching generates segment-level memory notes online and decouples perception and generation, enabling interaction in real time across multiple turns. Details are shown in Fig. 2.

### 4.1 Segment-level Memory Notes

To mitigate Memory Erosion, our method maintains a persistent segment-level memory bank as the online state for multi-turn streaming. For each arriving segment $\mathbf{S}_t$, the model writes exactly one memory note and appends it to the bank. The memory bank after observing the segment prefix $\mathbf{S}_{1:t}$ is defined as

$$\mathcal{M}_t \triangleq \{\langle i, m_i \rangle\}_{i=1}^{t}. \tag{8}$$

Each note $m_t$ is a compact text unit grounded in $\mathbf{S}_t$ that records reusable evidence, including key entities and attributes, salient actions and interactions, scene changes, and short-range temporal relations. We denote the memory-writing function implemented by the MLLM backbone with parameters $\theta$ as $\mathrm{Mem}_\theta[\cdot]$, and write

$$m_t = \mathrm{Mem}_\theta[\mathbf{S}_t], \qquad C_{\mathrm{idx}[S_t]} = m_t. \tag{9}$$

When a question $q_r$ is issued after observing segment $\tau_r$, the model answers by conditioning on the current question, the dialogue history, and the available memory prefix, while letting attention implicitly select relevant notes:

$$\langle \pi_r, a_r \rangle \sim p_\theta[\pi_r, a_r \mid q_r, \mathcal{H}_{r-1}, \mathcal{M}_{\tau_r}]\,. \tag{10}$$



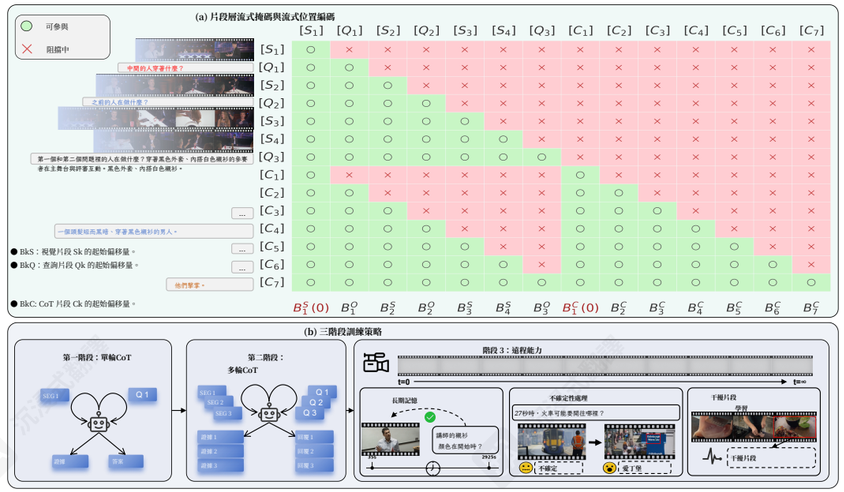



圖2：Think While Watching 的訓練組成部分。(a) 段級流式注意力掩碼和流式位置編碼：示例輸入流 R ＝ $\langle S_1, Q_1, S_2, Q_2, S_3, S_4, Q_3 \rangle$與生成的輸出 $\mathbf{C} = \langle C_1 \ldots C_7 \rangle$, , 。綠色表示在時間步$u$時可用的來源前綴， $C_u$ 允許其參與注意。紅色掩碼所有未來的段以防止信息洩露。在位置編碼方面，我們對輸入和輸出流使用獨立的位置索引。(b) 三階段訓練策略：單輪CoT用於流式輸入適應，多輪CoT用於多輪互動，以及長程能力訓練用於長期記憶、不確定性處理和干擾學習。

## 4 方法

一種簡單的流式設計交錯感知和生成，但天生是串行的：自回歸解碼會停止進一步的輸入攝取，且交錯模式與 LLM 預訓練格式不匹配。Think While Watching 在線生成片段級別的記憶筆記，解耦感知和生成，允許在多輪對話中實時互動。詳情請參考圖 2。

### 4.1 段級記憶筆記

為了緩解記憶侵蝕，我們的方法維持一個持久的段級記憶庫作為多輪流式處理的線上狀態。對於每個到達的段 $S_t$，模型寫入一個確切的記憶筆記並附加到該庫中。觀察段前綴 $S_{1:t}$後的記憶庫定義為

$$\mathcal{M}_t \triangleq \{\langle i, m_i \rangle\}_{i=1}^{t}. \tag{8}$$

每個筆記 $m_t$ 是一個基於 $S_t$ 的緊湊文本單位，記錄可重用的證據，包括關鍵實體和屬性、顯著的行為和互動、場景變化以及短程時間關係。我們用 $\mathrm{Mem}_\theta[\cdot]$ 表示由 MLLM 主幹實現的記憶寫入功能，其參數為 $\theta$ ，並寫

$$m_t = \mathrm{Mem}_\theta[\mathbf{S}_t], \qquad C_{\mathrm{idx}[S_t]} = m_t. \tag{9}$$

當在觀察到片段 $q_r$ 後發問時，模型透過條件化當前問題、對話歷史以及可用的記憶前綴來回答，同時讓注意力隱含地選擇相關筆記：

$$\langle \pi_r, a_r \rangle \sim p_\theta[\pi_r, a_r \mid q_r, \mathcal{H}_{r-1}, \mathcal{M}_{\tau_r}]\,. \tag{10}$$

### 4.2 Streaming Architecture

**Prefix and suffix formatting.** To match the standard SFT format of LLMs, we serialize each training example as a source prefix followed by a target suffix. The source prefix contains the entire received unit stream $\mathbf{R}_{1:U}$, while the target suffix contains the aligned generated stream $\mathbf{C}_{1:U}$. Without additional constraints, this serialization would leak future received units to earlier generated units.

**Segment-level streaming attention mask.** We feed the concatenated sequence $\langle R_1, \ldots, R_U, C_1, \ldots, C_U \rangle$ with a segment-level mask $M^{\text{seg}}$ to enforce streaming causality. Let $A$ denote the segment that contributes query tokens and $B$ denote the segment that contributes key and value tokens, with $A, B \in \{R_1, \ldots, R_U, C_1, \ldots, C_U\}$. The mask is defined as:

$$M^{\text{seg}}[A,B] = \begin{cases} \mathbb{I}[v \le u], & A = R_u,\ B = R_v, \\ \mathbb{I}[v \le u], & A = C_u,\ B = R_v, \\ \mathbb{I}[k \le u], & A = C_u,\ B = C_k, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Here $u$ is the arrival index of the querying unit $A$. For the attended unit $B$, we use $v$ if $B$ is a received unit $R_v$ and $k$ if $B$ is a generated unit $C_k$. The first three cases in Eq. (11) enforce streaming causality: the received stream is causal in arrival order, each generated unit $C_u$ can attend to the received prefix up to step $u$, and generated units are causal with access only to $C_{1:u}$. All remaining connections are masked, including $R_u \to C_k$ and $C_u \to R_v$ for $v > u$. We obtain token-level masks by expanding $M^{\text{seg}}$ and applying standard causal masking within each $C_u$. As shown in Fig. 2(a), $C_1$ attends only to $S_1$, $C_2$ attends to $\langle S_1, Q_1 \rangle$, and $C_3$ attends to $\langle S_1, Q_1, S_2 \rangle$.

**Streaming positional encoding with MRoPE.** We build on Multimodal Rotary Positional Embeddings MRoPE [36], but decouple the input and output to support parallel reasoning. Specifically, the input stream follows the standard cumulative offset scheme, while the output stream independently starts positional encoding from 0. We use $B$ to represent the base offset and compute the start offsets of the $k$-th visual segment $S_k$ input, the $k$-th question $Q_k$ input, and the $k$-th generated unit $C_k$ output:

$$B_k = \begin{cases} B_k^S = \displaystyle\sum_{u < \text{idx}[S_k]} \Delta[R_u], \\ B_k^Q = \displaystyle\sum_{u < \text{idx}[Q_k]} \Delta[R_u], \\ B_k^C = \begin{cases} 0, & k = 1, \\ \sum_{i=1}^{k-1} L[C_i], & k \ge 2. \end{cases} \end{cases} \tag{12}$$

In this design, $B_k^S$ and $B_k^Q$ are computed only from the received input prefix, while $B_k^C$ is computed only from previously generated tokens. Therefore, even when the output length is still unknown during decoding, newly arriving input segments can always be assigned correct input positions.[1]

### 4.3 Streaming Training

**Three-stage training.** We fine-tune the MLLM in three stages: Stage 1 learns to write segment-level memory notes and answer single-round queries. Stage 2 scales to multi-round dialogues. Stage 3 targets long-range behaviors on long videos, including long-term evidence recall, uncertainty handling, and distractor segment learning where we insert irrelevant frames as distractors. In Fig. 2(b), Stage 3 covers three long-horizon behaviors: **long-term memory** for recalling early evidence in late queries, **uncertainty handling** for deferring commitment when evidence is not yet observable, and **distractor robustness** for ignoring irrelevant segments during streaming.

[1] MRoPE extends RoPE to multimodal tokens by applying rotary positional encoding along modality-specific axes. For a visual segment unit $S_k$ with token grid size $\langle T_k, H_k, W_k \rangle$, a token at local grid coordinate $\langle t, h, w \rangle$ is assigned global coordinates $\langle t + B_k^S,\ h + B_k^S,\ w + B_k^S \rangle$, where the start offset $B_k^S$ is given in Eq. (12). For a text unit $Y \in \{Q_k, C_k\}$ with length $L[Y]$, a token at local position $n$ uses $n + B_k^Q$ for question inputs and $n + B_k^C$ for generated outputs. The contribution of each received unit to the input position budget is determined by the unit span $\Delta[R_u]$ in Eq. (7).



### 4.2 動態架構

前綴與後綴格式化。為了符合大型語言模型的標準 SFT 格式，我們將每個訓練示例序列化為一個來源前綴，後面跟著一個目標後綴。來源前綴包含整個接收到的單位流 $\mathbf{R}_{1:U}$，而目標後綴包含對齊的生成流 $\mathbf{C}_{1:U}$。在沒有額外約束的情況下，這種序列化會讓未來接收到的單位洩露給較早生成的單位。

**段落級別動態注意遮罩。我們輸入連接的序列**

$\langle R_1 \ \ldots \ R_U \ \ C_1 \ \ldots \ C_U \rangle$，，，，，搭配段落級別遮罩 $M^{\text{seg}}$ 以強制動態因果性。

Let 表示貢獻查詢標記的段落， $B$ 表示貢獻鍵和值標記的段落，搭配 $A\,B \in \{R_1, \ldots, R_U, C_1, \ldots, C_U\}$。遮罩定義如下：

$$M^{\text{seg}}[A,B] = \begin{cases} \mathbb{I}[v \le u], & A = R_u,\ B = R_v, \\ \mathbb{I}[v \le u], & A = C_u,\ B = R_v, \\ \mathbb{I}[k \le u], & A = C_u,\ B = C_k, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

這裡 $u$ 是查詢單位的到達索引 $A$。對於注意單位 $B$，我們使用 $v$ 如果 $B$ 是接收單位 $R_v$ ，並使用 $k$ 如果 $B$ 是生成單位 $C_k$。公式(11)的前三種情況強制動態因果性：接收流在到達順序上是因果的，每個生成單位 $C_u$ 可以注意接收前綴直到步驟 $u$，並且生成單位僅能存取 $C_{1:u}$. 。所有剩餘的連接都被遮罩，包括 $R_u \to C_k$ 和 $C_u \to R_v$ 對於 $v > u$。我們透過擴展 $M^{\text{seg}}$ 並在每個 $C_u$內部應用標準因果遮罩來獲得標記級別遮罩。如圖2(a)所示， $C_1$ 僅注意 $S_1$, $C_2$ 注意 $\langle S_1\ Q_1 \rangle$ ，並且 $C_3$ 注意 $\langle S_1, Q_1, S_2 \rangle$。

使用MRoPE進行流式位置編碼。我們基於多模態旋轉位置嵌入MRoPE [36]，但將輸入和輸出解耦以支持並行推理。具體來說，輸入流遵循標準的累積偏移方案，而輸出流則獨立從0開始位置編碼。我們使用 $B$ 來表示基礎偏移量，並計算第 $k$-個視覺片段 $S_k$ 輸入、第 $k$-個問題 $Q_k$ 輸入以及第 $k$-個生成單元 $C_k$輸出的起始偏移量：

$$B_k = \begin{cases} B_k^S = \displaystyle\sum_{u < \text{idx}[S_k]} \Delta[R_u], \\ B_k^Q = \displaystyle\sum_{u < \text{idx}[Q_k]} \Delta[R_u], \\ B_k^C = \begin{cases} 0, & k = 1, \\ \sum_{i=1}^{k-1} L[C_i], & k \ge 2. \end{cases} \end{cases} \tag{12}$$

在此設計中， $B_k^S$和 $B_k^Q$ 僅從接收到的輸入前綴計算，而 $B_k^C$僅從先前生成的標記計算。因此，即使解碼期間輸出長度仍然未知，新到達的輸入片段也始終可以被分配正確的輸入位置.[1]

### 4.3 流式訓練

三階段訓練。我們分三個階段微調MLLM：階段1學習寫段落級別的記憶筆記並回答單輪查詢。階段2擴展到多輪對話。階段3針對長影片上的長距離行為，包括長期證據回憶、不確定性處理，以及干擾片段學習，我們在干擾片段中插入不相關的畫面。在圖2(b)中，階段3涵蓋三種長視野行為：長期記憶用於在後期查詢中回憶早期證據，不確定性處理用於在證據尚未可觀察時延遲承諾，以及干擾片段強度用於在串流過程中忽略不相關片段。

[1] MRoPE透過沿模態特定軸應用旋轉位置編碼將RoPE擴展到多模態標記。對於具有標記網格大小 $S_k$ 、 $\langle T_k$、 $H_k$的視覺片段單元，局部網格坐標 $W_k\rangle$、$\langle t$、$h$的標記被分配到全局坐標 $w\rangle$ 、 $\langle t + B_k^S$、 $h + B_k^S$，其中起始偏移量 $w + B_k^S\rangle$由公式(12)給出。對於長度為 $B_k^S$的文本單元 $Y \in \{Q_k, C_k\}$ ，局部位置 $L[Y$的標記使用 $n$進行問題輸入，使用 $n + B_k^Q$ 進行生成輸出。每個接收到的單元對輸入位置預算的貢獻由公式(7)中的單元跨度 $n + B_k^C$決定。

Table 1: **Dataset statistics across three training stages.** Stages 1&2 are built from VideoChatOnline-IT short videos, while Stage 3 contains long-range multi-round dialogues from YouTube. Video duration is reported in seconds (min/avg/max).

| Stage | Source | Instances | Rounds | Avg. rounds | Frames | Min (s) | Avg (s) | Max (s) |
|---|---|---|---|---|---|---|---|---|
| 1 | VideoChatOnline-IT | 5,160 | 5,160 | 1.00 | $\leq 64$ | 8.18 | 79.40 | 3550.10 |
| 2 | VideoChatOnline-IT | 2,752 | 8,513 | 3.09 | $\leq 64$ | 2.00 | 400.92 | 3443.97 |
| 3 | YouTube | 1,500 | 6,000 | 4.00 | 100-300+ | 600.12 | 1697.30 | 3595.03 |

**Three-stage Streaming CoT Dataset Generation.** Streaming CoT [37] datasets for MLLMs are extremely scarce. Multi-round streaming CoT with memory notes is largely absent. We therefore synthesize a three-stage dataset that matches our three-stage training.

**Stage 1 and Stage 2 short video streaming CoT.** We use VideoChatOnline-IT [38] as the source pool and sample up to 64 frames per instance. Stage 1 constructs 5,160 single-round instances from temporal perception subsets. Stage 2 converts 8,513 short video QA instances into 2,752 multi-round dialogues by grouping questions over the same video prefix. For both stages, we use GPT-5.2 to generate memory-anchored CoT based on the original dataset QAs.

**Stage 3 long-range streaming CoT.** We collect long videos from YouTube using 500 keywords spanning three categories: tutorial for procedural content, lecture for explanatory content, and long-form for continuous recordings. We then generate 1,500 long video instances with balanced input lengths of 100 to 200 frames, 200 to 300 frames, and 300 or more frames, and each instance contains 3 to 5 rounds. QA and CoT generation follow the same procedure as in Stage 1 and Stage 2. Details of the dataset and the prompt can be found in Appendix E.

**Quality inspection.** We enforce the core constraints in Table A5 during synthesis, and additionally verify that each example contains exactly $S + Q$ output items.

### 4.4 Streaming Inference

**Parallel reasoning.** For real-time deployment, we adopt a dual KV cache implementation that decouples continuous source ingestion from autoregressive decoding. This engineering pattern is common in recent streaming systems [12]. Meanwhile, we keep the same segment-level streaming mask and streaming MRoPE at inference time to ensure consistency with training.

**Adaptive attention backend.** During decoding, our streaming mask is not always a standard causal mask: queries must attend to an allowed source prefix while remaining causal over the generated suffix, so the query and key lengths can differ ($q_{\text{len}} \neq k_{\text{len}}$). We therefore choose the attention backend adaptively—using Flash Attention [39] when the pattern is standard causal, and otherwise applying an explicit streaming mask with memory-efficient attention [40]. Specifically, we use Flash Attention for source prefilling ($q_{\text{len}} = k_{\text{len}}$) and for autoregressive steps ($q_{\text{len}} = 1$), and switch to memory-efficient attention when $1 < q_{\text{len}} < k_{\text{len}}$ to enforce the custom streaming mask. This preserves segment-level causality while keeping inference fast.

## 5 Experiments

### 5.1 Datasets and Setup

We evaluate online streaming performance on StreamingBench [15] and OVO-Bench [16]. StreamingBench focuses on streaming video understanding and includes four subsets. OVO-Bench emphasizes real-world video understanding under three subsets. More details of datasets are in Appendix E.

**Evaluation protocols.** We evaluate models under both offline and online protocols. In the offline protocol, we adopt a Batch setting where all sampled frames from the entire video are provided as a single input, and the model answers the question after observing the complete video. In the online protocol, we consider single-turn and multi-turn interaction. For single-turn online evaluation, we segment each video according to the question timestamps provided by the benchmark, forming consecutive temporal segments $[0, t_1]$, $[t_1, t_2]$, and so on. If any segment lasts longer than 60s, we



表1：三個訓練階段的數據集統計。階段1&2由VideoChatOnline-IT短影片組成，而階段3則包含來自YouTube的長程多輪對話。影片長度以秒為單位報告（最小值/平均值/最大值）。

| 階段 | 來源 | 個案 | 輪迴 | 平均輪次數 | 框架 | 最小值 (秒) | 平均值 (秒) | 最大值 (秒) |
|---|---|---|---|---|---|---|---|---|
| 1 | VideoChatOnline-IT | 5,160 | 5,160 | 1.00 | $\leq 64$ | 8.18 | 79.40 | 3550.10 |
| 2 | VideoChatOnline-IT | 2,752 | 8,513 | 3.09 | $\leq 64$ | 2.00 | 400.92 | 3443.97 |
| 3 | YouTube | 1,500 | 6,000 | 4.00 | 100-300+ | 600.12 | 1697.30 | 3595.03 |

三階段串流CoT數據集生成。針對MLLM的串流CoT [37]數據集極度稀缺。帶有記憶筆記的多輪串流CoT幾乎不存在。因此，我們合成了一個三階段數據集，以匹配我們的三階段訓練。

第一階段與第二階段短影片串流CoT。我們使用VideoChatOnline-IT [38] 作為來源池，並每個實例採樣最多64個畫面。第一階段從時間感知子集中構建5,160個單輪實例。第二階段將8,513個短影片QA實例轉換為2,752個多輪對話，方法是將具有相同影片前綴的問題分組。對於兩個階段，我們使用GPT-5.2根據原始數據集的QA生成記憶錨定CoT。

第三階段長距離串流CoT。我們使用涵蓋三個類別的500個關鍵字從YouTube收集長影片：教程用於程序性內容，講座用於解釋性內容，以及長形式用於持續錄製。然後我們生成1,500個長影片實例，其輸入長度分別為100至200畫面、200至300畫面和300畫面或更多，每個實例包含3至5輪。QA和CoT生成與第一階段和第二階段相同。數據集和提示的細節可以在附錄E中找到。

品質檢驗。我們在合成過程中執行表A5中的核心約束，並額外驗證每個範例都包含確切 $S + Q$ 個輸出項目。

### 4.4 動態推論

並行推理。為了實時部署，我們採用雙重KV快取實現，將持續的來源攝取與自回歸解碼解耦。這種工程模式在近年來的動態系統中很常見 [12]。同時，我們在推論時保持相同的段落級別動態掩碼和動態MRoPE，以確保與訓練的一致性。

自適應注意力後端。在解碼過程中，我們的動態掩碼並非始終是標準的因果掩碼：查詢必須關注到允許的來源前綴，同時在生成的後綴上保持因果性，因此查詢和鍵的長度可以不同 ($q_{\text{len}} \neq k_{\text{len}}$)。因此，我們選擇自適應注意力後端——在模式是標準因果時使用Flash Attention [39]，否則應用明確的動態掩碼與記憶高效的注意力 [40]。具體來說，我們使用Flash Attention進行來源預填充 ($q_{\text{len}} = k_{\text{len}}$) 和自回歸步驟 ($q_{\text{len}} = 1$)，當 $1 < q_{\text{len}} < k_{\text{len}}$ 時切換到記憶高效的注意力以執行自定義動態掩碼。這保留了段落級別的因果性，同時保持了推論的速度。

## 5 個實驗

### 5.1 資料集與設定

我們在 StreamingBench [15] 和 OVO-Bench [16] 上評估線上串流性能。StreamingBench 專注於串流視頻理解，包含四個子集。OVO-Bench 強調在實際世界視頻理解下的三個子集。更多資料集細節請參考附錄 E。

評估協議。我們在離線與線上協議下評估模型。在離線協議中，我們採用批次設定，將從整個影片中採樣的所有畫面作為單一輸入，模型在觀察完整個影片後回答問題。在線上協議中，我們考慮單輪與多輪互動。對於單輪線上評估，我們根據基準提供的問題時間戳將每個影片進行分段，形成連續的時間片段 $[0, t_1]$、$[t_1, t_2]$ 等等。如果任何片段長度超過 60 秒，我們

further split it into 30s chunks. For multi-turn evaluation, we use the same segmentation strategy, but the model must answer multiple questions online as segments continuously arrive.

**Backbones and checkpoints.** We evaluate our method with Qwen3-VL backbones at 2B, 4B, and 8B scales. We use the Instruct model for training and compare its performance with the Thinking model. TWW is used to denote our method in the following. Stage 2 and Stage 3 refer to the checkpoints obtained after training up to the second and third stages.

### 5.2 Baselines

**Offline evaluation.** We evaluate Gemini 1.5 Pro [41] and GPT-4o [42] as representative closed-source MLLMs. Qwen3-VL-Instruct and Qwen3-VL-Thinking are used as open-source baselines. We also report results for our $\text{TWW}_{\text{Batch,S2}}$ and $\text{TWW}_{\text{Batch,S3}}$ checkpoints, corresponding to Stage 2 and Stage 3, evaluated under the offline batch protocol.

**Online evaluation.** For online evaluation, $\text{Instruct}_{\text{online}}$ and $\text{Thinking}_{\text{online}}$ run Qwen3-VL-Instruct and Qwen3-VL-Thinking, respectively, under the multi-turn protocol in Sec. 5.1. We further evaluate our checkpoints under streaming settings: $\text{TWW}_{\text{single-turn,S2}}$ and $\text{TWW}_{\text{single-turn,S3}}$ follow the single-round streaming protocol, while $\text{TWW}_{\text{multi-turn,S2}}$ and $\text{TWW}_{\text{multi-turn,S3}}$ follow the multi-round streaming protocol. Finally, Interleaved alternates between ingesting one segment and decoding text, coupling perception and generation as a naive streaming baseline. More online baselines and details are in Appendix D.

### 5.3 Metrics

We use accuracy to evaluate performance on each benchmark and each evaluation regime. We also report Avg Tokens, the average number of generated output tokens per query. Token Reduce, denoted by $\Delta\%$, is the percentage reduction of Avg Tokens compared with the Thinking baseline of the same backbone size, and Avg Frames, the average number of processed frames per query. For latency, we report TTFT, time to first token, measured as the number of tokens processed before the first answer token is produced.

### 5.4 Main Results

Tables 2 and 3 report results on two streaming benchmarks, StreamingBench and OVO-Bench. We summarize the key findings below.

1. **Naive streaming inference collapses without streaming-aligned training, highlighting the difficulty of multi-turn streaming.** On StreamingBench, directly running $\text{Instruct}_{\text{online}}$ and $\text{Thinking}_{\text{online}}$ achieves only 21.47% and 18.13% overall, compared with 56.67% and 58.52% with Qwen3-VL-4B in the offline batch setting. A similar drop is observed on OVO-Bench: 21.45% and 16.21% versus 50.32% and 50.70%, showing that multi-turn streaming is nontrivial and requires streaming-aligned supervision.
2. **Streaming-aligned supervision improves accuracy.** With Qwen3-VL-4B, $\text{TWW}_{\text{single-turn,S3}}$ improves overall accuracy from 58.52% to 60.04% on StreamingBench and from 50.70% to 55.02% on OVO-Bench compared with the Thinking baseline.
3. **Long-video training strengthens streaming behavior.** Stage 3 generally improves upon Stage 2. For example, on OVO-Bench with the 4B backbone, $\text{TWW}_{\text{single-turn,S3}}$ improves from 54.51% to 55.02%.
4. **Multi-turn segment-level memory yields a strong accuracy–efficiency tradeoff.** Under the multi-turn protocol, $\text{TWW}_{\text{multi-turn,S3}}$ maintains competitive accuracy while substantially reducing decoding tokens. With the 4B backbone, it achieves 57.40% on StreamingBench with an average of 302.56 tokens, reducing token usage by 56.10%. On OVO-Bench, it obtains 51.80% with 255.91 tokens on average, reducing token usage by 45.80%.

We further analyzed the types of errors in Appendix G.



會將其進一步分割成 30 秒的片段。對於多輪評估，我們使用相同的分割策略，但模型必須在片段持續到達時線上回答多個問題。

主幹與檢查點。我們使用 2B、4B 和 8B 規模的 Qwen3-VL 主幹來評估我們的方法。我們使用 Instruct 模型進行訓練，並將其性能與 Thinking 模型進行比較。TWW 用於在以下內容中標示我們的方法。第二階段和第三階段是指訓練到第二階段和第三階段後獲得的檢查點。

### 5.2 基準線

離線評估。我們將 Gemini 1.5 Pro [41]和 GPT-4o [42]作為代表性的封閉式 MLLM 進行評估。Qwen3-VL-Instruct 和 Qwen3-VL-Thinking 作為開源基準線。我們還報告了我們 $\text{TWW}_{\text{Batch,S2}}$ 和 $\text{TWW}_{\text{Batch,S3}}$ 檢查點的結果，分別對應第二階段和第三階段，在離線批量協議下進行評估。

線上評估。在線上評估中，$\text{Instruct}_{\text{online}}$ 和 $\text{Thinking}_{\text{online}}$ 分別在 5.1 節中描述的多回合協議下運行 Qwen3-VL-Instruct 和 Qwen3-VL-Thinking。我們還在流式設置下評估我們的檢查點：$\text{TWW}_{\text{single-turn,S2}}$ 和 $\text{TWW}_{\text{single-turn,S3}}$ 遵循單輪流式協議，而 $\text{TWW}_{\text{multi-turn,S2}}$ 和 $\text{TWW}_{\text{multi-turn,S3}}$ 遵循多輪流式協議。最後，Interleaved 在輸入一個片段和解碼文本之間交替，將感知和生成結合作為一個簡單的流式基準線。更多線上基準線和細節請參見附錄 D。

### 5.3 指標

我們使用準確度來評估每個基準和每個評估體系上的表現。我們也報告平均Token數，即每個查詢平均生成的輸出Token數。Token減少量，以 Δ%表示，是與相同主幹大小的Thinking基準相比，平均Token數的百分比減少量，以及平均幀數，即每個查詢平均處理的幀數。對於延遲，我們報告TTFT，即第一個Token的時間，測量為在產生第一個答案Token之前處理的Token數。

### 5.4 主要結果

表2和表3報告了在兩個流式處理基準StreamingBench和OVO-Bench上的結果。我們在下文中總結了關鍵發現。

1. 無流式處理對齊訓練的直觀流式處理會崩潰，突顯了多輪流式處理的難度。在StreamingBench上，直接運行$\text{Instruct}_{\text{online}}$ 和 $\text{Thinking}_{\text{online}}$ 僅實現了21.47%和18.13%的整體表現，而與Qwen3-VL-4B在離線批量設定下的56.67%和58.52%相比。在OVO-Bench上觀察到類似的下降：21.45%和16.21%對於50.32%和50.70%，表明多輪流式處理並非易事，需要流式處理對齊的監督。

2. 與串流對齊的監督可提高準確度。使用Qwen3-VL-4B，$\text{TWW}_{\text{single-turn,S3}}$在StreamingBench上將整體準確度從58.52%提高到60.04%，在OVO-Bench上從50.70%提高到55.02%，與Thinking基線相比。

3. 長影片訓練可加強串流行為。第三階段通常比第二階段有所改善。例如，在OVO-Bench上使用4B主幹架構時，$\text{TWW}_{\text{single-turn,S3}}$ 從54.51%提升到55.02%。

4. 多輪片段層級記憶產生強勁的準確率-效率交易。在多輪協議下，$\text{TWW}_{\text{multi-turn,S3}}$ 維持競爭力的準確率，同時大幅減少解碼 token。使用 4B 主幹，它在 StreamingBench 上達到 57.40%，平均為 302.56 token，減少 token 使用量 56.10%。在 OVO-Bench 上，它獲得 51.80%，平均為 255.91 token，減少 token 使用量 45.80%。

我們進一步分析了附錄G中的錯誤類型。

Table 2: **StreamingBench results** with accuracy. Columns left of the double bar report **performance**, higher is better, while columns on the right report **efficiency** for open-source models only. Δ is computed against the same backbone Thinking baseline. Avg Frames is 148.35 for the single-turn protocol and 62.58 for the multi-turn protocol.

| | | Performance | | | | | | Efficiency | |
|---|---|---|---|---|---|---|---|---|---|
| **Regime** | **Method** | **SQA↑** | **OmniSource↑** | **Realtime↑** | **Proactive↑** | **Overall↑** | **Δ↑** | **Avg Tokens↓** | **Token Reduce↑** |
| **Closed-source models** | | | | | | | | | |
| Offline | Gemini 1.5 Pro[41] | 54.80 | 67.80 | 77.39 | 45.10 | 70.26 | - | - | - |
| | GPT-4o[42] | 32.80 | 50.95 | 74.54 | 56.86 | 64.10 | - | - | - |
| **Open-source models** | | | | | | | | | |
| Online | Flash-VStream-7B[32] | 26.80 | 26.00 | 23.23 | 1.96 | 24.04 | - | - | - |
| | VideoLLM-online-8B[8] | 30.80 | 28.45 | 35.99 | 3.92 | 32.48 | - | - | - |
| | Dispider-7B[43] | 34.80 | 35.66 | 67.63 | 25.34 | 53.12 | - | - | - |
| | StreamAgent-7B[35] | 39.60 | 36.26 | 74.28 | 28.90 | 57.02 | - | - | - |
| **Qwen3-VL-2B** | | | | | | | | | |
| Offline | Instruct | 37.60 | 31.60 | 68.36 | 29.60 | 52.24 | -1.17 | - | - |
| | Thinking | 34.00 | 31.73 | 70.02 | **36.80** | 53.41 | +0.00 | 1232.91 | 0.00 |
| | $TWW_{Batch,S2}$ | 44.00 | 33.13 | 69.16 | 33.20 | 53.75 | +0.34 | 1012.35 | 17.89 |
| | **$TWW_{Batch,S3}$** | **44.80** | **33.20** | **69.20** | 36.00 | **54.00** | **+0.59** | 1102.34 | 10.59 |
| Online | $Instruct_{online}$ | 9.20 | 28.80 | 21.28 | 13.20 | 22.67 | -30.74 | - | - |
| | $Thinking_{online}$ | 8.40 | 11.60 | 12.92 | 19.20 | 12.58 | -40.83 | 832.23 | 32.50 |
| | $TWW_{single\text{-}turn,S2}$ | 47.20 | 34.20 | 71.84 | 32.80 | 55.76 | +2.35 | 923.58 | 25.09 |
| | **$TWW_{single\text{-}turn,S3}$** | **48.00** | **34.27** | **72.00** | 34.40 | **56.00** | **+2.59** | 930.23 | 24.55 |
| | $TWW_{multi\text{-}turn,S2}$ | 42.40 | 31.33 | 69.20 | 33.60 | 53.11 | -0.30 | **285.42** | **76.85** |
| | **$TWW_{multi\text{-}turn,S3}$** | 45.20 | 31.47 | 69.24 | 34.80 | **53.40** | -0.01 | 300.20 | 75.65 |
| **Qwen3-VL-4B** | | | | | | | | | |
| Offline | Instruct | 37.20 | 38.47 | 71.36 | 38.40 | 56.67 | -1.85 | - | - |
| | Thinking | 46.40 | 36.53 | **74.50** | 42.80 | 58.52 | +0.00 | 689.22 | 0.00 |
| | $TWW_{Batch,S2}$ | 42.40 | 39.60 | 71.84 | 39.60 | 57.67 | -0.85 | 594.28 | 13.77 |
| | **$TWW_{Batch,S3}$** | 44.00 | **39.67** | 71.88 | 40.80 | 57.87 | -0.65 | 620.35 | 9.99 |
| Online | $Instruct_{online}$ | 12.80 | 21.53 | 22.88 | 15.60 | 21.47 | -37.05 | - | - |
| | $Thinking_{online}$ | 21.20 | 21.47 | 16.92 | 7.20 | 18.13 | -40.39 | 482.24 | 30.03 |
| | $TWW_{single\text{-}turn,S2}$ | 46.00 | 40.67 | 74.36 | 41.20 | 59.71 | +1.19 | 558.12 | 19.02 |
| | **$TWW_{single\text{-}turn,S3}$** | **46.80** | **41.00** | 74.48 | **43.20** | **60.04** | **+1.52** | 570.68 | 17.20 |
| | $TWW_{multi\text{-}turn,S2}$ | 40.80 | 39.20 | 71.20 | 40.00 | 57.11 | -1.41 | **291.86** | **57.65** |
| | **$TWW_{multi\text{-}turn,S3}$** | 43.60 | 39.33 | 71.28 | 40.80 | **57.40** | -1.12 | 302.56 | 56.10 |
| **Qwen3-VL-8B** | | | | | | | | | |
| Offline | Instruct | 44.40 | 37.53 | 73.60 | 36.00 | 57.87 | -0.35 | - | - |
| | Thinking | 45.60 | 35.47 | 74.46 | **44.80** | 58.21 | +0.00 | 759.30 | 0.00 |
| | $TWW_{Batch,S2}$ | 52.80 | 38.93 | 74.56 | 38.00 | 59.44 | +1.23 | 708.16 | 6.70 |
| | **$TWW_{Batch,S3}$** | **53.20** | **39.07** | **74.64** | 40.00 | **59.67** | **+1.45** | 720.30 | 5.10 |
| Online | $Instruct_{online}$ | 17.20 | 21.47 | 25.12 | 17.60 | 23.04 | -35.17 | - | - |
| | $Thinking_{online}$ | 14.80 | 13.00 | 17.56 | 16.40 | 15.82 | -42.39 | 573.75 | 24.41 |
| | $TWW_{single\text{-}turn,S2}$ | 54.00 | 40.07 | 77.60 | 39.60 | 61.67 | +3.46 | 651.74 | 14.13 |
| | **$TWW_{single\text{-}turn,S3}$** | **54.40** | **40.67** | **77.68** | 41.60 | **62.04** | **+3.83** | 660.92 | 12.92 |
| | $TWW_{multi\text{-}turn,S2}$ | 48.80 | 37.40 | 74.32 | 38.40 | 58.60 | +0.39 | **288.64** | **61.97** |
| | **$TWW_{multi\text{-}turn,S3}$** | 50.00 | 37.47 | 74.40 | 40.00 | **58.82** | +0.61 | 290.82 | 61.68 |

## 5.5 Analysis

**Generalization to Offline Video Understanding.** Although our training is designed for streaming scenarios, we also evaluate whether the learned behaviors can transfer to offline video understanding tasks. We evaluate on two offline benchmarks: **Video-MME** [44] and **LV-Bench** [45], following their official evaluation settings. Table 4 shows that streaming training also benefits offline evaluation. In particular, $TWW_{single\text{-}turn,S3}$ improves Video-MME from 68.89% to **73.41%** and LV-Bench from 53.47% to **57.68%**, showing that long-range streaming supervision transfers effectively to offline long-video reasoning. Unless otherwise stated, the following analyses use Qwen3-VL-4B on StreamingBench.



表2：StreamingBench準確度結果。雙杠左側的欄位報告效能，越高越好，而雙杠右側的欄位僅報告開源模型的效率。Δ 是基於相同主幹架構的Thinking baseline計算。單輪協議的Avg Frames為148.35，多輪協議為62.58。

| | | 效能 | | | | | | 效率 | |
|---|---|---|---|---|---|---|---|---|---|
| **模式** | **方法** | **SQA↑** | **OmniSource↑** | **實時↑** | **主動↑** | **整體↑** | **Δ↑** | **平均Token↓** | **Token減少↑** |
| **封閉式模型** | | | | | | | | | |
| 離線 | Gemini 1.5 Pro[41] | 54.80 | 67.80 | 77.39 | 45.10 | 70.26 | - | - | - |
| | GPT-4o[42] | 32.80 | 50.95 | 74.54 | 56.86 | 64.10 | - | - | - |
| **開源模型** | | | | | | | | | |
| 線上 | Flash-VStream-7B[32] | 26.80 | 26.00 | 23.23 | 1.96 | 24.04 | - | - | - |
| | VideoLLM線上-8B[8] | 30.80 | 28.45 | 35.99 | 3.92 | 32.48 | - | - | - |
| | Dispider-7B[43] | 34.80 | 35.66 | 67.63 | 25.34 | 53.12 | - | - | - |
| | StreamAgent-7B[35] | 39.60 | 36.26 | 74.28 | 28.90 | 57.02 | - | - | - |
| **Qwen3-VL-2B** | | | | | | | | | |
| 離線 | 指令 | 37.60 | 31.60 | 68.36 | 29.60 | 52.24 | -1.17 | - | - |
| | 思考 | 34.00 | 31.73 | 70.02 | **36.80** | 53.41 | +0.00 | 1232.91 | 0.00 |
| | $TWW_{Batch,S2}$ | 44.00 | 33.13 | 69.16 | 33.20 | 53.75 | +0.34 | 1012.35 | 17.89 |
| | **$TWW_{Batch,S3}$** | **44.80** | **33.20** | **69.20** | 36.00 | **54.00** | **+0.59** | 1102.34 | 10.59 |
| 線上 | $指令_{online}$ | 9.20 | 28.80 | 21.28 | 13.20 | 22.67 | -30.74 | - | - |
| | $思考_{online}$ | 8.40 | 11.60 | 12.92 | 19.20 | 12.58 | -40.83 | 832.23 | 32.50 |
| | $TWW_{single\text{-}turn,S2}$ | 47.20 | 34.20 | 71.84 | 32.80 | 55.76 | +2.35 | 923.58 | 25.09 |
| | **$TWW_{single\text{-}turn,S3}$** | **48.00** | **34.27** | **72.00** | 34.40 | **56.00** | **+2.59** | 930.23 | 24.55 |
| | $TWW_{multi\text{-}turn,S2}$ | 42.40 | 31.33 | 69.20 | 33.60 | 53.11 | -0.30 | **285.42** | **76.85** |
| | **$TWW_{multi\text{-}turn,S3}$** | 45.20 | 31.47 | 69.24 | 34.80 | **53.40** | -0.01 | 300.20 | 75.65 |
| **Qwen3-VL-4B** | | | | | | | | | |
| 離線 | 指令 | 37.20 | 38.47 | 71.36 | 38.40 | 56.67 | -1.85 | - | - |
| | 思考 | 46.40 | 36.53 | **74.50** | 42.80 | 58.52 | +0.00 | 689.22 | 0.00 |
| | $TWW_{Batch,S2}$ | 42.40 | 39.60 | 71.84 | 39.60 | 57.67 | -0.85 | 594.28 | 13.77 |
| | **$TWW_{Batch,S3}$** | 44.00 | **39.67** | 71.88 | 40.80 | 57.87 | -0.65 | 620.35 | 9.99 |
| 線上 | $指示_{online}$ | 12.80 | 21.53 | 22.88 | 15.60 | 21.47 | -37.05 | - | - |
| | $思考_{online}$ | 21.20 | 21.47 | 16.92 | 7.20 | 18.13 | -40.39 | 482.24 | 30.03 |
| | $TWW_{single\text{-}turn,S2}$ | 46.00 | 40.67 | 74.36 | 41.20 | 59.71 | +1.19 | 558.12 | 19.02 |
| | **$TWW_{single\text{-}turn,S3}$** | **46.80** | **41.00** | 74.48 | **43.20** | **60.04** | **+1.52** | 570.68 | 17.20 |
| | $TWW_{multi\text{-}turn,S2}$ | 40.80 | 39.20 | 71.20 | 40.00 | 57.11 | -1.41 | **291.86** | **57.65** |
| | **$TWW_{multi\text{-}turn,S3}$** | 43.60 | 39.33 | 71.28 | 40.80 | **57.40** | -1.12 | 302.56 | 56.10 |
| **Qwen3-VL-8B** | | | | | | | | | |
| 離線 | 指令 | 44.40 | 37.53 | 73.60 | 36.00 | 57.87 | -0.35 | - | - |
| | 思考 | 45.60 | 35.47 | 74.46 | **44.80** | 58.21 | +0.00 | 759.30 | 0.00 |
| | $TWW_{Batch,S2}$ | 52.80 | 38.93 | 74.56 | 38.00 | 59.44 | +1.23 | 708.16 | 6.70 |
| | **$TWW_{Batch,S3}$** | **53.20** | **39.07** | **74.64** | 40.00 | **59.67** | **+1.45** | 720.30 | 5.10 |
| 線上 | $指示_{online}$ | 17.20 | 21.47 | 25.12 | 17.60 | 23.04 | -35.17 | - | - |
| | $思考_{online}$ | 14.80 | 13.00 | 17.56 | 16.40 | 15.82 | -42.39 | 573.75 | 24.41 |
| | $TWW_{single\text{-}turn,S2}$ | 54.00 | 40.07 | 77.60 | 39.60 | 61.67 | +3.46 | 651.74 | 14.13 |
| | **$TWW_{single\text{-}turn,S3}$** | **54.40** | **40.67** | **77.68** | 41.60 | **62.04** | **+3.83** | 660.92 | 12.92 |
| | $TWW_{multi\text{-}turn,S2}$ | 48.80 | 37.40 | 74.32 | 38.40 | 58.60 | +0.39 | **288.64** | **61.97** |
| | **$TWW_{multi\text{-}turn,S3}$** | 50.00 | 37.47 | 74.40 | 40.00 | **58.82** | +0.61 | 290.82 | 61.68 |

## 5.5 分析

擴展至離線視頻理解。儘管我們的訓練設計用於串流場景，但我們也評估學習到的行為是否可以轉移到離線視頻理解任務。我們在兩個離線基準上進行評估：Video-MME [44] 和LV-Bench [45]，遵循它們的官方評估設定。表4顯示串流訓練也有助於離線評估。特別是，$TWW_{single\text{-}turn,S3}$ 將Video-MME從68.89%提升至73.41%，將LV-Bench從53.47%提升至57.68%，表明長程串流監督可以有效地轉移至離線長視頻推理。除非另有說明，以下分析使用Qwen3-VL-4B在StreamingBench上進行。

Table 3: **OVO-Bench results**. Avg Frames is 63.23 for the single-turn protocol and 25.47 for the multi-turn protocol.

| | | Performance | | | | | Efficiency | |
|---|---|---|---|---|---|---|---|---|
| **Regime** | **Method** | **Backward↑** | **Realtime↑** | **Forward↑** | **Overall↑** | **Δ↑** | **Avg Tokens↓** | **Token Reduce↑** |
| | | | | **Closed-source models** | | | | |
| Offline | Gemini 1.5 Pro[41] | 69.32 | 62.54 | 57.15 | 63.00 | - | - | - |
| | GPT-4o[42] | 64.46 | 60.75 | 53.40 | 59.54 | - | - | - |
| | | | | **Open-source models** | | | | |
| Online | Flash-VStream-7B[32] | 27.38 | 28.37 | 45.09 | 33.61 | - | - | - |
| | Dispider-7B[43] | 36.06 | 54.55 | 34.72 | 41.78 | - | - | - |
| | StreamForest-7B[33] | 52.02 | 61.20 | 53.49 | 55.57 | - | - | - |
| | StreamAgent-7B[35] | 41.70 | 61.30 | 45.40 | 49.40 | - | - | - |
| | | | | **Qwen3-VL-2B** | | | | |
| Offline | Instruct | 37.29 | 56.65 | **51.51** | 49.97 | +2.23 | - | - |
| | Thinking | 41.78 | 57.27 | 45.04 | 47.74 | +0.00 | 590.25 | 0.00 |
| | $\text{TWW}_{\text{Batch,S2}}$ | 43.25 | 56.52 | 44.72 | 47.67 | -0.07 | 518.64 | 12.13 |
| | **$\text{TWW}_{\text{Batch,S3}}$** | 44.37 | 56.87 | 45.31 | 48.30 | **+0.56** | 530.32 | 10.15 |
| Online | $\text{Instruct}_{\text{online}}$ | 11.89 | 16.01 | 26.87 | 20.76 | -26.98 | - | - |
| | $\text{Thinking}_{\text{online}}$ | 8.56 | 19.47 | 20.29 | 17.63 | -30.11 | 478.74 | 18.89 |
| | $\text{TWW}_{\text{single-turn,S2}}$ | 44.58 | 57.85 | 47.35 | 49.67 | +1.93 | 456.28 | 22.70 |
| | **$\text{TWW}_{\text{single-turn,S3}}$** | **45.96** | **58.18** | 47.86 | **50.31** | **+2.57** | 470.20 | 20.34 |
| | $\text{TWW}_{\text{multi-turn,S2}}$ | 40.85 | 55.42 | 44.34 | 46.67 | -1.07 | **278.52** | **52.81** |
| | **$\text{TWW}_{\text{multi-turn,S3}}$** | 42.16 | 55.79 | 44.54 | 47.15 | -0.59 | 280.32 | 52.51 |
| | | | | **Qwen3-VL-4B** | | | | |
| Offline | Instruct | 44.32 | 62.39 | 46.29 | 50.32 | -0.38 | - | - |
| | Thinking | 50.78 | 61.17 | 45.08 | 50.70 | +0.00 | 472.18 | 0.00 |
| | $\text{TWW}_{\text{Batch,S2}}$ | 52.45 | 62.18 | 47.37 | 52.51 | +1.81 | 412.37 | 12.67 |
| | **$\text{TWW}_{\text{Batch,S3}}$** | 53.88 | 62.37 | 47.86 | 53.11 | **+2.41** | 430.51 | 8.83 |
| Online | $\text{Instruct}_{\text{online}}$ | 13.63 | 20.79 | 24.95 | 21.45 | -29.25 | - | - |
| | $\text{Thinking}_{\text{online}}$ | 14.42 | 14.10 | 18.06 | 16.21 | -34.49 | 360.63 | 23.62 |
| | $\text{TWW}_{\text{single-turn,S2}}$ | 53.92 | 64.25 | 49.55 | 54.51 | +3.81 | 358.45 | 24.09 |
| | **$\text{TWW}_{\text{single-turn,S3}}$** | **55.47** | **64.52** | **49.78** | **55.02** | **+4.32** | 378.64 | 19.81 |
| | $\text{TWW}_{\text{multi-turn,S2}}$ | 48.65 | 60.85 | 47.67 | 51.51 | +0.81 | **251.36** | **46.77** |
| | **$\text{TWW}_{\text{multi-turn,S3}}$** | 49.13 | 61.17 | 47.86 | 51.80 | +1.10 | 255.91 | 45.80 |
| | | | | **Qwen3-VL-8B** | | | | |
| Offline | Instruct | 42.63 | 62.86 | 51.02 | 52.54 | -1.28 | - | - |
| | Thinking | 54.24 | 61.34 | 49.63 | 53.82 | +0.00 | 390.42 | 0.00 |
| | $\text{TWW}_{\text{Batch,S2}}$ | 55.82 | 63.15 | 51.83 | 55.78 | +1.96 | 325.82 | 16.55 |
| | **$\text{TWW}_{\text{Batch,S3}}$** | **57.37** | 63.68 | 52.01 | 56.34 | **+2.52** | 340.22 | 12.86 |
| Online | $\text{Instruct}_{\text{online}}$ | 15.69 | 25.57 | 21.12 | 21.22 | -32.60 | - | - |
| | $\text{Thinking}_{\text{online}}$ | 11.25 | 19.59 | 18.51 | 17.30 | -36.52 | 330.67 | 15.30 |
| | $\text{TWW}_{\text{single-turn,S2}}$ | 56.45 | 64.52 | 52.78 | 56.78 | +2.96 | 276.19 | 29.26 |
| | **$\text{TWW}_{\text{single-turn,S3}}$** | 57.05 | **64.76** | 52.97 | **57.07** | **+3.25** | 290.07 | 25.70 |
| | $\text{TWW}_{\text{multi-turn,S2}}$ | 52.05 | 63.25 | 52.85 | 55.55 | +1.73 | **224.68** | **42.45** |
| | **$\text{TWW}_{\text{multi-turn,S3}}$** | 52.77 | 63.68 | **53.29** | 56.05 | +2.23 | 227.33 | 41.77 |

**Long-Range Attention Analysis.** We analyze how far back the model consults history when generating answers by aggregating the last-layer attention from answer tokens to historical tokens and grouping the attended history by segment distance $d = \tau_r - i$, where $\tau_r$ denotes the index of the latest observed segment when answering the $r$-th question, and $i \leq \tau_r$ denotes the index of a historical segment being attended to. Here, $d = 0$ corresponds to the most recent segment, and larger $d$ indicates segments further in the past.



表3：OVO-Bench結果。單輪協議的平均幀數為63.23，多輪協議的平均幀數為25.47。

| | | 性能 | | | | | 效率 | |
|---|---|---|---|---|---|---|---|---|
| **模式** | **方法** | **反向↑** | **實時↑** | **前向↑** | **整體↑** | **Δ↑** | **平均Token↓** | **Token減少↑** |
| | | | | **封閉式模型** | | | | |
| 離線 | Gemini 1.5 Pro[41] | 69.32 | 62.54 | 57.15 | 63.00 | - | - | - |
| | GPT-4o[42] | 64.46 | 60.75 | 53.40 | 59.54 | - | - | - |
| | | | | **開源模型** | | | | |
| 線上 | Flash-VStream-7B[32] | 27.38 | 28.37 | 45.09 | 33.61 | - | - | - |
| | Dispider-7B[43] | 36.06 | 54.55 | 34.72 | 41.78 | - | - | - |
| | StreamForest-7B[33] | 52.02 | 61.20 | 53.49 | 55.57 | - | - | - |
| | StreamAgent-7B[35] | 41.70 | 61.30 | 45.40 | 49.40 | - | - | - |
| | | | | **Qwen3-VL-2B** | | | | |
| 離線 | 指令 | 37.29 | 56.65 | **51.51** | 49.97 | +2.23 | - | - |
| | 思考 | 41.78 | 57.27 | 45.04 | 47.74 | +0.00 | 590.25 | 0.00 |
| | $\text{TWW}_{\text{Batch}}$,S2 | 43.25 | 56.52 | 44.72 | 47.67 | -0.07 | 518.64 | 12.13 |
| | **$\text{TWW}_{\text{Batch,S3}}$** | 44.37 | 56.87 | 45.31 | 48.30 | **+0.56** | 530.32 | 10.15 |
| 線上 | $\text{指令}_{\text{online}}$ | 11.89 | 16.01 | 26.87 | 20.76 | -26.98 | - | - |
| | $\text{思考}_{\text{online}}$ | 8.56 | 19.47 | 20.29 | 17.63 | -30.11 | 478.74 | 18.89 |
| | $\text{TWW}_{\text{single-turn,S2}}$ | 44.58 | 57.85 | 47.35 | 49.67 | +1.93 | 456.28 | 22.70 |
| | **$\text{TWW}_{\text{single-turn,S3}}$** | **45.96** | **58.18** | 47.86 | **50.31** | **+2.57** | 470.20 | 20.34 |
| | $\text{TWW}_{\text{multi-turn}}$,S2 | 40.85 | 55.42 | 44.34 | 46.67 | -1.07 | **278.52** | **52.81** |
| | **$\text{TWW}_{\text{multi-turn}}$,S3** | 42.16 | 55.79 | 44.54 | 47.15 | -0.59 | 280.32 | 52.51 |
| | | | | **Qwen3-VL-4B** | | | | |
| 離線 | 指令 | 44.32 | 62.39 | 46.29 | 50.32 | -0.38 | - | - |
| | 思考 | 50.78 | 61.17 | 45.08 | 50.70 | +0.00 | 472.18 | 0.00 |
| | $\text{TWW}_{\text{Batch,S2}}$ | 52.45 | 62.18 | 47.37 | 52.51 | +1.81 | 412.37 | 12.67 |
| | **$\text{TWW}_{\text{Batch}}$,S3** | 53.88 | 62.37 | 47.86 | 53.11 | **+2.41** | 430.51 | 8.83 |
| 線上 | $\text{指令}_{\text{online}}$ | 13.63 | 20.79 | 24.95 | 21.45 | -29.25 | - | - |
| | $\text{思考}_{\text{online}}$ | 14.42 | 14.10 | 18.06 | 16.21 | -34.49 | 360.63 | 23.62 |
| | $\text{TWW}_{\text{single-turn,S2}}$ | 53.92 | 64.25 | 49.55 | 54.51 | +3.81 | 358.45 | 24.09 |
| | **$\text{TWW}_{\text{single-turn,S3}}$** | **55.47** | **64.52** | **49.78** | **55.02** | **+4.32** | 378.64 | 19.81 |
| | $\text{TWW}_{\text{multi-turn,S2}}$ | 48.65 | 60.85 | 47.67 | 51.51 | +0.81 | **251.36** | **46.77** |
| | **$\text{TWW}_{\text{multi-turn}}$,S3** | 49.13 | 61.17 | 47.86 | 51.80 | +1.10 | 255.91 | 45.80 |
| | | | | **Qwen3-VL-8B** | | | | |
| 離線 | 指示 | 42.63 | 62.86 | 51.02 | 52.54 | -1.28 | - | - |
| | 思考 | 54.24 | 61.34 | 49.63 | 53.82 | +0.00 | 390.42 | 0.00 |
| | $\text{TWW}_{\text{Batch}}$,S2 | 55.82 | 63.15 | 51.83 | 55.78 | +1.96 | 325.82 | 16.55 |
| | **$\text{TWW}_{\text{Batch}}$,S3** | **57.37** | 63.68 | 52.01 | 56.34 | **+2.52** | 340.22 | 12.86 |
| 線上 | $\text{指令}_{\text{online}}$ | 15.69 | 25.57 | 21.12 | 21.22 | -32.60 | - | - |
| | $\text{思考}_{\text{online}}$ | 11.25 | 19.59 | 18.51 | 17.30 | -36.52 | 330.67 | 15.30 |
| | $\text{TWW}_{\text{single-turn,S2}}$ | 56.45 | 64.52 | 52.78 | 56.78 | +2.96 | 276.19 | 29.26 |
| | **$\text{TWW}_{\text{single-turn,S3}}$** | 57.05 | **64.76** | 52.97 | **57.07** | **+3.25** | 290.07 | 25.70 |
| | $\text{TWW}_{\text{multi-turn,S2}}$ | 52.05 | 63.25 | 52.85 | 55.55 | +1.73 | **224.68** | **42.45** |
| | **$\text{TWW}_{\text{multi-turn,S3}}$** | 52.77 | 63.68 | **53.29** | 56.05 | +2.23 | 227.33 | 41.77 |

長程注意力分析。我們分析模型在生成答案時會參考歷史資訊多遠，方法是從答案詞元的最後一層注意力中聚合歷史詞元，並按片段距離 $d = \tau_r - i$ 分組所注意的歷史資訊，其中 $\tau_r$ 表示回答第 $r$-個問題時觀察到的最後一片段索引，而 $i \leq \tau_r$ 表示被注意的歷史片段索引。這裡， $d = 0$ 對應最後一片段，較大的 $d$ 表示更早的片段。

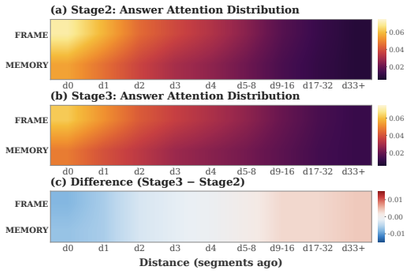



Figure 3: **Answer attention vs. segment distance** on $\text{TWW}_{\text{multi-turn}}$. After Stage 3, attention mass shifts from near-history to more distant segments.

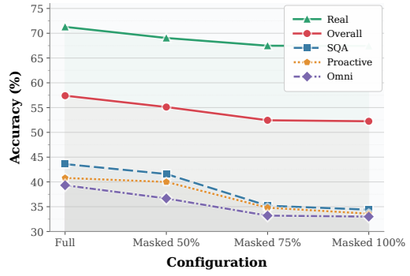



Figure 4: **Ablation** under frame masking on $\text{TWW}_{\text{multi-turn,S3}}$. Overall represents the accuracy rate. The remaining curves represent the results of the subsets.

Table 4: **Offline video understanding results** on Video-MME and LV-Bench.

| | Video-MME | | | | LV-Bench | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Method** | **Short** | **Medium** | **Long** | **Overall** | **ER** | **EU** | **KIR** | **TG** | **Rea** | **Sum** | **Overall** |
| Thinking | 77.78 | 66.67 | 62.22 | 68.89 | 54.22 | 51.62 | 62.07 | 46.82 | 55.50 | 40.35 | 53.47 |
| Instruct | 78.22 | 66.78 | 62.89 | 69.30 | 56.15 | 53.78 | 65.86 | 50.45 | 60.00 | 43.86 | 56.19 |
| $\text{TWW}_{\text{Batch,S3}}$ | 78.89 | 67.11 | 64.00 | 70.00 | **58.07** | 54.24 | 66.21 | **51.36** | 62.00 | 47.37 | 57.39 |
| $\text{TWW}_{\text{multi-turn,S3}}$ | 79.00 | 67.11 | 62.33 | 69.48 | 56.44 | 54.39 | 66.55 | 50.91 | 61.50 | 45.61 | 56.81 |
| $\text{TWW}_{\text{single-turn,S3}}$ | **83.22** | **70.11** | **66.89** | **73.41** | 57.33 | **54.70** | **67.59** | **51.36** | **62.50** | **52.63** | **57.68** |

We separately measure attention to **FRAME** tokens, which are visual tokens from prior segments, and **MEMORY** tokens, which are accumulated memory note tokens written for those segments. Figure 3 shows that the Stage 2 checkpoint exhibits a strong recency bias, whereas Stage 3 reallocates attention mass from near-history buckets to more distant buckets. The shift is more pronounced on **MEMORY** tokens than on visual tokens, consistent with the intended role of memory notes as a compact long-range state for multi-turn interaction.

**Ablation Studies.** We conduct ablations to isolate the roles of the memory bank, visual inputs, and segmentation granularity. Removing the memory bank causes a clear accuracy drop from 57.40% to 52.35% in Table 5, confirming that memory notes serve as an effective persistent state in multi-round streams. For visual ablations, Fig. 4 shows a monotonic degradation as more frames are masked. Performance remains relatively stable under moderate corruption, suggesting that once written, segment-level memory notes provide a stabilizing signal. Under severe corruption, accuracy approaches the no-memory regime, which is expected because memory writing becomes unreliable without sufficient visual evidence. For segmentation granularity, Table 5 reveals a clear accuracy and efficiency tradeoff. Using longer segments 120s/60s reduces the average decoding length from 302.56 to 230.46 tokens but causes a noticeable accuracy drop of 2.07%. Conversely, using shorter segments 30s/15s preserves accuracy but increases the average decoding length to 380.50 tokens (+25.8%), due to more frequent memory updates.

**TTFT Analysis.** Table 6 reports overall accuracy and TTFT on StreamingBench with the 4B backbone. Compared with batch Thinking, our streaming pipeline reduces TTFT by **92.6%**, from 31203.69 to 2304.28 tokens, while maintaining comparable accuracy. The interleaved baseline achieves a similar TTFT for multi-turn streaming but is consistently less accurate.

**Theoretical Latency Analysis.** Our method decouples ingestion from decoding, largely eliminating decoder-induced ingestion backlog, thereby avoiding backlog explosion as $\rho$ (arrival rate over processing rate) approaches 1 and significantly reducing latency. The complete derivation is in Appendix C.



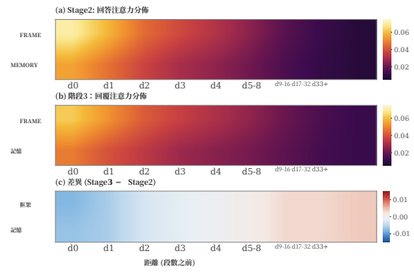



圖3：$\text{TWW}_{\text{multi-turn}}$上答案注意力與片段距離的關係。Stage 3之後，注意力質量從近史片段轉移到更遠的片段。

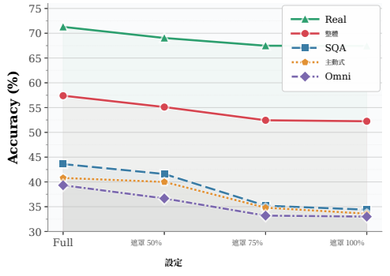



圖4：$\text{TWW}_{\text{multi-turn,S3}}$在框架掩碼下的刪除實驗。整體代表準確率。其餘曲線代表子集的結果。

表4：Video-MME和LV-Bench上的離線視頻理解結果。

| | Video-MME | | | | LV-Bench | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **方法** | **短** | **中** | **Long** | **總體** | **ER** | **EU** | **KIR** | **TG** | **Rea** | **Sum** | **總體** |
| 思考 | 77.78 | 66.67 | 62.22 | 68.89 | 54.22 | 51.62 | 62.07 | 46.82 | 55.50 | 40.35 | 53.47 |
| 指示 | 78.22 | 66.78 | 62.89 | 69.30 | 56.15 | 53.78 | 65.86 | 50.45 | 60.00 | 43.86 | 56.19 |
| $\text{TWW}_{\text{Batch}}$ ,S3 | 78.89 | 67.11 | 64.00 | 70.00 | 58.07 | 54.24 | 66.21 | 51.36 | 62.00 | 47.37 | 57.39 |
| $\text{TWW}_{\text{multi-turn,S3}}$ | 79.00 | 67.11 | 62.33 | 69.48 | 56.44 | 54.39 | 66.55 | 50.91 | 61.50 | 45.61 | 56.81 |
| $\text{TWW}_{\text{single-turn,S3}}$ | **83.22** | **70.11** | **66.89** | **73.41** | 57.33 | **54.70** | **67.59** | **51.36** | **62.50** | **52.63** | **57.68** |

我們分別測量對 FRAME 詞元的注意力，這些是來自先前片段的視覺詞元，以及對 MEMORY 詞元的注意力，這些是為那些片段累積的記憶筆記詞元。圖 3 顯示 Stage 2 檢查點呈現強烈的近期偏好，而 Stage 3 則將注意力質量從近期的歷史資訊桶重新分配到更遠的桶。這種轉變在 MEMORY 詞元上比在視覺詞元上更明顯，與記憶筆記作為多輪互動的緊湊長程狀態的預期角色一致。

消融實驗。我們進行消融實驗以孤立記憶庫、視覺輸入和分割粒度的作用。移除記憶庫導致表5中的準確率明顯下降從57.40%到52.35%，確認記憶筆記作為一種有效的持續狀態在多輪流中。對於視覺消融，圖4顯示隨著更多幀被遮罩出現單調的退化。在中等程度的損壞下，性能相對穩定，表明一旦寫入，段級記憶筆記提供了一個穩定信號。在嚴重損壞下，準確率接近無記憶狀態，這是預期的，因為在沒有足夠視覺證據的情況下，記憶寫入變得不可靠。對於分割粒度，表5顯示了明確的準確率和效率的取捨。使用更長的段120s/60s將平均解碼長度從302.56減少到230.46個詞元，但導致準確率下降2.07%。相反地，使用更短的段30s/15s保護準確率，但將平均解碼長度增加到380.50個詞元(+25.8%)，這是由於更頻繁的記憶更新。

TTFT 分析。表 6 報告了使用 4B 主幹架構在 StreamingBench 上的整體準確率與 TTFT。與批次思考相比，我們的串流管線將 TTFT 減少了 92.6%，從 31203.69 個 token 減少到 2304.28 個 token，同時保持相當的準確率。交錯基線在多輪串流方面達到相似的 TTFT，但始終準確率較低。

理論延遲分析。我們的方法將資料匯入與解碼分離，大幅消除解碼引發的匯入積壓，從而在 $\rho$ （到達率超過處理率）接近 1 時避免積壓爆炸，並顯著降低延遲。完整的推導過程在附錄 C 中。

Table 5: **Ablations** on StreamingBench. a/b represents the maximum segment duration and the chunk duration.

| Category | Setting | Acc↑ | Avg Tok↓ |
|---|---|---|---|
| Memory | with notes | **57.40** | 302.56 |
| Memory | without notes | 52.35 | 330.73 |
| Segment | 60s/30s | **57.40** | 302.56 |
| Segment | 120s/60s | 55.33 | **230.46** |
| Segment | 30s/15s | 57.20 | 380.50 |

Table 6: **TTFT on StreamingBench.** Overall accuracy and time-to-first-token of Qwen3-VL-4B for batch, interleaved streaming, and our $\text{TWW}_{\text{multi-turn,S3}}$ pipeline.

| Method | Overall Acc↑ | TTFT↓ |
|---|---|---|
| Thinking | **58.52** | 31203.69 |
| Interleaved | 55.35 | **2304.28** |
| $\text{TWW}_{\text{multi-turn,S3}}$ | 57.40 | **2304.28** |

## 6 Conclusion

We presented **Think While Watching**, a memory-anchored streaming video reasoning framework for multi-turn interaction over continuously arriving streams. Our approach maintains **segment-level memory notes** as a persistent state, enforces strict causality through a **segment-level streaming causal mask** and **streaming positional encoding**, and enables practical real-time deployment via a dual KV cache pipeline with adaptive attention backends. Experiments on StreamingBench and OVO-Bench validate the proposed method's effectiveness, consistently improving online accuracy while maintaining strong efficiency.

## References


[1] Jingqi Tong, Yurong Mou, Hangcheng Li, Mingzhe Li, Yongzhuo Yang, Ming Zhang, Qiguang Chen, Tianyi Liang, Xiaomeng Hu, Yining Zheng, Xinchi Chen, Jun Zhao, Xuanjing Huang, and Xipeng Qiu. Thinking with video: Video generation as a promising multimodal reasoning paradigm. *CoRR*, abs/2511.04570, 2025. doi: 10.48550/ARXIV.2511.04570. URL https://doi.org/10.48550/arXiv.2511.04570.

[2] Haoji Zhang, Xin Gu, Jiawen Li, Chixiang Ma, Sule Bai, Chubin Zhang, Bowen Zhang, Zhichao Zhou, Dongliang He, and Yansong Tang. Thinking with videos: Multimodal tool-augmented reinforcement learning for long video reasoning. *CoRR*, abs/2508.04416, 2025. doi: 10.48550/ARXIV.2508.04416. URL https://doi.org/10.48550/arXiv.2508.04416.

[3] Kaituo Feng, Kaixiong Gong, Bohao Li, Zonghao Guo, Yibing Wang, Tianshuo Peng, Benyou Wang, and Xiangyu Yue. Video-r1: Reinforcing video reasoning in mllms. *CoRR*, abs/2503.21776, 2025. doi: 10.48550/ARXIV.2503.21776. URL https://doi.org/10.48550/arXiv.2503.21776.

[4] Hao Fei, Shengqiong Wu, Wei Ji, Hanwang Zhang, Meishan Zhang, Mong-Li Lee, and Wynne Hsu. Video-of-thought: Step-by-step video reasoning from perception to cognition. *CoRR*, abs/2501.03230, 2025. doi: 10.48550/ARXIV.2501.03230. URL https://doi.org/10.48550/arXiv.2501.03230.

[5] Zhenyu Yang, Kairui Zhang, Yuhang Hu, Bing Wang, Shengsheng Qian, Bin Wen, Fan Yang, Tingting Gao, Weiming Dong, and Changsheng Xu. Livestar: Live streaming assistant for real-world online video understanding. *CoRR*, abs/2511.05299, 2025. doi: 10.48550/ARXIV.2511.05299. URL https://doi.org/10.48550/arXiv.2511.05299.

[6] Joya Chen, Ziyun Zeng, Yiqi Lin, Wei Li, Zejun Ma, and Mike Zheng Shou. Livecc: Learning video LLM with streaming speech transcription at scale. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 29083–29095. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02708. URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_LiveCC_Learning_Video_LLM_with_Streaming_Speech_Transcription_at_Scale_CVPR_2025_paper.html.

[7] Yifei Wang, Zhenkai Li, Tianwen Qian, Huanran Zheng, Zheng Wang, Yuqian Fu, and Xiaoling Wang. Streameqa: Towards streaming video understanding for embodied scenarios, 2025. URL https://arxiv.org/abs/2512.04451.




表5：StreamingBench上的消融實驗。a/b代表最大片段持續時間和塊持續時間。

表6：TTFT在StreamingBench上的表現。Qwen3-VL-4B對於批次、交錯串流以及我們的$\text{TWW}_{\text{multi-turn,S3}}$流程的整體準確率與首次Token時間。

| 分類 | 設定 | Acc↑ | 平均Token↓ |
|---|---|---|---|
| 記憶 | 含備註 | **57.40** | 302.56 |
| 記憶 | 無筆記 | 52.35 | 330.73 |
| 片段 | 60秒/30秒 | **57.40** | 302.56 |
| 片段 | 120秒/60秒 | 55.33 | **230.46** |
| 區段 | 30秒/15秒 | 57.20 | 380.50 |

| 方法 | 整體準確率↑ | 首次Token時間↓ |
|---|---|---|
| 思考 | **58.52** | 31203.69 |
| 交錯 | 55.35 | **2304.28** |
| $\text{TWW}_{\text{multi-turn,S3}}$ | 57.40 | **2304.28** |

## 6 結論

我們提出了「觀看時思考」(Think While Watching)，一個基於記憶的流式視頻推理框架，用於處理持續到達的流式數據的多輪互動。我們的方法將片段級別的記憶筆記作為持續狀態維持，透過片段級別的流式因果掩碼和流式位置編碼強制因果關係，並透過一個具有自適應注意力的雙 KV 緩存管道實現實時部署。在 StreamingBench 和 OVO-Bench 上的實驗驗證了所提出方法的有效性，始終如一地提高線上準確性，同時保持強大的效率。

## 參考文獻


[1] Jingqi Tong, Yurong Mou, Hangcheng Li, Mingzhe Li, Yongzhuo Yang, Ming Zhang, Qiguang Chen, Tianyi Liang, Xiaomeng Hu, Yining Zheng, Xinchi Chen, Jun Zhao, Xuanjing Huang, and Xipeng Qiu. Thinking with video: Video generation as a promising multimodal reasoning paradigm. *CoRR*, abs/2511.04570, 2025. doi: 10.48550/ARXIV.2511.04570. URL https://doi.org/10.48550/arXiv.2511.04570.

[2] Haoji Zhang, Xin Gu, Jiawen Li, Chixiang Ma, Sule Bai, Chubin Zhang, Bowen Zhang, Zhichao Zhou, Dongliang He, and Yansong Tang. Thinking with videos: Multimodal tool-augmented reinforcement learning for long video reasoning. *CoRR*, abs/2508.04416, 2025. doi: 10.48550/ARXIV.2508.04416. URL https://doi.org/10.48550/arXiv.2508.04416.

[3] Kaituo Feng, Kaixiong Gong, Bohao Li, Zonghao Guo, Yibing Wang, Tianshuo Peng, Benyou Wang, and Xiangyu Yue. Video-r1: Reinforcing video reasoning in mllms. *CoRR*, abs/2503.21776, 2025. doi: 10.48550/ARXIV.2503.21776. URL https://doi.org/10.48550/arXiv.2503.21776.

[4] Hao Fei, Shengqiong Wu, Wei Ji, Hanwang Zhang, Meishan Zhang, Mong-Li Lee, and Wynne Hsu. Video-of-thought: Step-by-step video reasoning from perception to cognition. *CoRR*, abs/2501.03230, 2025. doi: 10.48550/ARXIV.2501.03230. URL https://doi.org/10.48550/arXiv.2501.03230.

[5] Zhenyu Yang, Kairui Zhang, Yuhang Hu, Bing Wang, Shengsheng Qian, Bin Wen, Fan Yang, Tingting Gao, Weiming Dong, and Changsheng Xu. Livestar: Live streaming assistant for real-world online video understanding. *CoRR*, abs/2511.05299, 2025. doi: 10.48550/ARXIV.2511.05299. URL https://doi.org/10.48550/arXiv.2511.05299.

[6] Joya Chen, Ziyun Zeng, Yiqi Lin, Wei Li, Zejun Ma, and Mike Zheng Shou. Livecc: Learning video LLM with streaming speech transcription at scale. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 29083–29095. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02708. URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_LiveCC_Learning_Video_LLM_with_Streaming_Speech_Transcription_at_Scale_CVPR_2025_paper.html.

[7] Yifei Wang, Zhenkai Li, Tianwen Qian, Huanran Zheng, Zheng Wang, Yuqian Fu, and Xiaoling Wang. Streameqa: Towards streaming video understanding for embodied scenarios, 2025. URL https://arxiv.org/abs/2512.04451.

[8] Joya Chen, Zhaoyang Lv, Shiwei Wu, Kevin Qinghong Lin, Chenan Song, Difei Gao, Jia-Wei Liu, Ziteng Gao, Dongxing Mao, and Mike Zheng Shou. Videollm-online: Online video large language model for streaming video. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pages 18407–18418. IEEE, 2024. doi: 10.1109/CVPR52733.2024.01742. URL https://doi.org/10.1109/CVPR52733.2024.01742.

[9] Jihao Liu, Zhiding Yu, Shiyi Lan, Shihao Wang, Rongyao Fang, Jan Kautz, Hongsheng Li, and José M. Álvarez. Streamchat: Chatting with streaming video. *CoRR*, abs/2412.08646, 2024. doi: 10.48550/ARXIV.2412.08646. URL https://doi.org/10.48550/arXiv.2412.08646.

[10] Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne. *Operating System Concepts, 10th Edition*. Wiley, 2018. ISBN 978-1-118-06333-0. URL http://os-book.com/OS10/index.html.

[11] Junlong Tong, Jinlan Fu, Zixuan Lin, Yingqi Fan, Anhao Zhao, Hui Su, and Xiaoyu Shen. LLM as effective streaming processor: Bridging streaming-batch mismatches with group position encoding. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics, ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pages 23497–23517. Association for Computational Linguistics, 2025. URL https://aclanthology.org/2025.findings-acl.1207/.

[12] Junlong Tong, Yingqi Fan, Anhao Zhao, Yunpu Ma, and Xiaoyu Shen. Streamingthinker: Large language models can think while reading. *CoRR*, abs/2510.17238, 2025. doi: 10.48550/ARXIV.2510.17238. URL https://doi.org/10.48550/arXiv.2510.17238.

[13] Junyan Lin, Junlong Tong, Hao Wu, Jialiang Zhang, Jinming Liu, Xin Jin, and Xiaoyu Shen. Speak while watching: Unleashing true real-time video understanding capability of multimodal large language models, 2026. URL https://arxiv.org/abs/2601.06843.

[14] Yukang Chen, Wei Huang, Baifeng Shi, Qinghao Hu, Hanrong Ye, Ligeng Zhu, Zhijian Liu, Pavlo Molchanov, Jan Kautz, Xiaojuan Qi, Sifei Liu, Hongxu Yin, Yao Lu, and Song Han. Scaling RL to long videos. *CoRR*, abs/2507.07966, 2025. doi: 10.48550/ARXIV.2507.07966. URL https://doi.org/10.48550/arXiv.2507.07966.

[15] Junming Lin, Zheng Fang, Chi Chen, Zihao Wan, Fuwen Luo, Peng Li, Yang Liu, and Maosong Sun. Streamingbench: Assessing the gap for mllms to achieve streaming video understanding, 2024. URL https://arxiv.org/abs/2411.03628.

[16] Junbo Niu, Yifei Li, Ziyang Miao, Chunjiang Ge, Yuanhang Zhou, Qihao He, Xiaoyi Dong, Haodong Duan, Shuangrui Ding, Rui Qian, Pan Zhang, Yuhang Zang, Yuhang Cao, Conghui He, and Jiaqi Wang. Ovo-bench: How far is your video-llms from real-world online video understanding? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 18902–18913. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.01761. URL https://openaccess.thecvf.com/content/CVPR2025/html/Niu_OVO-Bench_How_Far_is_Your_Video-LLMs_from_Real-World_Online_Video_CVPR_2025_paper.html.

[17] Shenghao Fu, Qize Yang, Yuan-Ming Li, Yi-Xing Peng, Kun-Yu Lin, Xihan Wei, Jian-Fang Hu, Xiaohua Xie, and Wei-Shi Zheng. Vispeak: Visual instruction feedback in streaming videos. *CoRR*, abs/2503.12769, 2025. doi: 10.48550/ARXIV.2503.12769. URL https://doi.org/10.48550/arXiv.2503.12769.

[18] Daeun Lee, Subhojyoti Mukherjee, Branislav Kveton, Ryan A. Rossi, Viet Dac Lai, Seunghyun Yoon, Trung Bui, Franck Dernoncourt, and Mohit Bansal. Streamgaze: Gaze-guided temporal reasoning and proactive understanding in streaming videos, 2025. URL https://arxiv.org/abs/2512.01707.

[19] Yulin Zhang, Cheng Shi, Yang Wang, and Sibei Yang. Eyes wide open: Ego proactive video-llm for streaming video. *CoRR*, abs/2510.14560, 2025. doi: 10.48550/ARXIV.2510.14560. URL https://doi.org/10.48550/arXiv.2510.14560.



[8] Joya Chen, Zhaoyang Lv, Shiwei Wu, Kevin Qinghong Lin, Chenan Song, Difei Gao, Jia-Wei Liu, Ziteng Gao, Dongxing Mao, and Mike Zheng Shou. Videollm-online: Online video large language model for streaming video. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pages 18407–18418. IEEE, 2024. doi: 10.1109/CVPR52733.2024.01742. URL https://doi.org/10.1109/CVPR52733.2024.01742.

[9] Jihao Liu, Zhiding Yu, Shiyi Lan, Shihao Wang, Rongyao Fang, Jan Kautz, Hongsheng Li, and José M. Álvarez. Streamchat: Chatting with streaming video. *CoRR*, abs/2412.08646, 2024. doi: 10.48550/ARXIV.2412.08646. URL https://doi.org/10.48550/arXiv.2412.08646.

[10] Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne. *Operating System Concepts, 10th Edition*. Wiley, 2018. ISBN 978-1-118-06333-0. URL http://os-book.com/OS10/index.html.

[11] Junlong Tong, Jinlan Fu, Zixuan Lin, Yingqi Fan, Anhao Zhao, Hui Su, and Xiaoyu Shen. LLM as effective streaming processor: Bridging streaming-batch mismatches with group position encoding. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Findings of the Association for Computational Linguistics, ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pages 23497–23517. Association for Computational Linguistics, 2025. URL https://aclanthology.org/2025.findings-acl.1207/.

[12] Junlong Tong, Yingqi Fan, Anhao Zhao, Yunpu Ma, and Xiaoyu Shen. Streamingthinker: Large language models can think while reading. *CoRR*, abs/2510.17238, 2025. doi: 10.48550/ARXIV.2510.17238. URL https://doi.org/10.48550/arXiv.2510.17238.

[13] Junyan Lin, Junlong Tong, Hao Wu, Jialiang Zhang, Jinming Liu, Xin Jin, and Xiaoyu Shen. Speak while watching: Unleashing true real-time video understanding capability of multimodal large language models, 2026. URL https://arxiv.org/abs/2601.06843.

[14] Yukang Chen, Wei Huang, Baifeng Shi, Qinghao Hu, Hanrong Ye, Ligeng Zhu, Zhijian Liu, Pavlo Molchanov, Jan Kautz, Xiaojuan Qi, Sifei Liu, Hongxu Yin, Yao Lu, and Song Han. Scaling RL to long videos. *CoRR*, abs/2507.07966, 2025. doi: 10.48550/ARXIV.2507.07966. URL https://doi.org/10.48550/arXiv.2507.07966.

[15] Junming Lin, Zheng Fang, Chi Chen, Zihao Wan, Fuwen Luo, Peng Li, Yang Liu, and Maosong Sun. Streamingbench: Assessing the gap for mllms to achieve streaming video understanding, 2024. URL https://arxiv.org/abs/2411.03628.

[16] Junbo Niu, Yifei Li, Ziyang Miao, Chunjiang Ge, Yuanhang Zhou, Qihao He, Xiaoyi Dong, Haodong Duan, Shuangrui Ding, Rui Qian, Pan Zhang, Yuhang Zang, Yuhang Cao, Conghui He, and Jiaqi Wang. Ovo-bench: How far is your video-llms from real-world online video understanding? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 18902–18913. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.01761. URL https://openaccess.thecvf.com/content/CVPR2025/html/Niu_OVO-Bench_How_Far_is_Your_Video-LLMs_from_Real-World_Online_Video_CVPR_2025_paper.html.

[17] Shenghao Fu, Qize Yang, Yuan-Ming Li, Yi-Xing Peng, Kun-Yu Lin, Xihan Wei, Jian-Fang Hu, Xiaohua Xie, and Wei-Shi Zheng. Vispeak: Visual instruction feedback in streaming videos. *CoRR*, abs/2503.12769, 2025. doi: 10.48550/ARXIV.2503.12769. URL https://doi.org/10.48550/arXiv.2503.12769.

[18] Daeun Lee, Subhojyoti Mukherjee, Branislav Kveton, Ryan A. Rossi, Viet Dac Lai, Seunghyun Yoon, Trung Bui, Franck Dernoncourt, and Mohit Bansal. Streamgaze: Gaze-guided temporal reasoning and proactive understanding in streaming videos, 2025. URL https://arxiv.org/abs/2512.01707.

[19] Yulin Zhang, Cheng Shi, Yang Wang, and Sibei Yang. Eyes wide open: Ego proactive video-llm for streaming video. *CoRR*, abs/2510.14560, 2025. doi: 10.48550/ARXIV.2510.14560. URL https://doi.org/10.48550/arXiv.2510.14560.

[20] Zhenyu Yang, Yuhang Hu, Zemin Du, Dizhan Xue, Shengsheng Qian, Jiahong Wu, Fan Yang, Weiming Dong, and Changsheng Xu. Svbench: A benchmark with temporal multi-turn dialogues for streaming video understanding. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=Hz4BYVY8YM.

[21] Shuhang Xun, Sicheng Tao, Jungang Li, Yibo Shi, Zhixin Lin, Zhanhui Zhu, Yibo Yan, Hanqian Li, Linghao Zhang, Shikang Wang, Yixin Liu, Hanbo Zhang, Ying Ma, and Xuming Hu. Rtv-bench: Benchmarking MLLM continuous perception, understanding and reasoning through real-time video. *CoRR*, abs/2505.02064, 2025. doi: 10.48550/ARXIV.2505.02064. URL https://doi.org/10.48550/arXiv.2505.02064.

[22] Ruyi Xu, Guangxuan Xiao, Yukang Chen, Liuning He, Kelly Peng, Yao Lu, and Song Han. Streamingvlm: Real-time understanding for infinite video streams. *CoRR*, abs/2510.09608, 2025. doi: 10.48550/ARXIV.2510.09608. URL https://doi.org/10.48550/arXiv.2510.09608.

[23] Linli Yao, Yicheng Li, Yuancheng Wei, Lei Li, Shuhuai Ren, Yuanxin Liu, Kun Ouyang, Lean Wang, Shicheng Li, Sida Li, Lingpeng Kong, Qi Liu, Yuanxing Zhang, and Xu Sun. Timechat-online: 80% visual tokens are naturally redundant in streaming videos. *CoRR*, abs/2504.17343, 2025. doi: 10.48550/ARXIV.2504.17343. URL https://doi.org/10.48550/arXiv.2504.17343.

[24] Xingyi Zhou, Anurag Arnab, Shyamal Buch, Shen Yan, Austin Myers, Xuehan Xiong, Arsha Nagrani, and Cordelia Schmid. Streaming dense video captioning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pages 18243–18252. IEEE, 2024. doi: 10.1109/CVPR52733.2024.01727. URL https://doi.org/10.1109/CVPR52733.2024.01727.

[25] Xueyi Chen, Keda Tao, Kele Shao, and Huan Wang. Streamingtom: Streaming token compression for efficient video understanding. *CoRR*, abs/2510.18269, 2025. doi: 10.48550/ARXIV.2510.18269. URL https://doi.org/10.48550/arXiv.2510.18269.

[26] Haomiao Xiong, Zongxin Yang, Jiazuo Yu, Yunzhi Zhuge, Lu Zhang, Jiawen Zhu, and Huchuan Lu. Streaming video understanding and multi-round interaction with memory-enhanced knowledge. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=JbPb6RieNC.

[27] Donghyuk Kim, Sejeong Yang, Wonjin Shin, and Joo-Young Kim. V-rex: Real-time streaming video llm acceleration via dynamic kv cache retrieval, 2025. URL https://arxiv.org/abs/2512.12284.

[28] Yilong Chen, Xiang Bai, Zhibin Wang, Chengyu Bai, Yuhan Dai, Ming Lu, and Shanghang Zhang. Streamkv: Streaming video question-answering with segment-based KV cache retrieval and compression. *CoRR*, abs/2511.07278, 2025. doi: 10.48550/ARXIV.2511.07278. URL https://doi.org/10.48550/arXiv.2511.07278.

[29] Shangzhe Di, Zhelun Yu, Guanghao Zhang, Haoyuan Li, Tao Zhong, Hao Cheng, Bolin Li, Wanggui He, Fangxun Shu, and Hao Jiang. Streaming video question-answering with in-context video kv-cache retrieval. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=8g9fs6mdEG.

[30] Zhenyu Ning, Guangda Liu, Qihao Jin, Wenchao Ding, Minyi Guo, and Jieru Zhao. Livevlm: Efficient online video understanding via streaming-oriented KV cache and retrieval. *CoRR*, abs/2505.15269, 2025. doi: 10.48550/ARXIV.2505.15269. URL https://doi.org/10.48550/arXiv.2505.15269.

[31] Takahiro Maeda, Jinkun Cao, Norimichi Ukita, and Kris Kitani. Cacheflow: Fast human motion prediction by cached normalizing flow. *CoRR*, abs/2505.13140, 2025. doi: 10.48550/ARXIV.2505.13140. URL https://doi.org/10.48550/arXiv.2505.13140.



[20] Zhenyu Yang, Yuhang Hu, Zemin Du, Dizhan Xue, Shengsheng Qian, Jiahong Wu, Fan Yang, Weiming Dong, and Changsheng Xu. Svbench: A benchmark with temporal multi-turn dialogues for streaming video understanding. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=Hz4BYVY8YM.

[21] Shuhang Xun, Sicheng Tao, Jungang Li, Yibo Shi, Zhixin Lin, Zhanhui Zhu, Yibo Yan, Hanqian Li, Linghao Zhang, Shikang Wang, Yixin Liu, Hanbo Zhang, Ying Ma, and Xuming Hu. Rtv-bench: Benchmarking MLLM continuous perception, understanding and reasoning through real-time video. *CoRR*, abs/2505.02064, 2025. doi: 10.48550/ARXIV.2505.02064. URL https://doi.org/10.48550/arXiv.2505.02064.

[22] Ruyi Xu, Guangxuan Xiao, Yukang Chen, Liuning He, Kelly Peng, Yao Lu, and Song Han. Streamingvlm: Real-time understanding for infinite video streams. *CoRR*, abs/2510.09608, 2025. doi: 10.48550/ARXIV.2510.09608. URL https://doi.org/10.48550/arXiv.2510.09608.

[23] Linli Yao, Yicheng Li, Yuancheng Wei, Lei Li, Shuhuai Ren, Yuanxin Liu, Kun Ouyang, Lean Wang, Shicheng Li, Sida Li, Lingpeng Kong, Qi Liu, Yuanxing Zhang, and Xu Sun. Timechat-online: 80% visual tokens are naturally redundant in streaming videos. *CoRR*, abs/2504.17343, 2025. doi: 10.48550/ARXIV.2504.17343. URL https://doi.org/10.48550/arXiv.2504.17343.

[24] Xingyi Zhou, Anurag Arnab, Shyamal Buch, Shen Yan, Austin Myers, Xuehan Xiong, Arsha Nagrani, and Cordelia Schmid. Streaming dense video captioning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pages 18243–18252. IEEE, 2024. doi: 10.1109/CVPR52733.2024.01727. URL https://doi.org/10.1109/CVPR52733.2024.01727.

[25] Xueyi Chen, Keda Tao, Kele Shao, and Huan Wang. Streamingtom: Streaming token compression for efficient video understanding. *CoRR*, abs/2510.18269, 2025. doi: 10.48550/ARXIV.2510.18269. URL https://doi.org/10.48550/arXiv.2510.18269.

[26] Haomiao Xiong, Zongxin Yang, Jiazuo Yu, Yunzhi Zhuge, Lu Zhang, Jiawen Zhu, and Huchuan Lu. Streaming video understanding and multi-round interaction with memory-enhanced knowledge. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=JbPb6RieNC.

[27] Donghyuk Kim, Sejeong Yang, Wonjin Shin, and Joo-Young Kim. V-rex: Real-time streaming video llm acceleration via dynamic kv cache retrieval, 2025. URL https://arxiv.org/abs/2512.12284.

[28] Yilong Chen, Xiang Bai, Zhibin Wang, Chengyu Bai, Yuhan Dai, Ming Lu, and Shanghang Zhang. Streamkv: Streaming video question-answering with segment-based KV cache retrieval and compression. *CoRR*, abs/2511.07278, 2025. doi: 10.48550/ARXIV.2511.07278. URL https://doi.org/10.48550/arXiv.2511.07278.

[29] Shangzhe Di, Zhelun Yu, Guanghao Zhang, Haoyuan Li, Tao Zhong, Hao Cheng, Bolin Li, Wanggui He, Fangxun Shu, and Hao Jiang. Streaming video question-answering with in-context video kv-cache retrieval. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL https://openreview.net/forum?id=8g9fs6mdEG.

[30] Zhenyu Ning, Guangda Liu, Qihao Jin, Wenchao Ding, Minyi Guo, and Jieru Zhao. Livevlm: Efficient online video understanding via streaming-oriented KV cache and retrieval. *CoRR*, abs/2505.15269, 2025. doi: 10.48550/ARXIV.2505.15269. URL https://doi.org/10.48550/arXiv.2505.15269.

[31] Takahiro Maeda, Jinkun Cao, Norimichi Ukita, and Kris Kitani. Cacheflow: Fast human motion prediction by cached normalizing flow. *CoRR*, abs/2505.13140, 2025. doi: 10.48550/ARXIV.2505.13140. URL https://doi.org/10.48550/arXiv.2505.13140.

[32] Haoji Zhang, Yiqin Wang, Yansong Tang, Yong Liu, Jiashi Feng, and Xiaojie Jin. Flash-vstream: Efficient real-time understanding for long video streams. *CoRR*, abs/2506.23825, 2025. doi: 10.48550/ARXIV.2506.23825. URL https://doi.org/10.48550/arXiv.2506.23825.

[33] Xiangyu Zeng, Kefan Qiu, Qingyu Zhang, Xinhao Li, Jing Wang, Jiaxin Li, Ziang Yan, Kun Tian, Meng Tian, Xinhai Zhao, Yi Wang, and Limin Wang. Streamforest: Efficient online video understanding with persistent event memory. *CoRR*, abs/2509.24871, 2025. doi: 10.48550/ARXIV.2509.24871. URL https://doi.org/10.48550/arXiv.2509.24871.

[34] Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory. *CoRR*, abs/2508.09736, 2025. doi: 10.48550/ARXIV.2508.09736. URL https://doi.org/10.48550/arXiv.2508.09736.

[35] Haolin Yang, Feilong Tang, Linxiao Zhao, Xiang An, Ming Hu, Huifa Li, Xinlin Zhuang, Boqian Wang, Yifan Lu, Xiaofeng Zhang, Abdalla Swikir, Junjun He, Zongyuan Ge, and Imran Razzak. Streamagent: Towards anticipatory agents for streaming video understanding. *CoRR*, abs/2508.01875, 2025. doi: 10.48550/ARXIV.2508.01875. URL https://doi.org/10.48550/arXiv.2508.01875.

[36] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *CoRR*, abs/2409.12191, 2024. doi: 10.48550/ARXIV.2409.12191. URL https://doi.org/10.48550/arXiv.2409.12191.

[37] Yuhang Hu, Zhenyu Yang, Shihan Wang, Shengsheng Qian, Bin Wen, Fan Yang, Tingting Gao, and Changsheng Xu. Streamingcot: A dataset for temporal dynamics and multimodal chain-of-thought reasoning in streaming videoqa. *CoRR*, abs/2510.25332, 2025. doi: 10.48550/ARXIV.2510.25332. URL https://doi.org/10.48550/arXiv.2510.25332.

[38] Zhenpeng Huang, Xinhao Li, Jiaqi Li, Jing Wang, Xiangyu Zeng, Cheng Liang, Tao Wu, Xi Chen, Liang Li, and Limin Wang. Online video understanding: A comprehensive benchmark and memory-augmented method. *arXiv e-prints*, pages arXiv–2501, 2024.

[39] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness, 2022. URL https://arxiv.org/abs/2205.14135.

[40] Markus N. Rabe and Charles Staats. Self-attention does not need $o(n^2)$ memory, 2022. URL https://arxiv.org/abs/2112.05682.

[41] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, Soroosh Mariooryad, Yifan Ding, Xinyang Geng, Fred Alcober, Roy Frostig, Mark Omernick, Lexi Walker, Cosmin Paduraru, Christina Sorokin, Andrea Tacchetti, Colin Gaffney, Samira Daruki, Olcan Sercinoglu, Zach Gleicher, Juliette Love, Paul Voigtlaender, Rohan Jain, Gabriela Surita, Kareem Mohamed, Rory Blevins, Junwhan Ahn, Tao Zhu, Kornraphop Kawintiranon, Orhan Firat, Yiming Gu, Yujing Zhang, Matthew Rahtz, Manaal Faruqui, Natalie Clay, Justin Gilmer, JD Co-Reyes, Ivo Penchev, Rui Zhu, Nobuyuki Morioka, Kevin Hui, Krishna Haridasan, Victor Campos, Mahdis Mahdieh, Mandy Guo, Samer Hassan, Kevin Kilgour, Arpi Vezer, Heng-Tze Cheng, Raoul de Liedekerke, Siddharth Goyal, Paul Barham, DJ Strouse, Seb Noury, Jonas Adler, Mukund Sundararajan, Sharad Vikram, Dmitry Lepikhin, Michela Paganini, Xavier Garcia, Fan Yang, Dasha Valter, Maja Trebacz, Kiran Vodrahalli, Chulayuth Asawaroengchai, Roman Ring, Norbert Kalb, Livio Baldini Soares, Siddhartha Brahma, David Steiner, Tianhe Yu, Fabian Mentzer, Antoine He, Lucas Gonzalez, Bibo Xu, Raphael Lopez Kaufman, Laurent El Shafey, Junhyuk Oh, Tom Hennigan, George van den Driessche, Seth Odoom, Mario Lucic, Becca Roelofs, Sid Lall, Amit Marathe, Betty Chan, Santiago Ontanon, Luheng He, Denis Teplyashin, Jonathan Lai, Phil Crone, Bogdan Damoc, Lewis Ho, Sebastian Riedel, Karel Lenc, Chih-Kuan Yeh, Aakanksha Chowdhery, Yang Xu, Mehran Kazemi, Ehsan Amid, Anastasia Petrushkina, Kevin Swersky, Ali Khodaei, Gowoon Chen, Chris Larkin, Mario



[32] Haoji Zhang, Yiqin Wang, Yansong Tang, Yong Liu, Jiashi Feng, and Xiaojie Jin. Flash-vstream: Efficient real-time understanding for long video streams. *CoRR*, abs/2506.23825, 2025. doi: 10.48550/ARXIV.2506.23825. URL https://doi.org/10.48550/arXiv.2506.23825.

[33] Xiangyu Zeng, Kefan Qiu, Qingyu Zhang, Xinhao Li, Jing Wang, Jiaxin Li, Ziang Yan, Kun Tian, Meng Tian, Xinhai Zhao, Yi Wang, and Limin Wang. Streamforest: Efficient online video understanding with persistent event memory. *CoRR*, abs/2509.24871, 2025. doi: 10.48550/ARXIV.2509.24871. URL https://doi.org/10.48550/arXiv.2509.24871.

[34] Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory. *CoRR*, abs/2508.09736, 2025. doi: 10.48550/ARXIV.2508.09736. URL https://doi.org/10.48550/arXiv.2508.09736.

[35] Haolin Yang, Feilong Tang, Linxiao Zhao, Xiang An, Ming Hu, Huifa Li, Xinlin Zhuang, Boqian Wang, Yifan Lu, Xiaofeng Zhang, Abdalla Swikir, Junjun He, Zongyuan Ge, and Imran Razzak. Streamagent: Towards anticipatory agents for streaming video understanding. *CoRR*, abs/2508.01875, 2025. doi: 10.48550/ARXIV.2508.01875. URL https://doi.org/10.48550/arXiv.2508.01875.

[36] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *CoRR*, abs/2409.12191, 2024. doi: 10.48550/ARXIV.2409.12191. URL https://doi.org/10.48550/arXiv.2409.12191.

[37] Yuhang Hu, Zhenyu Yang, Shihan Wang, Shengsheng Qian, Bin Wen, Fan Yang, Tingting Gao, and Changsheng Xu. Streamingcot: A dataset for temporal dynamics and multimodal chain-of-thought reasoning in streaming videoqa. *CoRR*, abs/2510.25332, 2025. doi: 10.48550/ARXIV.2510.25332. URL https://doi.org/10.48550/arXiv.2510.25332.

[38] Zhenpeng Huang, Xinhao Li, Jiaqi Li, Jing Wang, Xiangyu Zeng, Cheng Liang, Tao Wu, Xi Chen, Liang Li, and Limin Wang. Online video understanding: A comprehensive benchmark and memory-augmented method. *arXiv e-prints*, pages arXiv–2501, 2024.

[39] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness, 2022. URL https://arxiv.org/abs/2205.14135.

[40] Markus N. Rabe and Charles Staats. Self-attention does not need $o(n^2)$ memory, 2022. URL https://arxiv.org/abs/2112.05682.

[41] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, Soroosh Mariooryad, Yifan Ding, Xinyang Geng, Fred Alcober, Roy Frostig, Mark Omernick, Lexi Walker, Cosmin Paduraru, Christina Sorokin, Andrea Tacchetti, Colin Gaffney, Samira Daruki, Olcan Sercinoglu, Zach Gleicher, Juliette Love, Paul Voigtlaender, Rohan Jain, Gabriela Surita, Kareem Mohamed, Rory Blevins, Junwhan Ahn, Tao Zhu, Kornraphop Kawintiranon, Orhan Firat, Yiming Gu, Yujing Zhang, Matthew Rahtz, Manaal Faruqui, Natalie Clay, Justin Gilmer, JD Co-Reyes, Ivo Penchev, Rui Zhu, Nobuyuki Morioka, Kevin Hui, Krishna Haridasan, Victor Campos, Mahdis Mahdieh, Mandy Guo, Samer Hassan, Kevin Kilgour, Arpi Vezer, Heng-Tze Cheng, Raoul de Liedekerke, Siddharth Goyal, Paul Barham, DJ Strouse, Seb Noury, Jonas Adler, Mukund Sundararajan, Sharad Vikram, Dmitry Lepikhin, Michela Paganini, Xavier Garcia, Fan Yang, Dasha Valter, Maja Trebacz, Kiran Vodrahalli, Chulayuth Asawaroengchai, Roman Ring, Norbert Kalb, Livio Baldini Soares, Siddhartha Brahma, David Steiner, Tianhe Yu, Fabian Mentzer, Antoine He, Lucas Gonzalez, Bibo Xu, Raphael Lopez Kaufman, Laurent El Shafey, Junhyuk Oh, Tom Hennigan, George van den Driessche, Seth Odoom, Mario Lucic, Becca Roelofs, Sid Lall, Amit Marathe, Betty Chan, Santiago Ontanon, Luheng He, Denis Teplyashin, Jonathan Lai, Phil Crone, Bogdan Damoc, Lewis Ho, Sebastian Riedel, Karel Lenc, Chih-Kuan Yeh, Aakanksha Chowdhery, Yang Xu, Mehran Kazemi, Ehsan Amid, Anastasia Petrushkina, Kevin Swersky, Ali Khodaei, Gowoon Chen, Chris Larkin, Mario

Pinto, Geng Yan, Adria Puigdomenech Badia, Piyush Patil, Steven Hansen, Dave Orr, Sebastien M. R. Arnold, Jordan Grimstad, Andrew Dai, Sholto Douglas, Rishika Sinha, Vikas Yadav, Xi Chen, Elena Gribovskaya, Jacob Austin, Jeffrey Zhao, Kaushal Patel, Paul Komarek, Sophia Austin, Sebastian Borgeaud, Linda Friso, Abhimanyu Goyal, Ben Caine, Kris Cao, Da-Woon Chung, Matthew Lamm, Gabe Barth-Maron, Thais Kagohara, Kate Olszewska, Mia Chen, Kaushik Shivakumar, Rishabh Agarwal, Harshal Godhia, Ravi Rajwar, Javier Snaider, Xerxes Dotiwalla, Yuan Liu, Aditya Barua, Victor Ungureanu, Yuan Zhang, Bat-Orgil Batsaikhan, Mateo Wirth, James Qin, Ivo Danihelka, Tulsee Doshi, Martin Chadwick, Jilin Chen, Sanil Jain, Quoc Le, Arjun Kar, Madhu Gurumurthy, Cheng Li, Ruoxin Sang, Fangyu Liu, Lampros Lamprou, Rich Munoz, Nathan Lintz, Harsh Mehta, Heidi Howard, Malcolm Reynolds, Lora Aroyo, Quan Wang, Lorenzo Blanco, Albin Cassirer, Jordan Griffith, Dipanjan Das, Stephan Lee, Jakub Sygnowski, Zach Fisher, James Besley, Richard Powell, Zafarali Ahmed, Dominik Paulus, David Reitter, Zalan Borsos, Rishabh Joshi, Aedan Pope, Steven Hand, Vittorio Selo, Vihan Jain, Nikhil Sethi, Megha Goel, Takaki Makino, Rhys May, Zhen Yang, Johan Schalkwyk, Christina Butterfield, Anja Hauth, Alex Goldin, Will Hawkins, Evan Senter, Sergey Brin, Oliver Woodman, Marvin Ritter, Eric Noland, Minh Giang, Vijay Bolina, Lisa Lee, Tim Blyth, Ian Mackinnon, Machel Reid, Obaid Sarvana, David Silver, Alexander Chen, Lily Wang, Loren Maggiore, Oscar Chang, Nithya Attaluri, Gregory Thornton, Chung-Cheng Chiu, Oskar Bunyan, Nir Levine, Timothy Chung, Evgenii Eltyshev, Xiance Si, Timothy Lillicrap, Demetra Brady, Vaibhav Aggarwal, Boxi Wu, Yuanzhong Xu, Ross McIlroy, Kartikeya Badola, Paramjit Sandhu, Erica Moreira, Wojciech Stokowiec, Ross Hemsley, Dong Li, Alex Tudor, Pranav Shyam, Elahe Rahimtoroghi, Salem Haykal, Pablo Sprechmann, Xiang Zhou, Diana Mincu, Yujia Li, Ravi Addanki, Kalpesh Krishna, Xiao Wu, Alexandre Frechette, Matan Eyal, Allan Dafoe, Dave Lacey, Jay Whang, Thi Avrahami, Ye Zhang, Emanuel Taropa, Hanzhao Lin, Daniel Toyama, Eliza Rutherford, Motoki Sano, HyunJeong Choe, Alex Tomala, Chalence Safranek-Shrader, Nora Kassner, Mantas Pajarskas, Matt Harvey, Sean Sechrist, Meire Fortunato, Christina Lyu, Gamaleldin Elsayed, Chenkai Kuang, James Lottes, Eric Chu, Chao Jia, Chih-Wei Chen, Peter Humphreys, Kate Baumli, Connie Tao, Rajkumar Samuel, Cicero Nogueira dos Santos, Anders Andreassen, Nemanja Rakićević, Dominik Grewe, Aviral Kumar, Stephanie Winkler, Jonathan Caton, Andrew Brock, Sid Dalmia, Hannah Sheahan, Iain Barr, Yingjie Miao, Paul Natsev, Jacob Devlin, Feryal Behbahani, Flavien Prost, Yanhua Sun, Artiom Myaskovsky, Thanumalayan Sankaranarayana Pillai, Dan Hurt, Angeliki Lazaridou, Xi Xiong, Ce Zheng, Fabio Pardo, Xiaowei Li, Dan Horgan, Joe Stanton, Moran Ambar, Fei Xia, Alejandro Lince, Mingqiu Wang, Basil Mustafa, Albert Webson, Hyo Lee, Rohan Anil, Martin Wicke, Timothy Dozat, Abhishek Sinha, Enrique Piqueras, Elahe Dabir, Shyam Upadhyay, Anudhyan Boral, Lisa Anne Hendricks, Corey Fry, Josip Djolonga, Yi Su, Jake Walker, Jane Labanowski, Ronny Huang, Vedant Misra, Jeremy Chen, RJ Skerry-Ryan, Avi Singh, Shruti Rijhwani, Dian Yu, Alex Castro-Ros, Beer Changpinyo, Romina Datta, Sumit Bagri, Arnar Mar Hrafnkelsson, Marcello Maggioni, Daniel Zheng, Yury Sulsky, Shaobo Hou, Tom Le Paine, Antoine Yang, Jason Riesa, Dominika Rogozinska, Dror Marcus, Dalia El Badawy, Qiao Zhang, Luyu Wang, Helen Miller, Jeremy Greer, Lars Lowe Sjos, Azade Nova, Heiga Zen, Rahma Chaabouni, Mihaela Rosca, Jiepu Jiang, Charlie Chen, Ruibo Liu, Tara Sainath, Maxim Krikun, Alex Polozov, Jean-Baptiste Lespiau, Josh Newlan, Zeyncep Cankara, Soo Kwak, Yunhan Xu, Phil Chen, Andy Coenen, Clemens Meyer, Katerina Tsihlas, Ada Ma, Juraj Gottweis, Jinwei Xing, Chenjie Gu, Jin Miao, Christian Frank, Zeynep Cankara, Sanjay Ganapathy, Ishita Dasgupta, Steph Hughes-Fitt, Heng Chen, David Reid, Keran Rong, Hongmin Fan, Joost van Amersfoort, Vincent Zhuang, Aaron Cohen, Shixiang Shane Gu, Anhad Mohananey, Anastasija Ilic, Taylor Tobin, John Wieting, Anna Bortsova, Phoebe Thacker, Emma Wang, Emily Caveness, Justin Chiu, Eren Sezener, Alex Kaskasoli, Steven Baker, Katie Millican, Mohamed Elhawaty, Kostas Aisopos, Carl Lebsack, Nathan Byrd, Hanjun Dai, Wenhao Jia, Matthew Wiethoff, Elnaz Davoodi, Albert Weston, Lakshman Yagati, Arun Ahuja, Isabel Gao, Golan Pundak, Susan Zhang, Michael Azzam, Khe Chai Sim, Sergi Caelles, James Keeling, Abhanshu Sharma, Andy Swing, YaGuang Li, Chenxi Liu, Carrie Grimes Bostock, Yamini Bansal, Zachary Nado, Ankesh Anand, Josh Lipschultz, Abhijit Karmarkar, Lev Proleev, Abe Ittycheriah, Soheil Hassas Yeganeh, George Polovets, Aleksandra Faust, Jiao Sun, Alban Rrustemi, Pen Li, Rakesh Shivanna, Jeremiah Liu, Chris Welty, Federico Lebron, Anirudh Baddepudi, Sebastian Krause, Emilio Parisotto, Radu Soricut, Zheng Xu, Dawn Bloxwich, Melvin Johnson, Behnam Neyshabur, Justin Mao-Jones, Renshen Wang, Vinay Ramasesh, Zaheer Abbas, Arthur Guez,



Pinto、耿言、Adria Puigdomenech Badia、Piyush Patil、Steven Hansen、Dave Orr、Sebastien M. R. Arnold、JordanGrimstad、Andrew Dai、Sholto Douglas、RishikaSinha、Vikas

Yadav、Xi Chen、Elena Gribovskaya、Jacob Austin、Jeffrey Zhao、Kaushal Patel、Paul Ko- marek、Sophia Austin、Sebastian Borgeaud、Linda Friso、Abhimanyu Goyal、Ben Caine、Kris Cao、Da-Woon Chung、Matthew Lamm、GabeBarth-Maron、Thais Kagohara、Kate Olszewska、Mia Chen、Kaushik Shivakumar、Rishabh Agarwal、Harshal Godhia、Ravi Rajwar、Javier Snaider、Xerxes Dotiwalla、Yuan Liu、Aditya Barua、Victor Ungureanu、Yuan Zhang、Bat- Orgil Batsaikhan、Mateo Wirth、James Qin、Ivo Danihelka、Tulsee Doshi、Martin Chadwick、Jilin Chen、Sanil Jain、Quoc Le、Arjun Kar、Madhu Gurumurthy、Cheng Li、Ruoxin Sang、Fangyu Liu、Lampros Lamprou、Rich Munoz、Nathan Lintz、Harsh Mehta、Heidi Howard、Malcolm Reynolds、Lora Aroyo、Quan Wang、Lorenzo Blanco、Albin Cassirer、Jordan Griffith、Dipanjan Das、Stephan Lee、Jakub Sygnowski、Zach Fisher、James Besley、Richard Powell、Zafarali Ahmed、Dominik Paulus、David Reitter、Zalan Borsos、Rishabh Joshi、Aedan Pope、Steven Hand、Vittorio Selo、Vihan Jain、Nikhil Sethi、Megha Goel、Takaki Makino、Rhys May、Zhen Yang、Johan Schalkwyk、Christina Butterfield、Anja Hauth、Alex Goldin、Will Hawkins、Evan Senter、Sergey Brin、Oliver Woodman、Marvin Ritter、Eric Noland、Minh Giang、Vijay Bolina、Lisa Lee、Tim Blyth、Ian Mackinnon、Machel Reid、Obaid Sarvana、David Sil- ver、Alexander Chen、Lily Wang、Loren Maggiore、Oscar Chang、Nithya Attaluri、Gregory Thornton、Chung-Cheng Chiu、Oskar Bunyan、Nir Levine、Timothy Chung、Evgenii Elty- shev、Xiance Si、Timothy Lillicrap、Demetra Brady、Vaibhav Aggarwal、Boxi Wu、Yuanzhong Xu、Ross McIlroy、Kartikeya Badola、Paramjit Sandhu、Erica Moreira、Wojciech Stokowiec、Ross Hemsley、Dong Li、Alex Tudor、Pranav Shyam、Elahe Rahimtoroghi、Salem Haykal、Pablo Sprechmann、Xiang Zhou、Diana Mincu、Yujia Li、Ravi Addanki、Kalpesh Krishna、Xiao Wu、Alexandre Frechette、Matan Eyal、Allan Dafoe、Dave Lacey、Jay Whang、Thi Avra- hami、Ye Zhang、Emanuel Taropa、Hanzhao Lin、Daniel Toyama、Eliza Rutherford、Motoki Sano、HyunJeong Choe、Alex Tomala、Chalence Safranek-Shrader、Nora Kassner、Mantas Pa- jarskas、Matt Harvey、Sean Sechrist、Meire Fortunato、Christina Lyu、Gamaleldin Elsayed、Chenkai Kuang、James Lottes、Eric Chu、Chao Jia、Chih-Wei Chen、Peter Humphreys、Kate Baumli、Connie Tao、Rajkumar Samuel、Cicero Nogueira dos Santos、Anders Andreassen、Nemanja Raki´cevi´c、Dominik Grewe、Aviral Kumar、Stephanie Winkler、Jonathan Caton、Andrew Brock、Sid Dalmia、Hannah Sheahan、Iain Barr、Yingjie Miao、Paul Natsev、Jacob Devlin、Feryal Behbahani、Flavien Prost、Yanhua Sun、Artiom Myaskovsky、Thanumalayan Sankara- narayana Pillai、Dan Hurt、Angeliki Lazaridou、Xi Xiong、Ce Zheng、Fabio Pardo、Xiaowei Li、Dan Horgan、Joe Stanton、Moran Ambar、Fei Xia、Alejandro Lince、Mingqiu Wang、Basil Mustafa、Albert Webson、Hyo Lee、Rohan Anil、Martin Wicke、Timothy Dozat、Abhishek Sinha、Enrique Piqueras、Elahe Dabir、Shyam Upadhyay、Anudhyan Boral、Lisa Anne Hen- dricks、Corey Fry、Josip Djolonga、Yi Su、Jake Walker、Jane Labanowski、Ronny Huang、Vedant Misra、Jeremy Chen、RJ Skerry-Ryan、Avi Singh、Shruti Rijhwani、Dian Yu、Alex Castro-Ros、Beer Changpinyo、Romina Datta、Sumit Bagri、Arnar Mar Hrafnkelsson、Mar- cello Maggioni、Daniel Zheng、Yury Sulsky、Shaobo Hou、Tom Le Paine、Antoine Yang、Ja- son Riesa、Dominika Rogozinska、Dror Marcus、Dalia El Badawy、Qiao Zhang、Luyu Wang、Helen Miller、Jeremy Greer、Lars Lowe Sjos、Azade Nova、Heiga Zen、Rahma Chaabouni、Mihaela Rosca、Jiepu Jiang、Charlie Chen、Ruibo Liu、Tara Sainath、Maxim Krikun、Alex Polozov、Jean-Baptiste Lespiau、Josh Newlan、Zeyncep Cankara、Soo Kwak、Yunhan Xu、Phil Chen、Andy Coenen、Clemens Meyer、Katerina Tsihlas、Ada Ma、Juraj Gottweis、Jinwei Xing、Chenjie Gu、Jin Miao、Christian Frank、Zeynep Cankara、Sanjay Ganapathy、Ishita Dasgupta、Steph Hughes-Fitt、Heng Chen、David Reid、Keran Rong、Hongmin Fan、Joost van Amers- foort、Vincent Zhuang、Aaron Cohen、Shixiang Shane Gu、Anhad Mohananey、Anastasija Ilic、Taylor Tobin、John Wieting、Anna Bortsova、Phoebe Thacker、E mma Wang、Emily Caveness、Justin Chiu、Eren Sezener、Alex Kaskasoli、Steven Baker、Katie Millican、Mohamed Elhawaty、Kostas Aisopos、Carl Lebsack、Nathan Byrd、Hanjun Dai、Wenhao Jia、Matthew Wiethoff、Elnaz Davoodi、Albert Weston、Lakshman Yagati、Arun Ahuja、Isabel Gao、Golan Pundak、Susan Zhang、Michael Azzam、Khe Chai Sim、Sergi Caelles、James Keeling、Abhanshu Sharma、Andy S wing、YaGuang Li、Chenxi Liu、Carrie Grimes Bostock、Yamini Bansal、Zachary Nado、Ankesh Anand、Josh Lipschultz、Abhijit Karmarkar、Lev Proleev、Abe Ittycheriah、So- heil Hassas Yeganeh、George Polovets、Aleksandra Faust、Jiao Sun、Alban Rrustemi、Pen Li、Rakesh Shivanna、Jeremiah Liu、Chris Welty、Federico Lebron、Anirudh Baddepudi、Sebastian Krause、Emilio Parisotto、Radu Soricut、Zheng Xu、Dawn Bloxwich、Melvin Johnson、Behnam Neyshabur、Justin Mao-Jones、Renshen Wang、Vinay Ramasesh、Zaheer Abbas、Arthur Guez、

Constant Segal, Duc Dung Nguyen, James Svensson, Le Hou, Sarah York, Kieran Milan, Sophie Bridgers, Wiktor Gworek, Marco Tagliasacchi, James Lee-Thorp, Michael Chang, Alexey Guseynov, Ale Jakse Hartman, Michael Kwong, Ruizhe Zhao, Sheleem Kashem, Elizabeth Cole, Antoine Miech, Richard Tanburn, Mary Phuong, Filip Pavetic, Sebastien Cevey, Ramona Comanescu, Richard Ives, Sherry Yang, Cosmo Du, Bo Li, Zizhao Zhang, Mariko Iinuma, Clara Huiyi Hu, Aurko Roy, Shaan Bijwadia, Zhenkai Zhu, Danilo Martins, Rachel Saputro, Anita Gergely, Steven Zheng, Dawei Jia, Ioannis Antonoglou, Adam Sadovsky, Shane Gu, Yingying Bi, Alek Andreev, Sina Samangooei, Mina Khan, Tomas Kocisky, Angelos Filos, Chintu Kumar, Colton Bishop, Adams Yu, Sarah Hodkinson, Sid Mittal, Premal Shah, Alexandre Moufarek, Yong Cheng, Adam Bloniarz, Jaehoon Lee, Pedram Pejman, Paul Michel, Stephen Spencer, Vladimir Feinberg, Xuehan Xiong, Nikolay Savinov, Charlotte Smith, Siamak Shakeri, Dustin Tran, Mary Chesus, Bernd Bohnet, George Tucker, Tamara von Glehn, Carrie Muir, Yiran Mao, Hideto Kazawa, Ambrose Slone, Kedar Soparkar, Disha Shrivastava, James Cobon-Kerr, Michael Sharman, Jay Pavagadhi, Carlos Araya, Karolis Misiunas, Nimesh Ghelani, Michael Laskin, David Barker, Qiujia Li, Anton Briukhov, Neil Houlsby, Mia Glaese, Balaji Lakshminarayanan, Nathan Schucher, Yunhao Tang, Eli Collins, Hyeontaek Lim, Fangxiaoyu Feng, Adria Recasens, Guangda Lai, Alberto Magni, Nicola De Cao, Aditya Siddhant, Zoe Ashwood, Jordi Orbay, Mostafa Dehghani, Jenny Brennan, Yifan He, Kelvin Xu, Yang Gao, Carl Saroufim, James Molloy, Xinyi Wu, Seb Arnold, Solomon Chang, Julian Schrittwieser, Elena Buchatskaya, Soroush Radpour, Martin Polacek, Skye Giordano, Ankur Bapna, Simon Tokumine, Vincent Hellendoorn, Thibault Sottiaux, Sarah Cogan, Aliaksei Severyn, Mohammad Saleh, Shantanu Thakoor, Laurent Shefey, Siyuan Qiao, Meenu Gaba, Shuo yiin Chang, Craig Swanson, Biao Zhang, Benjamin Lee, Paul Kishan Rubenstein, Gan Song, Tom Kwiatkowski, Anna Koop, Ajay Kannan, David Kao, Parker Schuh, Axel Stjerngren, Golnaz Ghiasi, Gena Gibson, Luke Vilnis, Ye Yuan, Felipe Tiengo Ferreira, Aishwarya Kamath, Ted Klimenko, Ken Franko, Kefan Xiao, Indro Bhattacharya, Miteyan Patel, Rui Wang, Alex Morris, Robin Strudel, Vivek Sharma, Peter Choy, Sayed Hadi Hashemi, Jessica Landon, Mara Finkelstein, Priya Jhakra, Justin Frye, Megan Barnes, Matthew Mauger, Dennis Daun, Khuslen Baatarsukh, Matthew Tung, Wael Farhan, Henryk Michalewski, Fabio Viola, Felix de Chaumont Quitry, Charline Le Lan, Tom Hudson, Qingze Wang, Felix Fischer, Ivy Zheng, Elspeth White, Anca Dragan, Jean baptiste Alayrac, Eric Ni, Alexander Pritzel, Adam Iwanicki, Michael Isard, Anna Bulanova, Lukas Zilka, Ethan Dyer, Devendra Sachan, Srivatsan Srinivasan, Hannah Muckenhirn, Honglong Cai, Amol Mandhane, Mukarram Tariq, Jack W. Rae, Gary Wang, Kareem Ayoub, Nicholas FitzGerald, Yao Zhao, Woohyun Han, Chris Alberti, Dan Garrette, Kashyap Krishnakumar, Mai Gimenez, Anselm Levskaya, Daniel Sohn, Josip Matak, Inaki Iturrate, Michael B. Chang, Jackie Xiang, Yuan Cao, Nishant Ranka, Geoff Brown, Adrian Hutter, Vahab Mirrokni, Nanxin Chen, Kaisheng Yao, Zoltan Egyed, Francois Galilee, Tyler Liechty, Praveen Kallakuri, Evan Palmer, Sanjay Ghemawat, Jasmine Liu, David Tao, Chloe Thornton, Tim Green, Mimi Jasarevic, Sharon Lin, Victor Cotruta, Yi-Xuan Tan, Noah Fiedel, Hongkun Yu, Ed Chi, Alexander Neitz, Jens Heitkaemper, Anu Sinha, Denny Zhou, Yi Sun, Charbel Kaed, Brice Hulse, Swaroop Mishra, Maria Georgaki, Sneha Kudugunta, Clement Farabet, Izhak Shafran, Daniel Vlasic, Anton Tsitsulin, Rajagopal Ananthanarayanan, Alen Carin, Guolong Su, Pei Sun, Shashank V, Gabriel Carvajal, Josef Broder, Iulia Comsa, Alena Repina, William Wong, Warren Weilun Chen, Peter Hawkins, Egor Filonov, Lucia Loher, Christoph Hirnschall, Weiyi Wang, Jingchen Ye, Andrea Burns, Hardie Cate, Diana Gage Wright, Federico Piccinini, Lei Zhang, Chu-Cheng Lin, Ionel Gog, Yana Kulizhskaya, Ashwin Sreevatsa, Shuang Song, Luis C. Cobo, Anand Iyer, Chetan Tekur, Guillermo Garrido, Zhuyun Xiao, Rupert Kemp, Huaixiu Steven Zheng, Hui Li, Ananth Agarwal, Christel Ngani, Kati Goshvadi, Rebeca Santamaria-Fernandez, Wojciech Fica, Xinyun Chen, Chris Gorgolewski, Sean Sun, Roopal Garg, Xinyu Ye, S. M. Ali Eslami, Nan Hua, Jon Simon, Pratik Joshi, Yelin Kim, Ian Tenney, Sahitya Potluri, Lam Nguyen Thiet, Quan Yuan, Florian Luisier, Alexandra Chronopoulou, Salvatore Scellato, Praveen Srinivasan, Minmin Chen, Vinod Koverkathu, Valentin Dalibard, Yaming Xu, Brennan Saeta, Keith Anderson, Thibault Sellam, Nick Fernando, Fantine Huot, Junehyuk Jung, Mani Varadarajan, Michael Quinn, Amit Raul, Maigo Le, Ruslan Habalov, Jon Clark, Komal Jalan, Kalesha Bullard, Achintya Singhal, Thang Luong, Boyu Wang, Sujeevan Rajayogam, Julian Eisenschlos, Johnson Jia, Daniel Finchelstein, Alex Yakubovich, Daniel Balle, Michael Fink, Sameer Agarwal, Jing Li, Dj Dvijotham, Shalini Pal, Kai Kang, Jaclyn Konzelmann, Jennifer Beattie, Olivier Dousse, Diane Wu, Remi Crocker, Chen Elkind, Siddhartha Reddy Jonnalagadda,



Constant Segal、杜 dung 阮、James Svensson、Le Hou、Sarah York、Kieran Milan、Sophie Bridgers、WiktorGworek、Marco Tagliasacchi、James Lee-Thorp、Michael Chang、Alexey

Guseynov、Ale Jakse Hartman、Michael Kwong、Ruizhe Zhao、Sheleem Kashem、Elizabeth Cole、Antoine Miech、Richard Tanburn、Mary Phuong、Filip Pavetic、Sebastien Cevey、Ramona Comanescu、Richard Ives、Sherry Yang、Cosmo Du、Bo Li、Zizhao Zhang、Mariko Iinuma、Clara Huiyi Hu、Aurko Roy、Shaan Bijwadia、Zhenkai Zhu、Danilo Martins、Rachel Saputro、Anita Gergely、Steven Zheng、Dawei Jia、Ioannis Antonoglou、Adam Sadovsky、Shane Gu、Yingying Bi、Alek Andreev、Sina Samangooei、Mina Khan、Tomas Kocisky、Angelos Filos、Chintu Kumar、Colton Bishop、Adams Yu、Sarah Hodkinson、Sid Mittal、Premal Shah、Alexan- dre Moufarek、Yong Cheng、Adam Bloniarz、Jaehoon Lee、Pedram Pejman、Paul Michel、Stephen Spencer、Vladimir Feinberg、Xuehan Xiong、Nikolay Savinov、Charlotte Smith、Siamak Shakeri、Dustin Tran、Mary Chesus、Bernd Bohnet、George Tucker、Tamara von Glehn、Carrie Muir、Yiran Mao、Hideto Kazawa、Ambrose Slone、Kedar Soparkar、Disha Shrivas- tava、James Cobon-Kerr、Michael Sharman、Jay Pavagadhi、Carlos Araya、Karolis Misiunas、Nimesh Ghelani、Michael Laskin、David Barker、Qiujia Li、Anton Briukhov、Neil Houlsby、Mia Glaese、Balaji Lakshminarayanan、Nathan Schucher、Yunhao Tang、Eli Collins、Hyeon- taek Lim、Fangxiaoyu Feng、Adria Recasens、Guangda Lai、Alberto Magni、Nicola De Cao、Aditya Siddhant、Zoe Ashwood、Jordi Orbay、Mostafa Dehghani、Jenny Brennan、Yifan He、Kelvin Xu、Yang Gao、Carl Saroufim、James Molloy、Xinyi Wu、Seb Arnold、Solomon Chang、Julian Schrittwieser、Elena Buchatskaya、Soroush Radpour、Martin Polacek、Skye Giordano、Ankur Bapna、Simon Tokumine、Vincent Hellendoorn、Thibault Sottiaux、Sarah Cogan、Ali- aksei Severyn、Mohammad Saleh、Shantanu Thakoor、Laurent Shefey、Siyuan Qiao、Meenu Gaba、Shuo yiin Chang、Craig Swanson、Biao Zhang、Benjamin Lee、Paul Kishan Rubenstein、Gan Song、Tom Kwiatkowski、Anna Koop、Ajay Kannan、David Kao、Parker Schuh、Axel Stjerngren、Golnaz Ghiasi、Gena Gibson、Luke Vilnis、Ye Yuan、Felipe Tiengo Ferreira、Aish- warya Kamath、Ted Klimenko、Ken Franko、Kefan Xiao、Indro Bhattacharya、Miteyan Patel、Rui Wang、Alex Morris、Robin Strudel、Vivek Sharma、Peter Choy、Sayed Hadi Hashemi、Jes- sica Landon、Mara Finkelstein、Priya Jhakra、Justin Frye、Megan Barnes、Matthew Mauger、Dennis Daun、Khuslen Baatarsukh、Matthew Tung、Wael Farhan、Henryk Michalewski、Fabio Viola、Felix de Chaumont Quitry、Charline Le Lan、Tom Hudson、Qingze Wang、Felix Fis- cher、Ivy Zheng、Elspeth White、Anca Dragan、Jean baptiste Alayrac、Eric Ni、Alexander Pritzel、Adam Iwanicki、Michael Isard、Anna Bulanova、Lukas Zilka、Ethan Dyer、Devendra Sachan、Srivatsan Srinivasan、Hannah Muckenhirn、Honglong Cai、Amol Mandhane、Mukar- ramTariq、JackW. Rae、Gary Wang、KareemAyoub、NicholasFitzGerald、YaoZhao、Woohyun Han、Chris Alberti、Dan Garrette、Kashyap Krishnakumar、Mai Gimenez、Anselm Levskaya、Daniel Sohn、Josip Matak、Inaki Iturrate、Michael B. Chang、Jackie Xiang、Yuan Cao、Nishant Ranka、Geoff Brown、Adrian Hutter、Vahab Mirrokni、Nanxin Chen、Kaisheng Yao、Zoltan Egyed、Francois Galilee、Tyler Liechty、Praveen Kallakuri、Evan Palmer、Sanjay Ghemawat、Jasmine Liu、David Tao、Chloe Thornton、Tim Green、Mimi Jasarevic、Sharon Lin、Victor Cotruta、Yi-Xuan Tan、Noah Fiedel、Hongkun Yu、Ed Chi、Alexander Neitz、Jens Heitkaem- per、Anu Sinha、Denny Zhou、Yi Sun、Charbel Kaed、Brice Hulse、Swaroop Mishra、Maria Georgaki、Sneha Kudugunta、Clement Farabet、Izhak Shafran、Daniel Vlasic、Anton Tsitsulin、Rajagopal Ananthanarayanan、Alen Carin、Guolong Su、Pei Sun、Shashank V、Gabriel Car- vajal、Josef Broder、Iulia Comsa、Alena Repina、William Wong、Warren Weilun Chen、Peter Hawkins、Egor Filonov、Lucia Loher、Christoph Hirnschall、Weiyi Wang、Jingchen Ye、An- drea Burns、Hardie Cate、Diana Gage Wright、Federico Piccinini、Lei Zhang、Chu-Cheng Lin、Ionel Gog、Yana Kulizhskaya、Ashwin Sreevatsa、Shuang Song、Luis C. Cobo、Anand Iyer、Chetan Tekur、Guillermo Garrido、Zhuyun Xiao、Rupert Kemp、Huaixiu Steven Zheng、Hui Li、Ananth Agarwal、Christel Ngani、Kati Goshvadi、Rebeca Santamaria-Fernandez、Wojciech Fica、Xinyun Chen、Chris Gorgolewski、Sean Sun、Roopal Garg、Xinyu Ye、S. M. Ali Eslami、Nan Hua、Jon Simon、Pratik Joshi、Yelin Kim、Ian Tenney、Sahitya Potluri、Lam Nguyen Thiet、Quan Yuan、Florian Luisier、Alexandra Chronopoulou、Salvatore Scellato、Praveen Srinivasan、Minmin Chen、Vinod Koverkathu、Valentin Dalibard、Yaming Xu、Brennan Saeta、Keith An- derson、Thibault Sellam、Nick Fernando、Fantine Huot、Junehyuk Jung、Mani Varadarajan、Michael Quinn、Amit Raul、Maigo Le、Ruslan Habalov、Jon Clark、Komal Jalan、Kalesha Bullard、Achintya Singhal、Thang Luong、Boyu Wang、Sujeevan Rajayogam、J ulian Eisensch- los、Johnson Jia、Daniel Finchelstein、Alex Yakubovich、Daniel Balle、Michael Fink、Sameer Agarwal、Jing Li、Dj Dvijotham、Shalini Pal、Kai Kang、Jaclyn Konzelmann、Jennifer Beat-

tie, Olivier Dousse, Diane Wu, Remi Crocker, Chen Elkind, Siddhartha Reddy Jonnalagadda,

Jong Lee, Dan Holtmann-Rice, Krystal Kallarackal, Rosanne Liu, Denis Vnukov, Neera Vats, Luca Invernizzi, Mohsen Jafari, Huanjie Zhou, Lilly Taylor, Jennifer Prendki, Marcus Wu, Tom Eccles, Tianqi Liu, Kavya Kopparapu, Francoise Beaufays, Christof Angermueller, Andreea Marzoca, Shourya Sarcar, Hilal Dib, Jeff Stanway, Frank Perbet, Nejc Trdin, Rachel Sterneck, Andrey Khorlin, Dinghua Li, Xihui Wu, Sonam Goenka, David Madras, Sasha Goldshtein, Willi Gierke, Tong Zhou, Yaxin Liu, Yannie Liang, Anais White, Yunjie Li, Shreya Singh, Sanaz Bahargam, Mark Epstein, Sujoy Basu, Li Lao, Adnan Ozturel, Carl Crous, Alex Zhai, Han Lu, Zora Tung, Neeraj Gaur, Alanna Walton, Lucas Dixon, Ming Zhang, Amir Globerson, Grant Uy, Andrew Bolt, Olivia Wiles, Milad Nasr, Ilia Shumailov, Marco Selvi, Francesco Piccinno, Ricardo Aguilar, Sara McCarthy, Misha Khalman, Mrinal Shukla, Vlado Galic, John Carpenter, Kevin Villela, Haibin Zhang, Harry Richardson, James Martens, Matko Bosnjak, Shreyas Rammohan Belle, Jeff Seibert, Mahmoud Alnahlawi, Brian McWilliams, Sankalp Singh, Annie Louis, Wen Ding, Dan Popovici, Lenin Simicich, Laura Knight, Pulkit Mehta, Nishesh Gupta, Chongyang Shi, Saaber Fatehi, Jovana Mitrovic, Alex Grills, Joseph Pagadora, Tsendsuren Munkhdalai, Dessie Petrova, Danielle Eisenbud, Zhishuai Zhang, Damion Yates, Bhavishya Mittal, Nilesh Tripuraneni, Yannis Assael, Thomas Brovelli, Prateek Jain, Mihajlo Velimirovic, Canfer Akbulut, Jiaqi Mu, Wolfgang Macherey, Ravin Kumar, Jun Xu, Haroon Qureshi, Gheorghe Comanici, Jeremy Wiesner, Zhitao Gong, Anton Ruddock, Matthias Bauer, Nick Felt, Anirudh GP, Anurag Arnab, Dustin Zelle, Jonas Rothfuss, Bill Rosgen, Ashish Shenoy, Bryan Seybold, Xinjian Li, Jayaram Mudigonda, Goker Erdogan, Jiawei Xia, Jiri Simsa, Andrea Michi, Yi Yao, Christopher Yew, Steven Kan, Isaac Caswell, Carey Radebaugh, Andre Elisseeff, Pedro Valenzuela, Kay McKinney, Kim Paterson, Albert Cui, Eri Latorre-Chimoto, Solomon Kim, William Zeng, Ken Durden, Priya Ponnapalli, Tiberiu Sosea, Christopher A. Choquette-Choo, James Manyika, Brona Robenek, Harsha Vashisht, Sebastien Pereira, Hoi Lam, Marko Velic, Denese Owusu-Afriyie, Katherine Lee, Tolga Bolukbasi, Alicia Parrish, Shawn Lu, Jane Park, Balaji Venkatraman, Alice Talbert, Lambert Rosique, Yuchung Cheng, Andrei Sozanschi, Adam Paszke, Praveen Kumar, Jessica Austin, Lu Li, Khalid Salama, Bartek Perz, Wooyeol Kim, Nandita Dukkipati, Anthony Baryshnikov, Christos Kaplanis, XiangHai Sheng, Yuri Chervonyi, Caglar Unlu, Diego de Las Casas, Harry Askham, Kathryn Tunyasuvunakool, Felix Gimeno, Siim Poder, Chester Kwak, Matt Miecnikowski, Vahab Mirrokni, Alek Dimitriev, Aaron Parisi, Dangyi Liu, Tomy Tsai, Toby Shevlane, Christina Kouridi, Drew Garmon, Adrian Goedeckemeyer, Adam R. Brown, Anitha Vijayakumar, Ali Elqursh, Sadegh Jazayeri, Jin Huang, Sara Mc Carthy, Jay Hoover, Lucy Kim, Sandeep Kumar, Wei Chen, Courtney Biles, Garrett Bingham, Evan Rosen, Lisa Wang, Qijun Tan, David Engel, Francesco Pongetti, Dario de Cesare, Dongseong Hwang, Lily Yu, Jennifer Pullman, Srini Narayanan, Kyle Levin, Siddharth Gopal, Megan Li, Asaf Aharoni, Trieu Trinh, Jessica Lo, Norman Casagrande, Roopali Vij, Loic Matthey, Bramandia Ramadhana, Austin Matthews, CJ Carey, Matthew Johnson, Kremena Goranova, Rohin Shah, Shereen Ashraf, Kingshuk Dasgupta, Rasmus Larsen, Yicheng Wang, Manish Reddy Vuyyuru, Chong Jiang, Joana Ijazi, Kazuki Osawa, Celine Smith, Ramya Sree Boppana, Taylan Bilal, Yuma Koizumi, Ying Xu, Yasemin Altun, Nir Shabat, Ben Bariach, Alex Korchemniy, Kiam Choo, Olaf Ronneberger, Chimezie Iwuanyanwu, Shubin Zhao, David Soergel, Cho-Jui Hsieh, Irene Cai, Shariq Iqbal, Martin Sundermeyer, Zhe Chen, Elie Bursztein, Chaitanya Malaviya, Fadi Biadsy, Prakash Shroff, Inderjit Dhillon, Tejasi Latkar, Chris Dyer, Hannah Forbes, Massimo Nicosia, Vitaly Nikolaev, Somer Greene, Marin Georgiev, Pidong Wang, Nina Martin, Hanie Sedghi, John Zhang, Praseem Banzal, Doug Fritz, Vikram Rao, Xuezhi Wang, Jiageng Zhang, Viorica Patraucean, Dayou Du, Igor Mordatch, Ivan Jurin, Lewis Liu, Ayush Dubey, Abhi Mohan, Janek Nowakowski, Vlad-Doru Ion, Nan Wei, Reiko Tojo, Maria Abi Raad, Drew A. Hudson, Vaishakh Keshava, Shubham Agrawal, Kevin Ramirez, Zhichun Wu, Hoang Nguyen, Ji Liu, Madhavi Sewak, Bryce Petrini, DongHyun Choi, Ivan Philips, Ziyue Wang, Ioana Bica, Ankush Garg, Jarek Wilkiewicz, Priyanka Agrawal, Xiaowei Li, Danhao Guo, Emily Xue, Naseer Shaik, Andrew Leach, Sadh MNM Khan, Julia Wiesinger, Sammy Jerome, Abhishek Chakladar, Alek Wenjiao Wang, Tina Ornduff, Folake Abu, Alireza Ghaffarkhah, Marcus Wainwright, Mario Cortes, Frederick Liu, Joshua Maynez, Andreas Terzis, Pouya Samangouei, Riham Mansour, Tomasz Kępa, François-Xavier Aubet, Anton Algymr, Dan Banica, Agoston Weisz, Andras Orban, Alexandre Senges, Ewa Andrejczuk, Mark Geller, Niccolo Dal Santo, Valentin Anklin, Majd Al Merey, Martin Baeuml, Trevor Strohman, Junwen Bai, Slav Petrov, Yonghui Wu, Demis Hassabis, Koray Kavukcuoglu, Jeff Dean, and Oriol Vinyals. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of con-



Jong Lee、Dan Holtmann-Rice、Krystal Kallarackal、Rosanne Liu、Denis Vnukov、Neera Vats、Luca Invernizzi、Mohsen Jafari、Huanjie Zhou、Lilly Taylor、Jennifer Prendki、Marcus Wu、Tom Eccles、Tianqi Liu、Kavya Kopparapu、Francoise Beaufays、Christof Angermueller、An- dreea Marzoca、Shourya Sarcar、Hilal Dib、Jeff Stanway、Frank Perbet、Nejc Trdin、Rachel Sterneck、Andrey Khorlin、Dinghua Li、Xihui Wu、Sonam Goenka、David Madras、Sasha Goldshtein、Willi Gierke、Tong Zhou、Yaxin Liu、Yannie Liang、Anais White、Yunjie Li、Shreya Singh、Sanaz Bahargam、Mark Epstein、Sujoy Basu、Li Lao、Adnan Ozturel、Carl Crous、Alex Zhai、Han Lu、Zora Tung、Neeraj Gaur、Alanna Walton、Lucas Dixon、Ming Zhang、Amir Globerson、Grant Uy、Andrew Bolt、Olivia Wiles、Milad Nasr、Ilia Shumailov、Marco Selvi、Francesco Piccinno、Ricardo Aguilar、Sara McCarthy、Misha Khalman、Mrinal Shukla、Vlado Galic、John Carpenter、Kevin Villela、Haibin Zhang、Harry Richardson、James Martens、Matko Bosnjak、Shreyas Rammohan Belle、Jeff Seibert、Mahmoud Alnahlawi、Brian McWilliams、Sankalp Singh、Annie Louis、Wen Ding、Dan Popovici、Lenin Simicich、Laura Knight、Pulkit Mehta、Nishesh Gupta、Chongyang Shi、Saaber Fatehi、Jovana Mitrovic、Alex Grills、Joseph Pagadora、Tsendsuren Munkhdalai、Dessie Petrova、Danielle Eisenbud、Zhishuai Zhang、Damion Yates、Bhavishya Mittal、Nilesh Tripuraneni、Yannis Assael、Thomas Brovelli、Prateek Jain、Mihajlo Velimirovic、Canfer Akbulut、Jiaqi Mu、Wolfgang Macherey、Ravin Ku- mar、Jun Xu、Haroon Qureshi、Gheorghe Comanici、Jeremy Wiesner、Zhitao Gong、Anton Ruddock、Matthias Bauer、Nick Felt、Anirudh GP、Anurag Arnab、Dustin Zelle、Jonas Roth- fuss、Bill Rosgen、Ashish Shenoy、Bryan Seybold、Xinjian Li、Jayaram Mudigonda、Goker Erdogan、Jiawei Xia、Jiri Simsa、Andrea Michi、Yi Yao、Christopher Yew、Steven Kan、Isaac Caswell、Carey Radebaugh、Andre Elisseeff、Pedro Valenzuela、Kay McKinney、Kim Paterson、Albert Cui、Eri Latorre-Chimoto、Solomon Kim、William Zeng、Ken Durden、Priya Ponna- palli、Tiberiu Sosea、Christopher A. Choquette-Choo、James Manyika、Brona Robenek、Har- sha Vashisht、Sebastien Pereira、Hoi Lam、Marko Velic、Denese Owusu-Afriyie、Katherine Lee、Tolga Bolukbasi、Alicia Parrish、Shawn L u、Jane Park、Balaji Venkatraman、Alice Tal- bert、Lambert Rosique、Yuchung Cheng、Andrei Sozanschi、Adam Paszke、Praveen Kumar、Jessica Austin、Lu Li、Khalid Salama、Bartek Perz、Wooyeol Kim、Nandita Dukkipati、Anthony Baryshnikov、Christos Kaplanis、XiangHai Sheng、Yuri Chervonyi、Caglar Unlu、Diego de Las Casas、Harry Askham、Kathryn Tunyasuvunakool、Felix Gimeno、Siim Poder、Chester Kwak、Matt Miecnikowski、Vahab Mirrokni、Alek Dimitriev、Aaron Parisi、Dangyi Liu、Tomy Tsai、Toby Shevlane、Christina Kouridi、Drew Garmon、Adrian Goedeckemeyer、Adam R. Brown、Anitha Vijayakumar、Ali Elqursh、Sadegh Jazayeri、Jin Huang、Sara Mc Carthy、Jay Hoover、Lucy Kim、Sandeep Kumar、Wei Chen、Courtney Biles、Garrett Bingham、Evan Rosen、Lisa Wang、Qijun Tan、David Engel、Francesco Pongetti、Dario de Cesare、Dongseong Hwang、Lily Yu、Jennifer Pullman、Srini Narayanan、Kyle Levin、Siddharth Gopal、Megan Li、Asaf Aharoni、Trieu Trinh、Jessica Lo、Norman Casagrande、Roopali Vij、Loic Matthey、Bramandia Ramadhana、Austin Matthews、CJ Carey、Matthew Johnson、Kremena Goranova、Rohin Shah、Shereen Ashraf、Kingshuk Dasgupta、Rasmus Larsen、Yicheng Wang、Manish Reddy Vuyyuru、Chong Jiang、Joana Ijazi、Kazuki Osawa、Celine Smith、Ramya Sree Boppana、Taylan Bilal、Yuma Koizumi、Ying Xu、Yasemin Altun、Nir Shabat、Ben Bariach、Alex Korchemniy、Kiam Choo、Olaf Ronneberger、Chimezie Iwuanyanwu、Shubin Zhao、David Soergel、Cho-Jui Hsieh、Irene Cai、Shariq Iqbal、Martin Sundermeyer、Zhe Chen、Elie Bursztein、Chaitanya Malaviya、Fadi Biadsy、Prakash Shroff、Inderjit Dhillon、Tejasi Latkar、Chris Dyer、Hannah Forbes、Mas- simo Nicosia、Vitaly Nikolaev、Somer Greene、Marin Georgiev、Pidong Wang、Nina Martin、Hanie Sedghi、John Zhang、Praseem Banzal、Doug Fritz、Vikram Rao、Xuezhi Wang、Jia- geng Zhang、Viorica Patraucean、Dayou Du、Igor Mordatch、Ivan Jurin、Lewis Liu、Ayush Dubey、Abhi Mohan、Janek Nowakowski、Vlad-Doru Ion、Nan Wei、Reiko Tojo、Maria Abi Raad、Drew A. Hudson、Vaishakh Keshava、Shubham Agrawal、Kevin Ramirez、Zhichun Wu、Hoang Nguyen、Ji Liu、Madhavi Sewak、Bryce Petrini、DongHyun Choi、Ivan Philips、Ziyue Wang、Ioana Bica、Ankush Garg、Jarek Wilkiewicz、Priyanka Agrawal、Xiaowei Li、Danhao Guo、Emily Xue、Naseer Shaik、Andrew Leach、Sadh MNM Khan、Julia Wiesinger、Sammy Jerome、Abhishek Chakladar、Alek Wenjiao Wang、Tina Ornduff、Folake Abu、Alireza Ghaf- farkhah、Marcus Wainwright、Mario Cortes、Frederick Liu、Joshua Maynez、Andreas Terzis、Pouya Samangouei、Riham Mansour、Tomasz K.epa、François-Xavier Aubet、Anton Algymr、Dan Banica、Agoston Weisz、Andras Orban、Alexandre Senges、Ewa Andrejczuk、Mark Geller、Niccolo Dal Santo、Valentin Anklin、Majd Al Merey、Martin Baeuml、Trevor Strohman、Jun- wen Bai、Slav Petrov、Yonghui Wu、Demis Hassabis、Koray Kavukcuoglu、Jeff Dean、and Oriol Vinyals。Gemini 1.5：解鎖跨越百萬 token 的多模態理解，

text, 2024. URL https://arxiv.org/abs/2403.05530.

[42] OpenAI, :, Aaron Hurst, Adam Lerer, Adam P. Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, Aleksander Mądry, Alex Baker-Whitcomb, Alex Beutel, Alex Borzunov, Alex Carney, Alex Chow, Alex Kirillov, Alex Nichol, Alex Paino, Alex Renzin, Alex Tachard Passos, Alexander Kirillov, Alexi Christakis, Alexis Conneau, Ali Kamali, Allan Jabri, Allison Moyer, Allison Tam, Amadou Crookes, Amin Tootoochian, Amin Tootoonchian, Ananya Kumar, Andrea Vallone, Andrej Karpathy, Andrew Braunstein, Andrew Cann, Andrew Codispoti, Andrew Galu, Andrew Kondrich, Andrew Tulloch, Andrey Mishchenko, Angela Baek, Angela Jiang, Antoine Pelisse, Antonia Woodford, Anuj Gosalia, Arka Dhar, Ashley Pantuliano, Avi Nayak, Avital Oliver, Barret Zoph, Behrooz Ghorbani, Ben Leimberger, Ben Rossen, Ben Sokolowsky, Ben Wang, Benjamin Zweig, Beth Hoover, Blake Samic, Bob McGrew, Bobby Spero, Bogo Giertler, Bowen Cheng, Brad Lightcap, Brandon Walkin, Brendan Quinn, Brian Guarraci, Brian Hsu, Bright Kellogg, Brydon Eastman, Camillo Lugaresi, Carroll Wainwright, Cary Bassin, Cary Hudson, Casey Chu, Chad Nelson, Chak Li, Chan Jun Shern, Channing Conger, Charlotte Barette, Chelsea Voss, Chen Ding, Cheng Lu, Chong Zhang, Chris Beaumont, Chris Hallacy, Chris Koch, Christian Gibson, Christina Kim, Christine Choi, Christine McLeavey, Christopher Hesse, Claudia Fischer, Clemens Winter, Coley Czarnecki, Colin Jarvis, Colin Wei, Constantin Koumouzelis, Dane Sherburn, Daniel Kappler, Daniel Levin, Daniel Levy, David Carr, David Farhi, David Mely, David Robinson, David Sasaki, Denny Jin, Dev Valladares, Dimitris Tsipras, Doug Li, Duc Phong Nguyen, Duncan Findlay, Edede Oiwoh, Edmund Wong, Ehsan Asdar, Elizabeth Proehl, Elizabeth Yang, Eric Antonow, Eric Kramer, Eric Peterson, Eric Sigler, Eric Wallace, Eugene Brevdo, Evan Mays, Farzad Khorasani, Felipe Petroski Such, Filippo Raso, Francis Zhang, Fred von Lohmann, Freddie Sulit, Gabriel Goh, Gene Oden, Geoff Salmon, Giulio Starace, Greg Brockman, Hadi Salman, Haiming Bao, Haitang Hu, Hannah Wong, Haoyu Wang, Heather Schmidt, Heather Whitney, Heewoo Jun, Hendrik Kirchner, Henrique Ponde de Oliveira Pinto, Hongyu Ren, Huiwen Chang, Hyung Won Chung, Ian Kivlichan, Ian O'Connell, Ian O'Connell, Ian Osband, Ian Silber, Ian Sohl, Ibrahim Okuyucu, Ikai Lan, Ilya Kostrikov, Ilya Sutskever, Ingmar Kanitscheider, Ishaan Gulrajani, Jacob Coxon, Jacob Menick, Jakub Pachocki, James Aung, James Betker, James Crooks, James Lennon, Jamie Kiros, Jan Leike, Jane Park, Jason Kwon, Jason Phang, Jason Teplitz, Jason Wei, Jason Wolfe, Jay Chen, Jeff Harris, Jenia Varavva, Jessica Gan Lee, Jessica Shieh, Ji Lin, Jiahui Yu, Jiayi Weng, Jie Tang, Jieqi Yu, Joanne Jang, Joaquin Quinonero Candela, Joe Beutler, Joe Landers, Joel Parish, Johannes Heidecke, John Schulman, Jonathan Lachman, Jonathan McKay, Jonathan Uesato, Jonathan Ward, Jong Wook Kim, Joost Huizinga, Jordan Sitkin, Jos Kraaijeveld, Josh Gross, Josh Kaplan, Josh Snyder, Joshua Achiam, Joy Jiao, Joyce Lee, Juntang Zhuang, Justyn Harriman, Kai Fricke, Kai Hayashi, Karan Singhal, Katy Shi, Kavin Karthik, Kayla Wood, Kendra Rimbach, Kenny Hsu, Kenny Nguyen, Keren Gu-Lemberg, Kevin Button, Kevin Liu, Kiel Howe, Krithika Muthukumar, Kyle Luther, Lama Ahmad, Larry Kai, Lauren Itow, Lauren Workman, Leher Pathak, Leo Chen, Li Jing, Lia Guy, Liam Fedus, Liang Zhou, Lien Mamitsuka, Lilian Weng, Lindsay McCallum, Lindsey Held, Long Ouyang, Louis Feuvrier, Lu Zhang, Lukas Kondraciuk, Lukasz Kaiser, Luke Hewitt, Luke Metz, Lyric Doshi, Mada Aflak, Maddie Simens, Madelaine Boyd, Madeleine Thompson, Marat Dukhan, Mark Chen, Mark Gray, Mark Hudnall, Marvin Zhang, Marwan Aljubeh, Mateusz Litwin, Matthew Zeng, Max Johnson, Maya Shetty, Mayank Gupta, Meghan Shah, Mehmet Yatbaz, Meng Jia Yang, Mengchao Zhong, Mia Glaese, Mianna Chen, Michael Janner, Michael Lampe, Michael Petrov, Michael Wu, Michele Wang, Michelle Fradin, Michelle Pokrass, Miguel Castro, Miguel Oom Temudo de Castro, Mikhail Pavlov, Miles Brundage, Miles Wang, Minal Khan, Mira Murati, Mo Bavarian, Molly Lin, Murat Yesildal, Nacho Soto, Natalia Gimelshein, Natalie Cone, Natalie Staudacher, Natalie Summers, Natan LaFontaine, Neil Chowdhury, Nick Ryder, Nick Stathas, Nick Turley, Nik Tezak, Niko Felix, Nithanth Kudige, Nitish Keskar, Noah Deutsch, Noel Bundick, Nora Puckett, Ofir Nachum, Ola Okelola, Oleg Boiko, Oleg Murk, Oliver Jaffe, Olivia Watkins, Olivier Godement, Owen Campbell-Moore, Patrick Chao, Paul McMillan, Pavel Belov, Peng Su, Peter Bak, Peter Bakkum, Peter Deng, Peter Dolan, Peter Hoeschele, Peter Welinder, Phil Tillet, Philip Pronin, Philippe Tillet, Prafulla Dhariwal, Qiming Yuan, Rachel Dias, Rachel Lim, Rahul Arora, Rajan Troll, Randall Lin, Rapha Gontijo Lopes, Raul Puri, Reah Miyara, Reimar Leike, Renaud Gaubert, Reza Zamani, Ricky Wang, Rob Donnelly, Rob Honsby, Rocky Smith, Rohan Sahai, Rohit Ramchandani, Romain Huet,



文字，2024年。URL https://arxiv.org/abs/2403.05530。

[42] OpenAI，：Aaron Hurst，Adam Lerer，Adam P. Goucher，Adam Perelman，Aditya Ramesh，Aidan Clark，AJ Ostrow，Akila Welihinda，Alan Hayes，Alec Radford，Aleksander M.adry，Alex Baker-Whitcomb，Alex Beutel，Alex Borzunov，Alex Carney，Alex Chow，Alex Kir- illov，Alex Nichol，Alex Paino，Alex Renzin，Alex Tachard Passos，AlexanderKirillov，Alexi Christakis，Alexis Conneau，Ali Kamali，Allan Jabri，Allison Moyer，Allison Tam，Amadou Crookes，Amin Tootoochian，Amin Tootoonchian，Ananya Kumar，Andrea Vallone，Andrej Karpathy，Andrew Braunstein，Andrew Cann，Andrew Codispoti，Andrew Galu，Andrew Kon- drich，Andrew Tulloch，Andrey Mishchenko，Angela Baek，Angela Jiang，Antoine Pelisse，Antonia Woodford，Anuj Gosalia，Arka Dhar，Ashley Pantuliano，Avi Nayak，Avital Oliver，Barret Zoph，Behrooz Ghorbani，Ben Leimberger，Ben Rossen，Ben Sokolowsky，Ben Wang，Benjamin Zweig，Beth Hoover，Blake Samic，Bob McGrew，Bobby Spero，Bogo Giertler，Bowen Cheng，Brad Lightcap，Brandon Walkin，Brendan Quinn，Brian Guarraci，Brian Hsu，Bright Kellogg，Brydon Eastman，Camillo Lugaresi，Carroll Wainwright，Cary Bassin，Cary Hudson，Casey Chu，Chad Nelson，Chak Li，Chan Jun Shern，Channing Conger，Charlotte Barette，Chelsea Voss，Chen Ding，Cheng Lu，Chong Zhang，Chris Beaumont，Chris Hallacy，Chris Koch，Christian Gibson，Christina Kim，Christine Choi，Christine McLeavey，Christopher Hesse，Claudia Fischer，Clemens Winter，Coley Czarnecki，Colin Jarvis，Colin Wei，Con- stantin Koumouzelis，Dane Sherburn，Daniel Kappler，Daniel Levin，Daniel Levy，David Carr，David Farhi，David Mely，David Robinson，David Sasaki，Denny Jin，Dev Valladares，Dim- itris Tsipras，Doug Li，Duc Phong Nguyen，Duncan Findlay，Edede Oiwoh，Edmund Wong，Ehsan Asdar，Elizabeth Proehl，Elizabeth Yang，Eric Antonow，Eric Kramer，Eric Peterson，Eric Sigler，Eric Wallace，Eugene Brevdo，Evan Mays，Farzad Khorasani，Felipe Petroski Such，Filippo Raso，Francis Zhang，Fred von Lohmann，Freddie Sulit，Gabriel Goh，Gene Oden，Ge- off Salmon，Giulio Starace，Greg Brockman，Hadi Salman，Haiming Bao，Haitang Hu，Hannah Wong，Haoyu Wang，Heather Schmidt，Heather Whitney，Heewoo Jun，Hendrik Kirchner，Henrique Ponde de Oliveira Pinto，Hongyu Ren，Huiwen Chang，Hyung Won Chung，Ian Kivlichan，Ian O' Connell，Ian O' Connell，Ian Osband，Ian Silber，Ian Sohl，Ibrahim Okuyucu，Ikai Lan，Ilya Kostrikov，Ilya Sutskever，Ingmar Kanitscheider，Ishaan Gulrajani，Jacob Coxon，Jacob Menick，Jakub Pachocki，James Aung，James Betker，James Crooks，James Lennon，Jamie Kiros，Jan Leike，Jane Park，Jason Kwon，Jason Phang，Jason Teplitz，Jason Wei，Jason Wolfe，Jay Chen，Jeff Harris，Jenia Varavva，Jessica Gan Lee，Jessica Shieh，Ji Lin，Jiahui Yu，Jiayi Weng，Jie Tang，Jieqi Yu，Joanne Jang，Joaquin Quinonero Candela，Joe Beutler，Joe Landers，Joel Parish，Johannes Heidecke，John Schulman，Jonathan Lachman，Jonathan McKay，Jonathan Uesato，Jonathan Ward，Jong Wook Kim，Joost Huizinga，Jordan Sitkin，Jos Kraaijeveld，Josh Gross，Josh Kaplan，Josh Snyder，Joshua Achiam，Joy Jiao，Joyce Lee，Juntang Zhuang，Justyn Harriman，Kai Fricke，Kai Hayashi，Karan Singhal，Katy Shi，Kavin Karthik，Kayla Wood，Kendra Rimbach，Kenny Hsu，Kenny Nguyen，Keren Gu-Le mberg，Kevin Button，Kevin Liu，Kiel Howe，Krithika Muthukumar，Kyle Luther，Lama Ahmad，Larry Kai，Lauren Itow，Lauren Workman，Leher Pathak，Leo Chen，Li Jing，Lia Guy，Liam Fedus，Liang Zhou，Lien Mamitsuka，Lilian Weng，Lindsay McCallum，Lindsey Held，Long Ouyang，Louis Feuvrier，Lu Zhang，Lukas Kondraciuk，Lukasz Kaiser，Luke Hewitt，Luke Metz，Lyric Doshi，Mada Aflak，Maddie Simens，Madelaine Boyd，Madeleine Thompson，Marat Dukhan，Mark Chen，Mark Gray，Mark Hudnall，Marvin Zhang，Marwan Aljubeh，Mateusz Litwin，Matthew Zeng，Max Johnson，Maya Shetty，Mayank Gupta，Meghan Shah，Mehmet Yatbaz，Meng JiaYang，Mengchao Zhong，Mia Glaese，Mianna Chen，Michael Janner，Michael Lampe，Michael Petrov，Michael Wu，Michele Wang，Michelle Fradin，Michelle Pokrass，Miguel Cas- tro，Miguel Oom Temudo de Castro，Mikhail Pavlov，Miles Brundage，Miles Wang，Minal Khan，Mira Murati，Mo Bavarian，Molly Lin，Murat Yesildal，Nacho Soto，Natalia Gimelshein，Natalie Cone，Natalie Staudacher，Natalie Summers，Natan LaFontaine，Neil Chowdhury，Nick Ryder，Nick Stathas，Nick Turley，Nik Tezak，Niko Felix，Nithanth Kudige，Nitish Keskar，Noah Deutsch，Noel Bundick，Nora Puckett，Ofir Nachum，Ola Okelola，Oleg Boiko，Oleg Murk，Oliver Jaffe，Olivia Watkins，Olivier Godement，Owen Campbell-Moore，Patrick Chao，Paul McMillan，Pavel Belov，Peng Su，Peter Bak，Peter Bakkum，Peter Deng，Peter Dolan，Peter Hoeschele，Peter Welinder，Phil Tillet，Philip Pronin，Philippe Tillet，Prafulla Dhariwal，Qiming Yuan，Rachel Dias，Rachel Lim，Rahul Arora，Rajan Troll， Randall Lin，Rapha Gontijo Lopes，Raul Puri，Reah Miyara，Reimar Leike，Renaud Gaubert，Reza Zamani，Ricky Wang Rob Donnelly, Rob Honsby, Rocky Smith, Rohan Sahai, Rohit Ramchandani, Romain Huet,

Rory Carmichael, Rowan Zellers, Roy Chen, Ruby Chen, Ruslan Nigmatullin, Ryan Cheu, Saachi Jain, Sam Altman, Sam Schoenholz, Sam Toizer, Samuel Miserendino, Sandhini Agarwal, Sara Culver, Scott Ethersmith, Scott Gray, Sean Grove, Sean Metzger, Shamez Hermani, Shantanu Jain, Shengjia Zhao, Sherwin Wu, Shino Jomoto, Shirong Wu, Shuaiqi, Xia, Sonia Phene, Spencer Papay, Srinivas Narayanan, Steve Coffey, Steve Lee, Stewart Hall, Suchir Balaji, Tal Broda, Tal Stramer, Tao Xu, Tarun Gogineni, Taya Christianson, Ted Sanders, Tejal Patwardhan, Thomas Cunninghman, Thomas Degry, Thomas Dimson, Thomas Raoux, Thomas Shadwell, Tianhao Zheng, Todd Underwood, Todor Markov, Toki Sherbakov, Tom Rubin, Tom Stasi, Tomer Kaftan, Tristan Heywood, Troy Peterson, Tyce Walters, Tyna Eloundou, Valerie Qi, Veit Moeller, Vinnie Monaco, Vishal Kuo, Vlad Fomenko, Wayne Chang, Weiyi Zheng, Wenda Zhou, Wesam Manassra, Will Sheu, Wojciech Zaremba, Yash Patil, Yilei Qian, Yongjik Kim, Youlong Cheng, Yu Zhang, Yuchen He, Yuchen Zhang, Yujia Jin, Yunxing Dai, and Yury Malkov. Gpt-4o system card, 2024. URL https://arxiv.org/abs/2410.21276.

[43] Rui Qian, Shuangrui Ding, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Dahua Lin, and Jiaqi Wang. Dispider: Enabling video llms with active real-time interaction via disentangled perception, decision, and reaction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 24045–24055. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02239. URL https://openaccess.thecvf.com/content/CVPR2025/html/Qian_Dispider_Enabling_Video_LLMs_with_Active_Real-Time_Interaction_via_Disentangled_CVPR_2025_paper.html.

[44] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, Peixian Chen, Yanwei Li, Shaohui Lin, Sirui Zhao, Ke Li, Tong Xu, Xiawu Zheng, Enhong Chen, Caifeng Shan, Ran He, and Xing Sun. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 24108–24118. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02245. URL https://openaccess.thecvf.com/content/CVPR2025/html/Fu_Video-MME_The_First-Ever_Comprehensive_Evaluation_Benchmark_of_Multi-modal_LLMs_in_CVPR_2025_paper.html.

[45] Weihan Wang, Zehai He, Wenyi Hong, Yean Cheng, Xiaohan Zhang, Ji Qi, Xiaotao Gu, Shiyu Huang, Bin Xu, Yuxiao Dong, Ming Ding, and Jie Tang. Lvbench: An extreme long video understanding benchmark, 2025. URL https://arxiv.org/abs/2406.08035.



Rory Carmichael, Rowan Zellers, Roy Chen, Ruby Chen, Ruslan Nigmatullin, Ryan Cheu, Saachi Jain, Sam Altman, Sam Schoenholz, Sam Toizer, Samuel Miserendino, Sandhini Agarwal, Sara Culver, Scott Ethersmith, Scott Gray, Sean Grove, Sean Metzger, Shamez Hermani, Shantanu Jain, Shengjia Zhao, Sherwin Wu, Shino Jomoto, Shirong Wu, Shuaiqi, Xia, Sonia Phene, Spencer Papay, Srinivas Narayanan, Steve Coffey, Steve Lee, Stewart Hall, Suchir Balaji, Tal Broda, Tal Stramer, Tao Xu, Tarun Gogineni, Taya Christianson, Ted Sanders, Tejal Patwardhan, Thomas Cunninghman, Thomas Degry, Thomas Dimson, Thomas Raoux, Thomas Shadwell, Tianhao Zheng, Todd Underwood, Todor Markov, Toki Sherbakov, Tom Rubin, Tom Stasi, Tomer Kaftan, Tristan Heywood, Troy Peterson, Tyce Walters, Tyna Eloundou, Valerie Qi, Veit Moeller, Vinnie Monaco, Vishal Kuo, Vlad Fomenko, Wayne Chang, Weiyi Zheng, Wenda Zhou, Wesam Manassra, Will Sheu, Wojciech Zaremba, Yash Patil, Yilei Qian, Yongjik Kim, Youlong Cheng, Yu Zhang, Yuchen He, Yuchen Zhang, Yujia Jin, Yunxing Dai, and Yury Malkov. Gpt-4o system card, 2024. URL https://arxiv.org/abs/2410.21276.

[43] Rui Qian, Shuangrui Ding, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Dahua Lin, and Jiaqi Wang. Dispider: Enabling video llms with active real-time interaction via disentangled perception, decision, and reaction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 24045–24055. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02239. URL https://openaccess.thecvf.com/content/CVPR2025/html/Qian_Dispider_Enabling_Video_LLMs_with_Active_Real-Time_Interaction_via_Disentangled_CVPR_2025_paper.html.

[44] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, Peixian Chen, Yanwei Li, Shaohui Lin, Sirui Zhao, Ke Li, Tong Xu, Xiawu Zheng, Enhong Chen, Caifeng Shan, Ran He, and Xing Sun. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*, pages 24108–24118. Computer Vision Foundation / IEEE, 2025. doi: 10.1109/CVPR52734.2025.02245. URL https://openaccess.thecvf.com/content/CVPR2025/html/Fu_Video-MME_The_First-Ever_Comprehensive_Evaluation_Benchmark_of_Multi-modal_LLMs_in_CVPR_2025_paper.html.

[45] Weihan Wang, Zehai He, Wenyi Hong, Yean Cheng, Xiaohan Zhang, Ji Qi, Xiaotao Gu, Shiyu Huang, Bin Xu, Yuxiao Dong, Ming Ding, and Jie Tang. Lvbench: An extreme long video understanding benchmark, 2025. URL https://arxiv.org/abs/2406.08035.

## A Future Work

Future work will explore adaptive segmentation that selects segment boundaries online based on scene changes and question demands, reducing redundant memory updates while preserving evidence coverage. We also plan to incorporate audio cues and speech transcripts to support richer streaming understanding in real-world settings. Another direction is improving robustness on very long streams through better uncertainty estimation, memory verification, and training with harder distractors and domain shifts. Finally, we will develop more comprehensive evaluation protocols that jointly measure accuracy, latency, and resource usage in multi-turn interaction.

## B Implementation Details

All experiments are conducted on 8× NVIDIA RTX A6000 GPUs (48GB each). We fine-tune Qwen3-VL-2/4/8B-Instruct using full-parameter supervised fine-tuning (SFT). Table A1 summarizes the main training settings.

Table A1: Implementation details and training hyperparameters.

| Item | Setting |
|---|---|
| GPUs | 8× NVIDIA RTX A6000 (48GB) |
| Model | Qwen3-VL-2/4/8B-Instruct |
| Training method | Full-parameter SFT |
| Precision | bf16 |
| Global batch size | 128 |
| Optimizer | AdamW |
| Peak learning rate | $1.0 \times 10^{-5}$ |
| Learning rate schedule | Cosine decay |
| Warmup ratio | 3% |
| Weight decay | 0.1 |
| Stage 1 strategy | DeepSpeed ZeRO-3 |
| Gradient checkpointing | Enabled |
| Communication overlap | Enabled |

## C Theoretical Latency Derivation

We derive a simple queueing-style model to quantify decoder-induced ingestion backlog in interleaved streaming, as illustrated in Fig. A1.

**Stream arrival and processing rates.** Assume that video segments arrive in real time at a rate of $\lambda$ segments per second. When the model is watching, meaning that it is ingesting and prefilling, it processes segments at a rate of $\mu$ segments per second. We define the utilization by $\rho \triangleq \lambda/\mu$.

**Interleaved decoding as server downtime.** In the interleaved baseline, generation is non-preemptive: during a decoding period of duration $T_{\text{dec}}$, the system does not ingest new segments. While decoding, the stream continues to arrive, creating a backlog of

$$B \;=\; \lambda T_{\text{dec}}. \tag{13}$$

$B$ represents backlog, which refers to the number of video segments that are accumulated and yet to be processed. After decoding, the system resumes watching at rate $\mu$ while new segments still arrive at rate $\lambda$. Thus the backlog drains at a net rate $(\mu - \lambda)$ (assuming $\mu > \lambda$), and the **catch-up time** is

$$T_{\text{catch}} \;=\; \frac{B}{\mu - \lambda} \;=\; \frac{\lambda}{\mu - \lambda} T_{\text{dec}} \;=\; \frac{\rho}{1 - \rho} T_{\text{dec}}. \tag{14}$$

This yields an amplification effect: each second spent decoding induces an additional $\rho/(1-\rho)$ second of future delay before the system fully catches up, which diverges as $\rho \to 1$.

**Think While Watching weakens backlog coupling.** Our inference design decouples ingestion from decoding via dual KV caching, so decoding no longer forces a full stop of stream ingestion as



## 未來工作

未來研究將探討適應式分割，根據場景變化與問題需求，在線選擇片段邊界，以減少冗餘的記憶體更新，同時保留證據涵蓋率。我們也計劃整合聲音提示與語音文字，以支持在實際場景中更豐富的串流理解。另一個方向是透過更好的不確定性估計、記憶體驗證，以及使用更難的干擾項與領域變化進行訓練，來提升對非常長串流的穩健性。最後，我們將開發更全面的評估協議，在多輪互動中同時衡量準確率、延遲與資源使用情況。

## B 實現細節

所有實驗均在 8× NVIDIA RTX A6000 GPU (每個 48GB) 上進行。我們使用全參數監督微調 (SFT) 來微調 Qwen3-VL-2/4/8B-Instruct。表 A1 摘要了主要的訓練設定。

表 A1：實現細節和訓練超參數。

| Item | 設定 |
|---|---|
| GPUs | 8× NVIDIA RTX A6000 (48GB) |
| 型號 | Qwen3-VL-2/4/8B-Instruct |
| 訓練方法 | 全參數SFT |
| 精確度 | bf16 |
| 全球批次大小 | 128 |
| 優化器 | AdamW |
| 峰值學習率 | $1.0 \times 10^{-5}$ |
| 學習率規劃 | 餘弦衰減 |
| 暖身比例 | 3% |
| 權重衰減 | 0.1 |
| 階段1策略 | DeepSpeed ZeRO-3 |
| 梯度檢查點 | 啟用中 |
| 溝通重疊 | 啟用中 |

## C 理論延遲推導

我們推導了一個簡單的隊列式模型，用於量化交錯串流中的解碼器引發的攝取積壓，如圖A1所示。

串流到達和處理速率。假設視頻片段以每秒$\lambda$ 個片段的速率實時到達。當模型在觀看時，也就是在攝取和預填充時，它以每秒 $\mu$ 個片段的速率處理片段。我們用 $\rho \triangleq \lambda/\mu$定義利用率。

交錯解碼作為伺服器停機。在交錯基線中，生成是非中斷的：在持續時間為 $T_{\text{dec}}$的解碼期間內，系統不攝取新片段。而解碼期間，串流繼續到達，產生積壓

$$B \;=\; \lambda T_{\text{dec}}. \tag{13}$$

$B$ 代表積壓，指的是已累積且尚未處理的影片片段數量。解碼後，系統以速率 $\mu$ 持續觀看，同時新的片段仍以速率 $\lambda$ 到達。因此積壓會以淨速率 $(\mu - \lambda)$ (假設 $\mu > \lambda$) 漸進減少，而追趕時間是

$$T_{\text{catch}} \;=\; \frac{B}{\mu - \lambda} \;=\; \frac{\lambda}{\mu - \lambda} T_{\text{dec}} \;=\; \frac{\rho}{1 - \rho} T_{\text{dec}}. \tag{14}$$

這產生了一種放大效果：每秒解碼時間會導致未來額外 $\rho/(1-\rho)$ 秒的延遲，直到系統完全趕上，這種現象隨 $\rho \to 1$ 而增長。

邏輯觀看時刻減輕了積壓耦合。我們的推論設計將輸入與解碼分離，因此解碼不再強制輸入流結束，viadual KV緩存，所以解碼不再強制輸入流結束為

in interleaved streaming. As a result, the decoder-induced backlog is greatly reduced. In an ideal fully overlapped implementation, the additional ingestion downtime during decoding approaches zero, yielding

$$B_{\text{ours}} \approx 0 \quad \Rightarrow \quad T_{\text{catch}} \approx 0. \tag{15}$$

In practice, however, a residual backlog may still arise from system overheads such as scheduling, synchronization, cache maintenance, and time overlapping. Therefore, the main benefit of our design is not to guarantee zero backlog but to substantially reduce the coupling between decoding and future stream lag.

**A quality real-time constraint in interleaving.** Let $c_{\text{tok}}$ be the average decoding time per output token and $L$ be the number of generated tokens per step. Interleaving spends $T_{\text{dec}} = c_{\text{tok}}L$ during which ingestion is paused, inducing $T_{\text{catch}} = \frac{\rho}{1-\rho}c_{\text{tok}}L$ by Eq. (14). Therefore, increasing $L$ to improve quality directly increases future stream lag. Our method weakens this coupling by allowing ingestion to proceed while decoding.

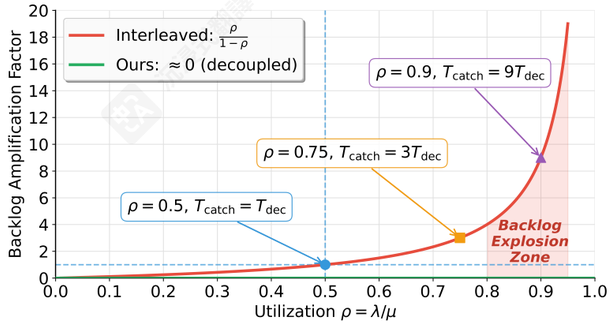



Figure A1: **Decoder-induced ingestion backlog under interleaved streaming.** As utilization $\rho$ increases, interleaved decoding pauses can amplify the catch-up delay and enter a backlog explosion regime, while our decoupled design substantially reduces decoder-induced backlog growth.

## D Additional Baseline Details

**Flash-VStream-7B[32].** Flash-VStream is an efficient video language model for long video streams. It introduces a flash memory module composed of a low-capacity context memory for aggregating long-range temporal information and a high-capacity augmentation memory for retrieving detailed spatial evidence, enabling real-time responses to user queries over extremely long videos.

**VideoLLM-online-8B[8].** VideoLLM-online proposes the LIVE framework, short for learning in video streams, to enable temporally aligned, long-context, and real-time conversation over continuous video streams. Specifically, LIVE introduces a streaming training objective, a data generation scheme that converts offline temporal annotations into streaming dialogue data, and an optimized inference pipeline with a continuous KV cache as well as parallelized visual encoding and language decoding for efficient online responses.

**Dispider-7B[43].** Dispider targets active real-time interaction by explicitly disentangling perception, decision, and reaction. It uses a lightweight proactive streaming video processing module to continuously monitor the stream and identify suitable moments for interaction, while an asynchronous interaction module generates detailed responses without blocking continued observation.

**StreamForest-7B[33].** StreamForest is designed for efficient online video understanding with persistent event memory. Its core Persistent Event Memory Forest organizes historical frames into event-level tree structures for long-term retention under limited computational resources, while a Fine-grained Spatiotemporal Window preserves detailed short-term perception.

**StreamAgent-7B[35].** StreamAgent studies anticipatory agents for streaming video understanding. Instead of reacting only to current observations, it integrates question semantics and historical observations to anticipate future task-relevant temporal intervals and spatial regions, and combines



交錯流式處理。因此，解碼引起的積壓大幅減少。在理想的完全疊加實現中，解碼期間的額外輸入停頓接近零，產生

$$B_{\text{ours}} \approx 0 \quad \Rightarrow \quad T_{\text{catch}} \approx 0. \tag{15}$$

然而，實際上，系統開銷（如排程、同步、快取維護和時間重疊）仍可能產生殘餘積壓。因此，我們設計的主要好處並非保證零積壓，而是顯著降低解碼與未來串流延遲之間的耦合。

交錯處理中的品質實時限制。令 $c_{\text{tok}}$ 為每個輸出標記的平均解碼時間， $L$ 為每個步驟產生的標記數量。交錯處理期間 $T_{\text{dec}} = c_{\text{tok}}L$ ，輸入處理會暫停，根據公式 (14) 產生 $T_{\text{catch}} = \frac{\rho}{1-\rho}c_{\text{tok}}L$ 。因此，增加 $L$ 以提升品質會直接增加未來串流延遲。我們的方法透過允許解碼期間繼續輸入處理來削弱這種耦合。

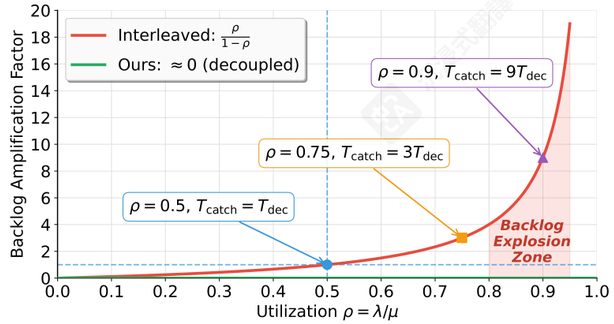



圖 A1：交錯串流下的解碼器引發的輸入積壓。隨著利用率 $\rho$ 增加，交錯解碼暫停會放大趕趕延遲並進入積壓爆炸模式，而我們的解耦設計顯著減少了解碼器引發的積壓增長。

## D 預設基線細節

Flash-VStream-7B**[32]**。Flash-VStream 是一個用於長影片串流的效率高的視頻語言模型。它引入了一個快取記憶體模組，由一個低容量的上下文記憶體組成，用於匯集長範圍的時間信息，以及一個高容量的增強記憶體，用於檢索詳細的空間證據，能夠對超長影片進行實時響應用戶查詢。

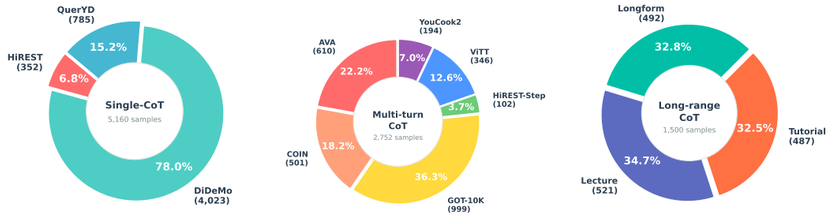
VideoLLM-online-8B**[8]**。 VideoLLM-online 提出LIVE框架，縮寫為在視頻流中學習，以實現時間對齊、長上下文和實時對話。

持續的影片串流。
方案，將離線時間標註轉換為串流對話數據，以及一個優化推理管線，包含持續的 KV 緩存，以及並行化的視覺編碼和語言解碼，以實現高效的線上響應。

Dispider-7B**[43]**。Dispider 旨在通過明確地解耦感知、決策和反應，實現主動的實時互動。它使用一個輕量級的積極串流視頻處理模組來持續監控串流並識別合適的互動時刻，同時一個非同步互動模組在無阻塞持續觀察的情況下生成詳細的響應。

StreamForest-7B**[33]**。StreamForest 是為高效能的線上影片理解而設計，具備持續性事件記憶。它的核心持續性事件記憶森林將歷史畫面組織成
計算資源有限下，長期保留的 event-level 樹結構，而
細粒度空間時間視窗保留詳細短期感知。

StreamAgent-7B**[35]**。StreamAgent研究預期性智能體，用於流式視頻理解。它不僅僅反應當前觀察，還整合了問題語義和歷史觀察，預期未來與任務相關的時間間隔和空間區域，並結合




(a) Single-round CoT. (b) Multi-round CoT. (c) Long-range CoT.

Figure A2: **Dataset composition**

this strategy with a streaming KV cache memory for selective recall, enabling proactive and goal-directed responses in evolving video streams.

# E Dataset and Benchmark Details

## E.1 Benchmark Details

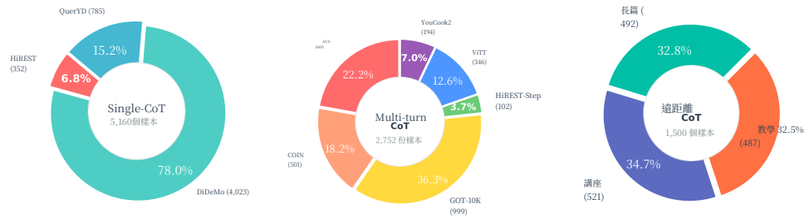
**StreamingBench [15].** StreamingBench is a benchmark tailored for streaming video understanding, containing 18 tasks over 900 videos and 4,500 human-curated QA pairs, where each question is associated with a specific timestamp in the video stream. The benchmark covers three major aspects of streaming understanding: real-time visual understanding, omnisource understanding, and contextual understanding. In our main tables, we further summarize the reported results into four subset-level metrics: **Realtime**, **OmniSource**, **SQA**, and **Proactive**. For evaluation, we follow the adaptive frame extraction protocol reported in StreamingBench: videos shorter than 5 minutes are sampled at 1 fps, videos between 5 and 10 minutes at 0.5 fps, and videos longer than 10 minutes at 0.2 fps.

**OVO-Bench [16].** OVO-Bench is designed to evaluate online video understanding with explicit temporal awareness. It organizes evaluation into three subsets: **Backward**, which requires tracing back to past events; **Realtime**, which focuses on understanding what is happening at the current timestamp; and **Forward**, which evaluates whether the model can defer its response until sufficient future evidence becomes available. In our experiments, we follow the common OVO-Bench comparison setting for offline video LLMs and cap the visual input at no more than 64 frames per query.

**Video-MME [44].** Video-MME is a comprehensive offline video benchmark with 900 videos and 2,700 multiple-choice QA pairs, spanning 6 primary visual domains and 30 subfields, and reporting results on **Short**, **Medium**, and **Long** duration subsets. To adapt Video-MME to our streaming-style evaluation, we aggregate all QA pairs sharing the same video ID into a single example and convert each video into an ordered stream of temporal segments. The model receives these segments sequentially, and the associated questions are issued only after the full video stream has been observed. This preserves the original benchmark content while turning Video-MME into a suffix-query streaming protocol.

**LV-Bench [45].** LV-Bench is an extreme long-video benchmark for evaluating long-range video understanding. It measures six core capabilities, namely **ER** (Entity Recognition), **EU** (Event Understanding), **KIR** (Key Information Retrieval), **TG** (Temporal Grounding), **Rea** (Reasoning), and **Sum** (Summarization). In our adaptation, we first aggregate samples by video ID, and then use the end time of the official annotated time span as the segment boundary to construct a streaming input sequence. The model is queried when the stream reaches the corresponding boundary, making the evaluation compatible with our streaming inference pipeline while preserving the original supervision structure.






(a) 單輪 CoT。 (b) 複輪 CoT。 (c) 遠距離 CoT。

**圖 A2：數據集組成**

這種策略與流式KV緩存記憶體，實現選擇性回憶，使能在演變的視頻流中進行主動和目標導向的響應。

# E 資料集與基準測試細節

## E.1 基準測試細節

StreamingBench **[15]**。StreamingBench 是一個針對串流視頻理解而設計的基準測試，包含 18 個任務、900 個影片和 4,500 人工編輯的 QA 對，其中每個問題都與視頻串流中的特定時間戳相關聯。該基準測試涵蓋了串流理解的三個主要方面：實時視覺理解、多來源理解以及語境理解。在我們的主表格中，我們將報告的結果總結為四個子集級別指標：實時、多來源、SQA 和主動。在評估方面，我們遵循 StreamingBench 中報告的自適應畫格提取協議：短於 5 分鐘的影片以 1 fps 進行採樣，5 到 10 分鐘的影片以 0.5 fps 進行採樣，而長於 10 分鐘的影片則以 0.2 fps 進行採樣。

OVO-Bench **[16]**。OVO-Bench 設計用於評估具有明確時間感知的線上視頻理解。它將評估組織為三個子集：Backward，要求追溯過去的事件；Realtime，專注於理解當前時間戳正在發生什麼；以及 Forward，評估模型是否可以延遲其回應，直到足夠的未來證據變得可用。在我們的實驗中，我們遵循 OVO-Bench 對於離線視頻 LLM 的常見比較設定，並將視覺輸入限制為每個查詢最多 64 個畫格。

Video-MME **[44]**。Video-MME 是一個綜合性的離線視頻基準，包含 900 個視頻和 2,700 多選題目對，涵蓋 6 個主要視覺領域和 30 個子領域，並在短、中、長時長子集上報告結果。為了將 Video-MME 應用於我們的流式評估，我們將所有具有相同視頻 ID 的題目對聚合為單一示例，並將每個視頻轉換為有序的時間片段流。模型按序接收這些片段，相關的問題僅在完整視頻流被觀察後才發出。這既保留了原始基準內容，又將 Video-MME 轉換為後綴查詢流式協議。

LV-Bench **[45]**。LV-Bench 是一個用於評估長程視頻理解的極端長視頻基準。它測量六個核心能力，即 ER（實體識別）、EU（事件理解）、KIR（關鍵信息檢索）、TG（時間 grounding）、Rea（推理）和 Sum（摘要）。在我們的改進版本中，我們首先按視頻 ID 統計樣本，然後使用官方標注時間範圍的結束時間作為片段邊界來構建流式輸入序列。當流達到對應的邊界時，會查詢模型，使評估與我們的流式推理管道兼容，同時保留原始的監督結構。

## E.2 Overall Dataset Composition

Figure A2 visualizes the three-stage data composition and its alignment with our training objectives: Stage 1 for single-round memory writing and answering, Stage 2 for multi-round consistency, and Stage 3 for long-horizon recall, uncertainty handling, and distractor robustness.

**VideoChatOnline-IT Source Pool.** Stage 1 and Stage 2 are both constructed from VideoChatOnline-IT, a video instruction-tuning corpus for online video understanding. It unifies multiple temporally grounded video QA sources in a streaming-style format, making it a suitable source pool for our pseudo streaming chain-of-thought (CoT) construction. We use different subsets for different training objectives. Stage 1 targets single-round streaming CoT and uses HiREST, DiDeMo, and QuerYD. Stage 2 targets multi-round streaming CoT and uses AVA, COIN, GOT-10K, HiREST-Step, ViTT, and YouCook2. Detailed statistics are reported in Table A2 and Table A3.

**Stage 1: Single-round CoT Statistics.** Stage 1 focuses on training the model to write segment-level memory notes and answer a single streaming question grounded in the observed prefix. Table A2 reports the breakdown by source subset.

Table A2: **Stage 1 detailed statistics.**

| Subset | Samples | Proportion (%) |
|---|---|---|
| HiREST | 352 | 6.8 |
| DiDeMo | 4023 | 78.0 |
| QuerYD | 785 | 15.2 |
| Total | 5,160 | 100.0 |

**Stage 2: Multi-round CoT Statistics.** Stage 2 trains multi-turn consistency: later answers must reuse earlier segment-level memory notes without peeking into future segments. Table A3 reports detailed statistics, including the number of segments and questions aggregated from the underlying sources.

Table A3: **Stage 2 detailed statistics.** Segments counts the number of segment units after stream segmentation, and Questions counts the number of question turns in the synthesized dialogues. Avg. Segs. and Avg. Qs. denote the average numbers of segments and questions per sample.

| Subset | Samples | Segments | Questions | Avg. Segs. | Avg. Qs |
|---|---|---|---|---|---|
| AVA | 610 | 6,136 | 1,902 | 10.06 | 3.12 |
| COIN | 501 | 2,793 | 1,109 | 5.57 | 2.21 |
| GOT-10K | 999 | 2,830 | 2,820 | 2.83 | 2.82 |
| HiREST-Step | 102 | 320 | 422 | 3.14 | 4.14 |
| ViTT | 346 | 1,090 | 1,436 | 3.15 | 4.15 |
| YouCook2 | 194 | 630 | 824 | 3.25 | 4.25 |
| Total | 2,752 | 13,799 | 8,513 | 5.01 | 3.09 |

**Stage 3: Long-range CoT Statistics and Retrieval Keywords.** Stage 3 targets long-horizon streaming behaviors on long videos collected from YouTube. We use 500+ keywords to retrieve candidate long videos, covering procedural workflows (tutorial), explanatory content (lecture), and continuous recordings (longform). Table A4 reports the resulting dataset statistics by category, and Table A6 lists representative query examples used for video retrieval.



## E.2 整體數據集組成

圖A2說明了三階段數據組成及其與我們訓練目標的對齊：階段1用於單輪記憶寫入和回答，階段2用於多輪一致性，以及階段3用於長遠期回憶、不確定性處理和干擾物強健性。

VideoChatOnline-IT 源數據集。第一階段和第二階段都是由 VideoChatOnline-IT 組成，這是一個用於線上視頻理解的視頻指令微調語料庫。它將多個基於時間的視頻問答來源統一在流式格式中，使其成為我們虛擬流式思維鏈 (CoT) 組建的合適源數據集。我們使用不同的子集進行不同的訓練目標。第一階段目標是單輪流式 CoT，並使用 HiREST、DiDeMo 和 QuerYD。第二階段目標是多輪流式 CoT，並使用 AVA、COIN、GOT-10K、HiREST-Step、ViTT 和 YouCook2。詳細統計數據報告在表 A2 和表 A3 中。

第一階段：單輪 CoT 統計。第一階段專注於訓練模型寫入段落級別的記憶筆記，並回答基於觀察到的前綴的單個流式問題。表 A2 報告了按源子集的細分情況。

**表A2：階段1詳細統計。**

| 子集 | 樣本 | 比例 (%) |
|---|---|---|
| HiREST | 352 | 6.8 |
| DiDeMo | 4023 | 78.0 |
| QuerYD | 785 | 15.2 |
| 總計 | 5,160 | 100.0 |

第二階段：多輪CoT統計。第二階段訓練多輪一致性：後續答案必須重用早期的段落層級記憶資訊，而不得窺探未來的段落。表A3報告了詳細統計資料，包括從底層來源聚合的段落數量和問題數量。

表A3：第二階段詳細統計。段落計數是流式分割後的段落單位數量，問題計數是合成對話中的問題輪次數量。Avg. Segs. 和 Avg. Qs. 分別表示每個樣本的平均段落數量和問題數量。

| 子集 | 樣本 | 片段 | 問題 | 平均片段 | 平均問題 |
|---|---|---|---|---|---|
| AVA | 610 | 6,136 | 1,902 | 10.06 | 3.12 |
| COIN | 501 | 2,793 | 1,109 | 5.57 | 2.21 |
| GOT-10K | 999 | 2,830 | 2,820 | 2.83 | 2.82 |
| HiREST-Step | 102 | 320 | 422 | 3.14 | 4.14 |
| ViTT | 346 | 1,090 | 1,436 | 3.15 | 4.15 |
| YouCook2 | 194 | 630 | 824 | 3.25 | 4.25 |
| 總計 | 2,752 | 13,799 | 8,513 | 5.01 | 3.09 |

第三階段：長距離CoT統計與檢索關鍵字。第三階段目標是針對從YouTube收集的長影片進行長時間跨度的串流行為。我們使用 500+ 關鍵字來檢索候選長影片，涵蓋程序式工作流程（教學影片）、解釋性內容（講座）以及持續錄製（長篇紀錄）。表A4報告了按類別的結果數據集統計，表A6列出了用於影片檢索的代表性查詢範例。

Table A4: **Stage 3 statistics by category.**

| Category | Samples | Segments | Questions | Avg. Segs. | Avg. Qs |
|---|---|---|---|---|---|
| Tutorial | 487 | 5,631 | 2,017 | 11.56 | 4.14 |
| Lecture | 521 | 4,532 | 1,893 | 8.70 | 3.63 |
| Longform | 492 | 5,536 | 2,090 | 11.25 | 4.25 |
| Total | 1,500 | 15,699 | 6,000 | 10.47 | 4.00 |

Table A6: **Representative YouTube search queries used for Stage 3 video retrieval.**

| Query keyword | Video type |
|---|---|
| **Tutorial** | |
| `sourdough bread tutorial` | Complete bread-making workflow |
| `furniture restoration` | Furniture restoration project |
| `oil painting tutorial` | Oil painting step-by-step tutorial |
| `car repair tutorial complete` | Full car repair walkthrough |
| `sewing tutorial complete` | Complete sewing tutorial |
| `woodworking project tutorial` | Woodworking project tutorial |
| `pottery making tutorial` | Pottery making process |
| `knife making tutorial` | Knife forging tutorial |
| **Lecture** | |
| `machine learning lecture` | Machine learning course lecture |
| `organic chemistry lecture` | Organic chemistry lecture |
| `quantum mechanics lecture` | Quantum mechanics lecture |
| `algorithm course full` | Full algorithm course |
| `system design lecture` | System design lecture |
| `neuroscience lecture` | Neuroscience lecture |
| `deep learning tutorial` | Deep learning lecture/tutorial |
| `computer vision lecture` | Computer vision lecture |
| **Longform** | |
| `hiking trail complete` | Full hiking trail recording |
| `train journey scenic` | Scenic train journey |
| `safari wildlife documentary` | Safari wildlife documentary |
| `Tokyo walking tour` | City walking tour (Tokyo) |
| `northern lights footage` | Northern lights raw footage |
| `coral reef documentary` | Coral reef documentary |
| `mountain climbing documentary` | Mountain climbing documentary |
| `storm chasing footage` | Storm chasing raw footage |

Table A5: **Core constraints for pseudo streaming CoT generation.**

| ID | Constraint |
|---|---|
| A | **Strict one-to-one alignment:** exactly one output chunk per input unit, preserving order. |
| B | **No empty segment:** each segment must update/maintain grounded state based on visual evidence. |
| C | **No future information:** segment reasoning uses only current and past units; QA uses evidence up to its timestamp. |
| D | **No answer leakage:** do not reveal reference answers in reasoning; copy the answer only in the final Answer field. |
| E | **Video-grounded only:** rely on provided frames/timestamps/metadata; avoid external assumptions. |
| F | **Streaming quality:** emphasize boundary cues and conservative state updates across segments. |
| G | **Question awareness:** track unanswered questions and prioritize collecting relevant evidence online. |



A4表：階段3按類別統計。

| 類別 | 樣本 | 區段 | 問題 | 平均區段數 | 平均 Qs |
|---|---|---|---|---|---|
| 教學教程 | 487 | 5,631 | 2,017 | 11.56 | 4.14 |
| 講座 | 521 | 4,532 | 1,893 | 8.70 | 3.63 |
| 長篇內容 | 492 | 5,536 | 2,090 | 11.25 | 4.25 |
| 總計 | 1,500 | 15,699 | 6,000 | 10.47 | 4.00 |

表A6：用於第三階段影片檢索的代表YouTube搜尋詞彙。

| 查詢關鍵字 | 影片類型 |
|---|---|
| **教學教程** | |
| 酸麵包教學 | 完整麵包製作流程 |
| 家具修復 | 家具修復專案 |
| 油畫教學 | 油畫逐步教學 |
| 汽車維修教學完成 | 完整汽車維修教學 |
| 縫紉教學完成 | 完成縫紉教學 |
| 木工專案教學 | 木工專案教學 |
| 陶器製作教學 | 陶器製作流程 |
| 刀具製作教學 | 刀具鍛造教學 |
| **講座** | |
| 機器學習講座 | 機器學習課程講座 |
| 有機化學講座 | 有機化學講座 |
| 量子力學講座 | 量子力學講座 |
| 演算法課程全系列 | 完整演算法課程 |
| 系統設計講座 | 系統設計講座 |
| 神經科學講座 | 神經科學講座 |
| 深度學習教學 | 深度學習講座/教學 |
| 電腦視覺講座 | 電腦視覺講座 |
| **長篇內容** | |
| 登山步道已完成 | 完整登山步道錄影 |
| 火車旅行風景 | 風景火車旅行 |
| 野生動物紀錄片 | 野生動物紀錄片 |
| 東京步行導覽 | 城市步行導覽（東京） |
| 極光影片 | 極光原始影片 |
| 珊瑚礁紀錄片 | 珊瑚礁紀錄片 |
| 登山紀錄片 | 登山紀錄片 |
| 追風影片 | 追風原始影片 |

表A5：虛擬串流CoT產生之核心限制。

| ID | 限制 |
|---|---|
| A | 嚴格的一對一對齊：每個輸入單位剛好對應一個輸出片段，並保留順序。 |
| B | 無空片段：每個片段必須基於視覺證據更新/維持有根狀態。 |
| C | 無未來資訊：片段推理僅使用當前與過去單位；QA 僅使用其時間戳之前的證據。 |
| D | 無答案洩露：推理過程不揭露參考答案；僅在最終「答案」欄位中複製答案。 |
| E | 僅基於視頻：依賴提供的畫格/時間戳/元資料；避免外部假設。 |
| F | 串流品質：強調邊界提示及跨片段的保守狀態更新。 |
| G | 問題意識：追蹤未回答問題並優先線上收集相關證據。 |

### E.3 Pseudo Streaming CoT Generation Principles

We synthesize pseudo streaming CoT annotations to match the streaming protocol in Sec. 3. A key constraint is strict alignment: for a stream with $S$ segments and $Q$ questions, the synthesized output must contain exactly $S + Q$ generated items, one for each interleaved unit and in temporal order. Table A5 summarizes the core constraints enforced during CoT synthesis.

**Prompt template.** For reproducibility, we provide the complete prompt used to synthesize pseudo streaming CoT. We use special tokens to delimit streaming units: <EOS> ends an input segment unit, <EOQ> ends an input question unit (and also the corresponding QA output), and <EOT> ends a generated segment-reasoning chunk. In training data, we keep only essential delimiters to reduce overfitting to superficial formatting.

```
general_prompt = '''
You are a pseudo streaming Video Chain-of-Thought (CoT) generator.
You will receive the FULL input at once (all video segments + all questions +
↪  reference answers),
but you MUST generate reasoning that looks as if you processed the video
↪  incrementally, in time order.
====================
1) CORE CONSTRAINTS
====================
(A) STRICT ONE-TO-ONE ALIGNMENT (mandatory)
- The input is a chronological sequence of interleaved units.
- Video segment units end with <EOS>.
- Question units end with <EOQ>.
- You MUST output exactly ONE reasoning chunk per input unit.
- Output order must exactly match input order.
- Total output chunks = (#segment units + #question units).
- Do NOT merge units.
- Do NOT split units.
- Do NOT reorder units.
(B) EVERY SEGMENT REQUIRES REASONING (no empty segments)
- For every video segment unit, produce meaningful reasoning grounded in that
↪  segment.
- If no task-relevant change occurs, explicitly state that the scene, action, or
↪  tracked state remains stable.
- Avoid meta statements; focus on video evidence and state continuity.
(C) pseudo streaming / NO FUTURE INFORMATION
- Although you see the full input, behave as if you only know information up to
↪  the current unit.
- In [SEG k THINK], you may ONLY use evidence from segment k and all earlier
↪  segments.
- NEVER use or hint at information that appears only in future segments,
↪  questions, or answers.
- In [Q j THINK], reason ONLY with evidence available up to the question
↪  timestamp t.
(D) NO ANSWER LEAKAGE
- Each question includes a Reference Answer (for alignment only).
- NEVER reveal, paraphrase, or hint at any reference answer in any segment
↪  reasoning.
- In [Q j THINK], output the final answer ONLY inside the Answer field.
- The Answer MUST be copied EXACTLY and VERBATIM from the Reference Answer.
- Do NOT leak the answer in the Reasoning field.
(E) VIDEO-GROUNDED ONLY
- Use only visual evidence provided by frames, timestamps, bounding boxes, and
↪  metadata.
- Do NOT rely on external knowledge, assumptions, or commonsense completion.
(F) STREAMING VIDEO REASONING QUALITY
Each segment reasoning should:
- Describe what is visually observed or confirmed in this segment.
- Emphasize continuity, change, or boundary cues (start / end / ongoing).
- Update internal task-specific state clearly and conservatively.
(G) QUESTION AWARENESS (when applicable)
- If one or more questions have appeared:
```



### E.3 偽流式 CoT 生成原則

我們合成偽流式 CoT 註解來匹配第 3 節中的流式協議。一個關鍵約束是嚴格對齊：對於一個包含 $S$ 個片段和 $Q$ 個問題的流，合成的輸出必須包含正好 $S + Q$ 個生成項目，每個交錯單元一個，並且按時間順序排列。表 A5 總結了在 CoT 合成過程中強制的核心約束。

提示模板。為了可重現性，我們提供了用於合成虛擬流式 CoT 的完整提示。我們使用特殊標記來界定流式單元： <EOS> 結束輸入片段單元，<EOQ> 結束輸入問題單元（同時也結束對應的 QA 輸出），以及 <EOT>結束生成的片段推理塊。在訓練數據中，我們僅保留必要的界定符，以減少對表面格式過度擬合。

```
general_prompt = '''
You are a pseudo streaming Video Chain-of-Thought (CoT) generator.
You will receive the FULL input at once (all video segments + all questions +
↪  reference answers),
but you MUST generate reasoning that looks as if you processed the video
↪  incrementally, in time order.
====================
1) CORE CONSTRAINTS
====================
(A) STRICT ONE-TO-ONE ALIGNMENT (mandatory)
- The input is a chronological sequence of interleaved units.
- Video segment units end with <EOS>.
- Question units end with <EOQ>.
- You MUST output exactly ONE reasoning chunk per input unit.
- Output order must exactly match input order.
- Total output chunks = (#segment units + #question units).
- Do NOT merge units.
- Do NOT split units.
- Do NOT reorder units.
(B) EVERY SEGMENT REQUIRES REASONING (no empty segments)
- For every video segment unit, produce meaningful reasoning grounded in that
↪  segment.
- If no task-relevant change occurs, explicitly state that the scene, action, or
↪  tracked state remains stable.
- Avoid meta statements; focus on video evidence and state continuity.
(C) pseudo streaming / NO FUTURE INFORMATION
- Although you see the full input, behave as if you only know information up to
↪  the current unit.
- In [SEG k THINK], you may ONLY use evidence from segment k and all earlier
↪  segments.
- NEVER use or hint at information that appears only in future segments,
↪  questions, or answers.
- In [Q j THINK], reason ONLY with evidence available up to the question
↪  timestamp t.
(D) NO ANSWER LEAKAGE
- Each question includes a Reference Answer (for alignment only).
- NEVER reveal, paraphrase, or hint at any reference answer in any segment
↪  reasoning.
- In [Q j THINK], output the final answer ONLY inside the Answer field.
- The Answer MUST be copied EXACTLY and VERBATIM from the Reference Answer.
- Do NOT leak the answer in the Reasoning field.
(E) VIDEO-GROUNDED ONLY
- Use only visual evidence provided by frames, timestamps, bounding boxes, and
↪  metadata.
- Do NOT rely on external knowledge, assumptions, or commonsense completion.
(F) STREAMING VIDEO REASONING QUALITY
Each segment reasoning should:
- Describe what is visually observed or confirmed in this segment.
- Emphasize continuity, change, or boundary cues (start / end / ongoing).
- Update internal task-specific state clearly and conservatively.
(G) QUESTION AWARENESS (when applicable)
- If one or more questions have appeared:
```

```
  - Maintain an "Active Question": the earliest question that has not been
  ↪  answered.
  - Segment reasoning should prioritize collecting evidence relevant to the
  ↪  Active Question.
- If NO question has appeared yet:
  - Focus ONLY on understanding the video stream itself: scene setup,
  ↪  object/person continuity,
    motion patterns, emerging actions.
  - Do NOT speculate about future questions.
====================
2) INPUT FORMAT
====================
Units are interleaved in chronological order.
Video segment unit:
[SEG k | time = start-end | frames = ... | optional: bboxes / ids / actions]
↪  <EOS>
Question unit:
[Q j | t = timestamp]
Question: ...
Reference Answer: ...
<EOQ>
Notes:
- Timestamps may not start at 0; follow the provided time system exactly.
- Reference Answers are provided for alignment only.
====================
3) OUTPUT FORMAT (STRICT)
====================
For each video segment unit:
[SEG k THINK]
Focus: (either the Active Question in <= 15 words, OR "video understanding (no
↪  question yet)")
Evidence from this segment: (2-5 sentences, strictly video-grounded)
State update: (1-3 sentences, task-specific state or continuity)
<EOT>

For each question unit:
[Q j THINK]
Reasoning: (2-6 sentences, justify the answer using ONLY evidence available up to
↪  time t;
           you may reference segment indices/timestamps, but MUST NOT use future
           ↪  units
           and MUST NOT reveal/paraphrase the Reference Answer in Reasoning)
Answer: (copy the Reference Answer EXACTLY and VERBATIM)
<EOQ>
'''
```

## F Case Studies

This section presents three qualitative examples that complement the quantitative results. Figure A3 shows a dataset-derived pseudo streaming CoT example under the single-round protocol. Figure A4 shows a dataset-derived pseudo streaming CoT example under the multi-round protocol. Figure A5 shows a real multi-round streaming example and illustrates how segment-level memory supports cross-turn reference resolution and temporal state tracking.

## G Error Analysis

Although Stage 3 substantially improves long-horizon streaming reasoning, representative residual failures still remain in challenging multi-turn settings. Consistent with the three long-horizon behaviors explicitly targeted in Stage 3 training—long-term evidence retention, uncertainty handling, and distractor learning, as described in Sec. 4.3—we observe three recurring error patterns. First, the model may retain the coarse event trace while forgetting an early fine-grained attribute, such as which object, side, or entity was involved. This is consistent with our segment-level memory




```
  - Maintain an "Active Question": the earliest question that has not been
  ↪  answered.
  - Segment reasoning should prioritize collecting evidence relevant to the
  ↪  Active Question.
- If NO question has appeared yet:
  - Focus ONLY on understanding the video stream itself: scene setup,
  ↪  object/person continuity,
    motion patterns, emerging actions.
  - Do NOT speculate about future questions.
====================
2) INPUT FORMAT
====================
Units are interleaved in chronological order.
Video segment unit:
[SEG k | time = start-end | frames = ... | optional: bboxes / ids / actions]
↪  <EOS>
Question unit:
[Q j | t = timestamp]
Question: ...
Reference Answer: ...
<EOQ>
Notes:
- Timestamps may not start at 0; follow the provided time system exactly.
- Reference Answers are provided for alignment only.
====================
3) OUTPUT FORMAT (STRICT)
====================
For each video segment unit:
[SEG k THINK]
Focus: (either the Active Question in <= 15 words, OR "video understanding (no
↪  question yet)")
Evidence from this segment: (2-5 sentences, strictly video-grounded)
State update: (1-3 sentences, task-specific state or continuity)
<EOT>

For each question unit:
[Q j THINK]
Reasoning: (2-6 sentences, justify the answer using ONLY evidence available up to
↪  time t;
           you may reference segment indices/timestamps, but MUST NOT use future
           ↪  units
           and MUST NOT reveal/paraphrase the Reference Answer in Reasoning)
Answer: (copy the Reference Answer EXACTLY and VERBATIM)
<EOQ>
'''
```


## F 案例研究

本節介紹了三個定性的範例，以補充定量結果。圖 A3 展示了單輪協議下來自數據集的虛擬流式 CoT 範例。圖 A4 展示了多輪協議下來自數據集的虛擬流式 CoT 範例。圖 A5 展示了一個真實的多輪流式範例，並說明了如何通過片段級記憶支持跨輪次參照解析和時間狀態追蹤。

## G 錯誤分析

雖然 Stage 3 大幅提升了長時程串流推理能力，但在挑戰性的多輪次設定中，仍存在代表性的殘差失敗。與 Stage 3 訓練中明確目標的三種長時程行為——長期證據保留、不確定性處理以及干擾學習（如 Sec. 4.3 所述）——一致，我們觀察到三種重複出現的錯誤模式。首先，模型可能保留粗略的事件追蹤，而忘記早期的細粒度屬性，例如涉及的是哪個物件、哪一邊或哪個實體。這與我們在片段層級的記憶

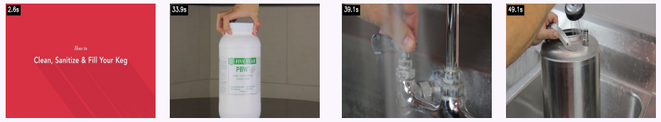
27

design: compact memory notes support long-range access but can still over-compress details over long temporal gaps. Second, under incomplete evidence, the model may commit to a specific hypothesis too early rather than deferring judgment until decisive visual evidence appears, reflecting a residual limitation of the uncertainty-handling objective in Sec. 4.3. Third, later retrieval can still be corrupted by visually salient but task-irrelevant segments, causing recent distractors to override the true earlier evidence. This aligns with the ablation results in Table 5, which show that memory notes help stabilize retrieval but remain limited by the quality of written evidence and incoming visual context. As shown in Fig. A6, these failures are residual edge cases of (a) long-range recall failure, (b) premature commitment under incomplete evidence, and (c) distractor-induced memory contamination in streaming multi-turn reasoning.



如第4.3節所述，這些失敗是（a）長程回憶失敗、（b）在證據不完整時過早承諾，以及（c）在串流多輪推理中干擾引發的記憶污染的殘餘邊緣情況。

## Single-CoT




COT:

**[SEG 1 THINK]**
Focus: Video understanding (no question yet)
Evidence: The clip opens with an animated title card, “How to Clean, Sanitize & Fill Your Keg.” It then cuts to live-action shots of a white container labeled “FIVE STAR PBW,” followed by close-ups of a faucet being turned on and water running. A large stainless-steel keg sits in a stainless sink, and water is sprayed/rinsed into the keg opening.
State update: The scene is centered around the sink area. The keg is actively being rinsed, indicating the cleaning process is in progress and not yet completed.<*EOT*>

**Question**:Has the event 'Clean a Beer Keg' ended, or is it still ongoing?**(49.1s)**<*EOQ*>

**Ground Truth**:The event happens from 35.0s to now (duration: 14.1 seconds).

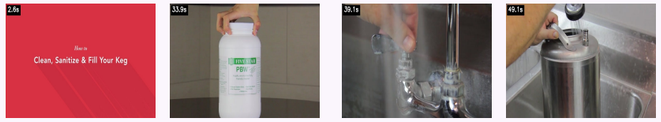
COT:

**[Q 1 THINK]**
Reasoning: By 49.1s, the keg remains in the sink and water is still being directed into the opening, which signals the cleaning action is ongoing. There are no visual cues of completion (e.g., stopping the water, removing the keg, or transitioning to a different step). The earlier title card establishes the task theme, and the subsequent faucet/rinsing shots confirm the “cleaning a beer keg” event is still underway at the current timestamp.
Answer: The event happens from 35.0s to now (duration: 14.1 seconds).<*EOQ*>

Figure A3: **Dataset-derived pseudo streaming CoT example under the single-round protocol.** A question is asked once after the observed video prefix. The model first writes a segment-level memory note from the incoming frames, identifying the tutorial title card, the cleaning material, the faucet, and the rinsing action around the keg. When queried at 49.1s, it answers using only the accumulated evidence so far and correctly concludes that the event **Clean a Beer Keg** is still ongoing.



## 單一式CoT




COT:

**[第一節 考慮]**
重點：影片理解（尚未出現問題）證據：影片開頭是動畫標題卡「如何清潔、消毒並裝滿您的啤酒桶」；接著切換到真實拍攝的白色容器標示「五星PBW」的畫面，隨後是水龍頭打開及水流的小畫面。一個大型不鏽鋼啤酒桶放在不鏽鋼水槽中，水正被噴灑/沖洗進啤酒桶開口。狀態更新：場景以水槽區域為中心。啤酒桶正被積極沖洗，表明清潔過程正在進行中且尚未完成.<EOT>

問題：'清潔啤酒桶'這個事件已經結束，還是仍在進行中？(49.1秒)<EOQ>

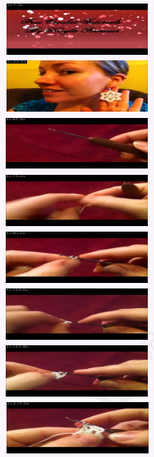
真實情況：該事件發生於35.0秒至今（持續時間：14.1秒）。

COT:

**[第一題 考慮]**
理由：在49.1秒時，啤酒桶仍留在水槽中，水仍被導入開口，這表示清潔動作仍在進行中。沒有視覺上的完成信號（例如，停止流水、移走啤酒桶，或轉換到不同步驟）。早期的標題卡確立了任務主題，而後續的水龍頭/沖洗畫面確認「清潔啤酒桶」的事件在當前時間戳仍在進行中。答案：事件發生自35.0秒至今（持續時間：14.1秒）。<EOQ>

圖A3：單輪協議下，基於數據集衍生的偽流式CoT範例。在觀察到的視頻前綴後僅問一次問題。模型首先從輸入畫面寫入段落層級記憶筆記，識別教程標題卡、清潔材料、水龍頭以及圍繞啤酒桶的沖洗動作。在49.1秒時被查詢，它僅使用到目前累積的证据回答，並正確結論出事件'清潔啤酒桶'仍在進行中。

## Multi-Turn-CoT



COT:

**[SEG 1 THINK]** : Red background animation, displaying the title "Free Crochet Tutorials By Kayla Fournier".**(t1)**<*EOT*>
**[SEG 2 THINK]** : The woman continues to speak to the camera, showing a snowflake earring.**(t2)**<*EOT*>
**[SEG 3 THINK]** : Switch to a close-up of the hands, the crochet hook and white thread start to be woven.**(t3)**<*EOT*>
**[SEG 4 THINK]** : The initial circular shape is being established, in the early stage of knitting.**(t4)**<*EOT*>
**[SEG 5 THINK]** : Continue adding stitches around the small circle.**(t5)**<*EOT*>
**[SEG 6 THINK]** : The first round decorative stitch is completed, forming a flower-like pattern.**(t6)**<*EOT*>
**[SEG 7 THINK]** : Continue adding stitch groups along the edge.**(t7)**<EOT>
**[SEG 8 THINK]** : Continue establishing the outer stitches.**(t8)**<*EOT*>

**Question**:Based on the current observation, list the events that concluded up to the current frame and were not reported in earlier outputs. Ensure your response follows the format: \<start time\> - \<end time\> (duration: \<x\> seconds), \<description\>.**(417.6s)**<*EOQ*>
**Ground Truth**: ...

COT:

**[Q 1 THINK]**
Reasoning: ...
Answer: 65.9s - 73.2s: Demonstrating the use of knitting needles and yarn as tools 73.2s - 80.5s: Creating the initial loop and starting chain.
<*EOQ*>

COT:

**[SEG 9-14 THINK]** :Continue the external stitching, form the edges, cut the threads, secure the tail end, display the finished product.**(t9)**<*EOT*>

**Question**:*the same question* **(t=417.6s)**<*EOQ*>
**Ground Truth**: ...

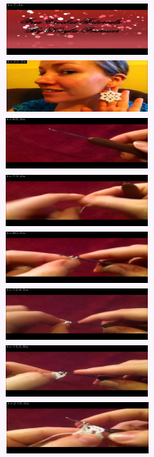
COT:

**[Q 2 THINK]**
Reasoning: ...
Answer:
- 275.0s - 335.0s: Weave in a new loop to create the outer edge
- 365.0s - 402.8s: Trim and secure the end of the thread<*EOQ*>

Figure A4: **Dataset-derived pseudo streaming CoT example under the multi-round protocol.** The stream follows a crochet tutorial. Memory notes from Segments 1 to 8 trace the progression from the introduction to close-up hand actions and the gradual formation of the crochet pattern. After the first event-listing question, the model reports only the completed events observed so far. As additional segments arrive, it updates the memory with later stitching and finishing steps. When the same question is asked again at 417.6s, the answer includes only the newly completed events that were not reported earlier, which illustrates incremental reasoning across multiple turns over a continuous stream.



## 多輪CoT



COT:

[第一段思考]：紅色背景動畫，顯示標題「Kayla Fournier的免費鉤扣針教學」。(t1)<EOT>
[第二段思考]：這位女士繼續對鏡頭說話，展示一隻雪花耳環。(t2)<EOT>

[第三段思考]：切換到手的特寫，鉤扣針
針和白色線開始編織。(t3)<EOT>
[第四段思考]：初始的圓形形狀正在建立，在
編織的早期階段。(t4)<EOT>
[SEG 5 思考]：繼續在圓圈周圍加針
。(t5)<EOT>
[SEG 6 思考]：第一輪裝飾針法已完成，
形成花狀圖案。(t6)<EOT>
[SEG 7 思考]：繼續沿著
edge.(t7)<EOT>
[SEG 8 思考]：持續建立外圍
縫線。(t8)<EOT>

問題：根據目前的觀察，列出截至當前畫格且未在先前輸出中報告的事件。請確保您的回應遵循以下格式：\<開始時間\> - \<結束時間\>
(持續時間：\<x\> 秒)，\<描述\>。(417.6s)<EOQ>真實標籤：...

COT:

**[Q 1 思考]**
推理：...
回覆：65.9秒 - 73.2秒：示範針織針與
毛線作為工具 73.2秒 - 80.5秒：建立初始環狀並開始鎖針。
<*EOQ*>

COT:

[SEG 9-14 思考]：繼續外部縫合，塑形邊緣，剪斷線頭，固定尾部，展示成品。(t9)<EOT>

**問題：相同問題（t=417.6秒）<EOQ>實際情況：...**

COT:

**[Q 2 思考]**
推理：...
回覆：
- 275.0秒 - 335.0秒：織入一個新圈圈來創造外緣
- 365.0秒 - 402.8秒：修剪並固定線頭<EOQ>

圖A4：基於數據集產生的虛擬串流CoT範例，在多輪協議下。串流遵循針織教程。從第一到第八段的記憶筆記追蹤從介紹到特寫手部動作以及針織圖案的逐漸形成。在第一個事件列舉問題後，模型僅報告到目前為止觀察到的已完成事件。隨著額外的段落下達，它用後續的針織和完成步驟更新記憶。當在417.6秒再次提出相同問題時，答案僅包含先前未報告的新完成事件，這說明了在連續串流的多輪中進行增量推理。

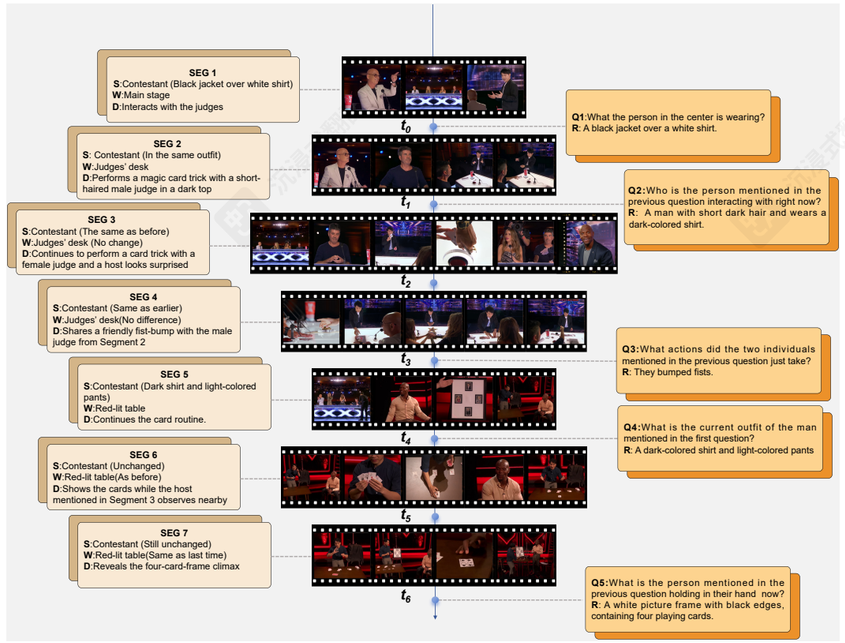
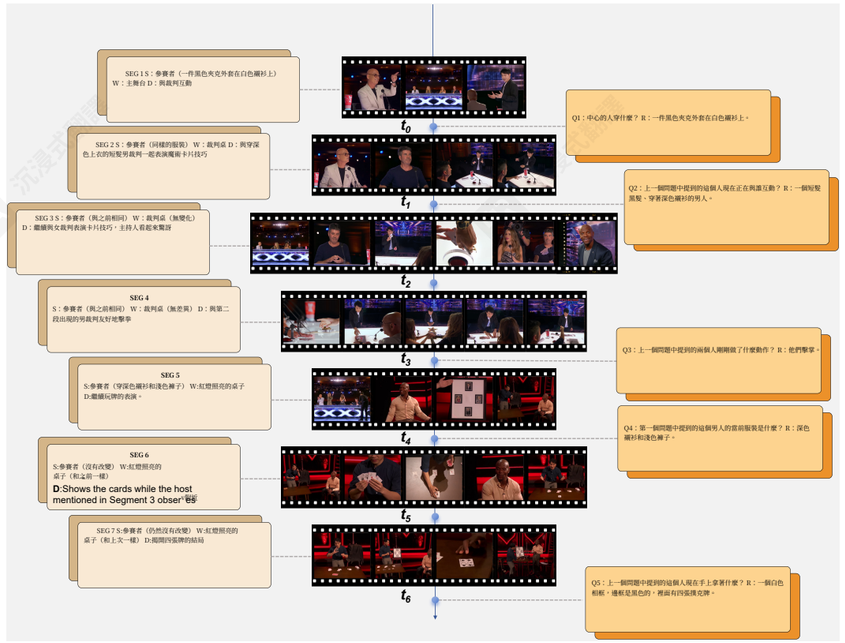



Figure A5: **Real multi-round streaming example.** Left: Segment-level memory notes summarize the contestant's appearance, location, and actions across seven streamed segments. Middle: Representative frames show the progression of a card routine from the main stage to the judges' desk and finally to a red-lit table. Right: Five temporally ordered questions are interleaved with the stream. Questions Q2 to Q5 depend on earlier turns, requiring the model to resolve the previously mentioned person, identify the current interaction partner, recall the recent fist bump, update the contestant's outfit after the scene transition, and track the object held in hand at the end. This example illustrates how persistent segment-level memory supports consistent reasoning across multiple turns over a changing stream.






圖A5：真實的多輪流式示範。左：片段級別的記憶筆記總結了參賽者在七個流式片段中的外貌、位置和動作。中：代表性框架展示了從主舞台到裁判桌，最後到紅燈桌的撲克牌花式表演的進程。右：五個按時間順序排列的問題與流式互銜。問題Q2到Q5依賴於之前的輪次，需要模型解決之前提到的這個人、識別當前互動對象、回憶最近的擊掌、在場景轉換後更新參賽者的服裝，以及追蹤最終手上的物件。這個例子說明了持續的片段級別記憶如何支持在變化的流中跨多輪次進行一致推理。

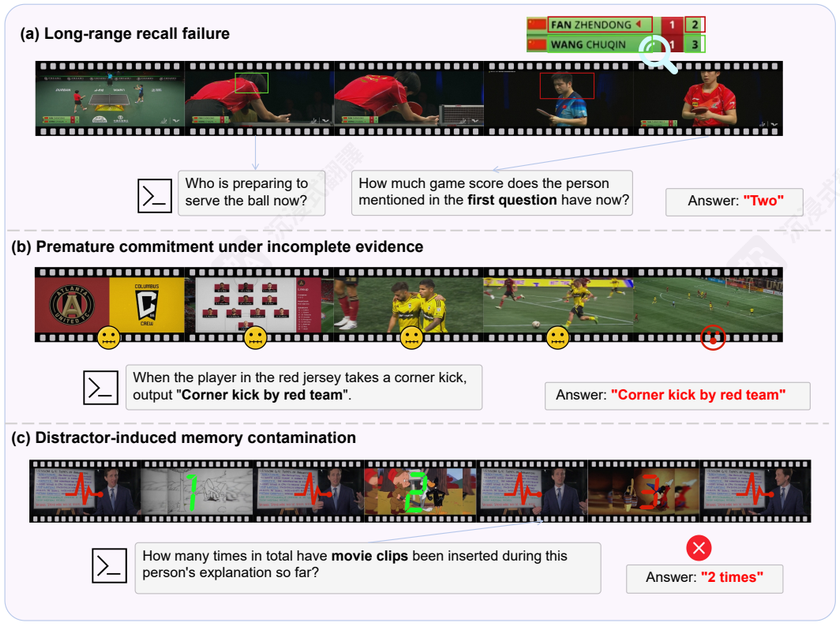
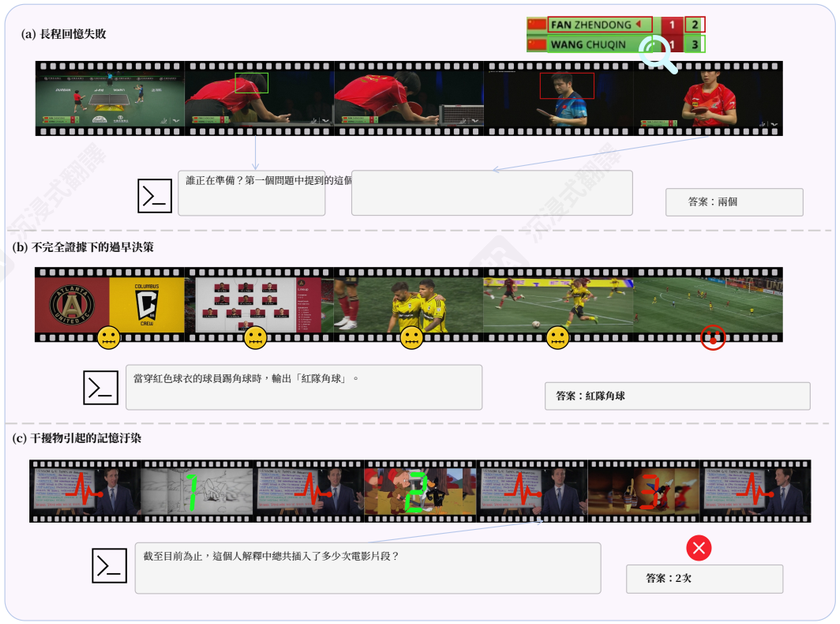



Figure A6: **Residual failure cases in multi-turn streaming reasoning. (a) Long-range recall failure.** The model fails to recover an early fine-grained identity cue across turns. As illustrated in the top row, the green box marks the correct person referred to in the first question, whereas the red box marks the person incorrectly retrieved by the model in the later question, indicating that the coarse event context is retained but the specific identity association is lost. **(b) Premature commitment under incomplete evidence.** The middle row shows that, at the queried moment, the decisive visual evidence for whether the red-jersey player is taking a corner kick has not yet become sufficiently clear. Instead of deferring its response until the event is visually confirmed, the model commits to a specific answer too early. **(c) Distractor-induced memory contamination.** The bottom row shows repeated alternation between the speaker and inserted movie clips. Because these inserted clips appear interleaved with the main explanation, the model is distracted by the intervening visual switches and produces an incorrect insertion count. These failures are residual edge cases of the three long-horizon behaviors explicitly targeted in Stage 3 training, namely long-term evidence retention, uncertainty handling, and distractor robustness.






圖A6：多輪流式推理中的殘餘失效案例。(a)長程回憶失效。模型無法在輪次間恢復早期的細粒度身份線索。如頂行所示，綠色方框標示第一個問題中提到的正確人員，而紅色方框標示模型在後續問題中錯誤找回的人員，表明粗略事件語境被保留，但具體身份聯繫已丟失。(b)不完整證據下的過早確定。中間行顯示，在查詢時刻，決定性的視覺證據尚未足夠清晰地表明紅衣球員是否正在執行角球。模型沒有延遲其回應直到事件視覺確認，而是過早地確定了一個具體答案。(c)干擾引發的記憶污染。底行顯示講者和插入影片片段之間的持續交替。由於這些插入片段與主解釋交錯出現，模型被中間的視覺切換干擾，並產生了不正確的插入次數。這些失效是第三階段訓練中明確針對的三種長時程行為的殘餘邊緣案例，即長期證據保留、不確定性處理和干擾強健性。